

ACADEMIC REGULATIONS, COURSE STRUCTURE

AND

DETAILED SYLLABI (A22 Regulations)

FOR

B. Tech – CSE I to IV Year

(Applicable for the Batches admitted from 2022 -2023)



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SREENIDHI INSTITUTE OF SCIENCE AND TECHNOLOGY

(An Autonomous Institution approved by UGC and affiliated to JNTUH)

Yamnampet, Ghatkesar, Hyderabad - 501 301

B.Tech (Computer Science and Engineering)



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Professor and Head – Department of CSE

It is with great pleasure that I welcome you to the Computer Science and Engineering (CSE) department at our esteemed college. As the Head of the Department, I am privileged to lead a team of distinguished faculty members who are not only educators but also mentors and researchers and experts in their respective fields. Our curriculum is meticulously designed and undergoes periodic updates to align with the ever-evolving demands of the software industry. This ensures that our graduates are well-equipped with the latest skills and knowledge, making them valuable contributors to the workforce.

In addition to academic rigor, the CSE department encourages a culture of research and innovation. Our students and faculty actively engage in cutting-edge research projects, contributing to advancements in various domains of computer science. We foster an environment that promotes creativity, critical thinking, and problem-solving skills, preparing our students to face the challenges of the digital era. Beyond the classroom, the department organizes workshops, seminars, and industry interactions to provide students with real-world insights and opportunities to interact with industry leaders. We also encourage participation in hackathons, coding competitions, and open-source projects, fostering a spirit of collaboration and hands-on learning. At the CSE department, we understand the importance of practical skills in today's competitive landscape. Therefore, we provide numerous opportunities for hands-on learning, industry exposure, and real-world applications of theoretical knowledge. Our goal is not just to produce graduates but to nurture individuals who can confidently tackle challenges and lead in the ever-changing world of technology.

Vision of the Department - B.Tech (CSE)

To emerge as a leading department in Technical Education and Research in India in Computer Science and Engineering with focus to produce professionally competent and socially sensitive engineers capable of working in global environment.

Mission of the Department - B.Tech (CSE)

1. To prepare Computer Science and Engineering graduates to be a lifelong learner with competence in basic science & engineering and professional core, multidisciplinary areas, with continuous update of the syllabus, so that they can succeed in industry as an individual and as a team or to pursue higher studies or to become an entrepreneur.
2. To enable the graduates to use modern tools, design and create novelty based products required for the society and communicate effectively with professional ethics.
3. To continuously engage in research and projects development with financial management to promote scientific temper in the graduates and attain sustainability.

Program Educational Objectives (PEOs) - B.Tech (CSE)

- I. Graduates will have a strong foundation in fundamentals of mathematics, basic sciences, computer science and engineering sciences with abilities to analyze problems, design and development of optimal solutions to address societal problems.
- II. Apply knowledge of diverse and emerging areas of computer science with modern tools to solve the complex problems; also be sensitive towards health, safety, legal, environmental and sustainable issues to be professionally competent and globally employable.
- III. Develop an ability to work effectively as an individual/team member or a leader or pursue entrepreneurial skills with good communications and ethical ideologies, practice project and finance management skills.
- IV. Acquire research temperament to pursue higher studies in reputed national and international institutes and become life-long learners.

Program Outcomes (POs) - B.Tech (CSE)

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

Correlation between the PEOs and the POs

PEO's	Program Outcomes (PO's 1-12)											
	1	2	3	4	5	6	7	8	9	10	11	12
I	H	M	M			L						
II				H	H	H	M					
III								H	H	H	M	
IV				L								H

Program Specific Outcomes (PSOs) - B.Tech (CSE)

On completion of the B.Tech.(Computer Science and Engineering) program the graduates will be able to:

1. Apply the knowledge of computer architecture, software development life cycle, networking and database concepts with emphasis on data structures and algorithms to analyze the engineering problems.
2. Design and develop programs and projects using software engineering practices, mathematical models, data mining techniques and algorithms to solve the societal problems with modern tools, web technologies and appropriate programming languages.
3. Solve complex engineering problems using disruptive technologies like Cloud Computing, Internet of Things (IoT), Data Science, Artificial Intelligence, Machine Learning, Cyber Security and Blockchain.

Academic Rules and Regulations (A22) for B.Tech. students (w.e.f. 2022-23)

1.0 Under Graduate Degree Programme in Engineering

- 1.1 Sreenidhi Institute of Science and Technology (SNIST) offers a 4-year (8 semesters) **Bachelor of Technology** (B.Tech.) degree programme under the Choice Based Credit System (CBCS) in the following branches of Engineering.

Sl. No.	Branch	Branch code
1	Civil Engineering (CE)	01
2	Computer Science and Engineering (CSE)	05
3	CSE-Artificial Intelligence and Machine Learning (AI&ML)	66
4	CSE-Cyber Security (CS)	62
5	CSE-Data Science (DS)	67
6	CSE-Internet of Things (IoT)	69
7	Electrical and Electronics Engineering (EEE)	02
8	Electronics and Communication Engineering (ECE)	04
9	Information Technology (IT)	12
10	Mechanical Engineering (ME)	03

- 1.2 The medium of instruction is **English**.

2.0 Eligibility for admission

Admission to all B.Tech. programmes will be made either on the basis of the rank of the candidate in the entrance test conducted by the Telangana State Government (EAMCET) or on the basis of any other order of merit approved by the affiliated University, subject to the reservations as prescribed by the Government from time to time. However, admissions under the management/NRI category will be made based on the relevant orders issued by the Government of Telangana from time to time.

3.0 B.Tech. Programme structure

- 3.1. A student after securing admission will pursue the B.Tech. programme for a minimum of 8 academic semesters, and a maximum of 16 academic semesters starting from the commencement of the 1st year 1st semester, failing which the student shall forfeit his/her seat in the B.Tech. programme. However, the student can take four more semesters to appear for the examinations to clear the backlog courses.

The Joint Board of Studies and the Academic Council of the Institute has approved the total number of credits to be 160 for the A22 regulation.

In the 1st year, the course work is structured to provide 37 credits, followed by 44, 42 and 37 credits in the 2nd, 3rd and 4th years, respectively. Each student will secure 160 credits with CGPA ≥ 5 for

the completion of the B.Tech. programme and the award of the degree. However, any revision made in this regard and approved by the Academic Council of the college and by the affiliated University will be implemented from the date of the revision.

3.2 UGC/AICTE specified definitions/descriptions are adopted appropriately for various terms and abbreviations stated below.

3.2.1 Semester scheme

Duration of the B.Tech. programme is of 8 academic semesters of 22 weeks each (with a minimum of 90 instructional days). Each semester has Continuous Internal Evaluation (CIE) and Semester End Examination (SEE). Choice Based Credit System (CBCS) and Credit Based Semester System (CBSS) are followed as suggested by AICTE.

3.2.2 Credit courses

A student in each semester has to earn credits which is as- signed to each course in an L T P C (Lecture, Tutorial, Practical, Credits) structure based on the following general pattern:

- for a theory course, one credit for one Lecture (L)/Tutorial (T) period per week.
- for a Workshop, Drawing or Laboratory course, one credit for two periods per week.

3.2.3 Non-Credit (Mandatory) courses

- Induction programme recommended by AICTE
- Environmental Science in the second year
- Other courses, if any

These courses will be reflected in the consolidated course memo along with other credit courses.

3.2.4 Course Classification

All courses offered for the B.Tech. programme are broadly classified into groups, as given in the table below.

Cat.	CSE	CSE AIML	CSE CS	CSE DS	CSE IoT	IT
HSS M	12.5	12.5	12.5	12.5	15	12.5
BSM	24	24	24	24	21	24
ES	29	28.5	28.5	28.5	27	29
PCC	57.5	58	58	58	59	57.5
PEC	15	15	15	15	15	15
OE	9	9	9	9	9	9
PSI	13	13	13	13	14	13
MC	NC	NC	NC	NC	NC	NC
Total	160	160	160	160	160	160

Cat.	CE	ECE	ECM	EEE	ME
HSS M	12.5	12.5	12.5	12.5	12.5
BSM	23	24	25.5	25.5	21
ES	25.5	24.5	25.5	18.5	28
PCC	59	62	59.5	67.5	60.5
PEC	15	15	15	15	15
OE	9	9	9	9	9
PSI	16	13	13	12	14
MC	NC	NC	NC	NC	NC
Total	160	160	160	160	160

HSS M	:	Humanities and Social Sciences including Management courses
BSM	:	Basic Sciences including Mathematics courses
ES	:	Engineering Science courses including workshop, drawing, basic electrical/electronics, mechanical engineering course as well as various computer courses offered for non-IT branches
PCC	:	Professional Core Courses
PE	:	Professional Elective (five courses) relevant to the chosen specialization/branch
OE	:	Open Electives (3 courses) offered by any other engineering department or School of Management.
PSI	:	Project work, seminar and internship in industry or elsewhere
MC	:	Mandatory courses (NC: No Credit)

4.0 Course registration

- 4.1 A faculty advisor will advise the student on the UG programme, its course structure and curriculum, choice/option for the professional and open electives based on their employment potential or further studies.
- 4.2 The student will progress from semester to semester, as the Institute is following a cohort system to satisfy the conditions of promotion semester-wise.
- 4.3 In the present regulation (A22), there will be 5 professional electives (PE) and 3 open electives (OE). A student will opt for a stream of PE/OE, which should be submitted to the faculty advisor with a copy to the Controller of Examination (CoE) through the HoD.
- 4.4 The student can take one extra course in each semester and complete the programme in 7 semesters. However, the original degree will be awarded after 8 semesters.
- 4.5 An *honors degree* will be awarded if a student acquires 20 credits more than the credits required by the regulation.
- 4.6 A *minor degree* in an offered specialization will be awarded if a student acquires 18 credits more than the credits required by the regulation, in the specified subjects for that specialization.
- 4.7 Changing of elective courses may be permitted, after obtaining a prior approval from the faculty advisor and the HoD, within 2 weeks from the beginning of the semester.

5.0 Condition for Electives to be offered

An Elective may be offered to the students only if a minimum of 30 students opts for it. The maximum strength of a section is limited to 80.

6.0 Attendance requirements

- 6.1 A student will be eligible to appear for the semester end examinations if the student acquires a minimum of 75% of attendance in aggregate for all the courses in that semester (including mandatory courses).

- 6.2 Shortage of attendance in aggregate up to 10% in each semester may be condoned by the college academic council on genuine and valid grounds, based on the student's representation with supporting evidence.
- 6.3 A stipulated fee will be payable towards condoning of shortage of attendance as decided by academic council from time to time.
- 6.4 Shortage of attendance below 65% in aggregate shall in NO CASE be condoned.
- 6.5 Students whose shortage of attendance is not condoned in any semester are not eligible to take their semester end examinations of that semester. They get detained and their admission for that semester will stand cancelled. They will not be promoted to the next semester. They may seek re-admission into that semester as and when offered.
- 6.6 Institute may introduce additional attendance requirements, as part of CIE, from time to time, for improving the academic performance of the students with prior information to the students.

7.0 Academic requirements

The following academic requirements have to be satisfied, in addition to the attendance requirements mentioned in item no.6.

- 7.1 A student will be deemed to have satisfied the academic requirements and earned the credits allotted to each course, if a student secures a minimum of
- (a) 35% marks (14 out of 40 marks) in CIE
 - (b) 35% marks (21 out of 60 marks) in SEE, and
 - (c) 40% marks (40 out of 100 marks) in sum total (CIE+SEE).
- 7.2 A student will be deemed to have satisfied the academic requirements and earned the credits allotted to technical seminar, comprehensive viva-voce, internship and project if a student secures not less than 40% marks in each of them.

The student would be treated as failed, if the student

- i. does not submit a report of internship, technical seminar and project, or does not make a presentation of the same before the evaluation committee as per schedule.
- ii. secures less than 40% marks in the comprehensive viva-voce, technical seminar, internship and project.

7.3 Promotion rules based upon credits

Promotion	Conditions to be fulfilled
1 st year 1 st semester to 1 st year 2 nd semester	Regular course of study in the 1 st semester and not detained due to any reason
1 st year 2 nd semester to 2 nd year 1 st semester	i. Regular course of study in the 1 st and the 2 nd semesters of the 1 st year. ii. Must have secured at least 50% of the total credits for the 1 st year from all the relevant regular and supplementary examinations.
2 nd year 1 st semester to 2 nd year 2 nd semester	Regular course of study in the 2 nd year 1 st semester and not detained due to any reason.
2 nd year 2 nd semester to 3 rd year 1 st semester	i. Regular course of study in the 1 st and 2 nd semesters of the 2 nd year ii. Must have secured at least 60% of the total credits up to the 2 nd year 2 nd semester from all the relevant regular and supplementary examinations.
3 rd year 1 st semester to 3 rd year 2 nd semester	Regular course of study in the 3 rd year 1 st semester and not detained due to any reasons.
3 rd year 2 nd semester to 4 th year 1 st semester	i. Regular course of study in the 1 st and 2 nd semesters of the 3 rd year ii. Must have secured at least 60% of the total credits up to the 3 rd year 2 nd semester from all the relevant regular and supplementary examinations.
4 th year 1 st semester to 4 th year 2 nd semester	Regular course of study in the 4 th year 1 st semester and not detained due to any reasons.

7.4 To successfully complete the B.Tech. programme, a student will

- i. attend all the courses covering 160 credits as specified and listed in the course structure,
- ii. fulfill all the attendance requirements,
- iii. earn all 160 credits by securing an SGPA ≥ 5 in each semester and a CGPA ≥ 5 , and
- iv. pass all the mandatory courses

The performance of the student in these 160 credits will be taken into account for the calculation of the final CGPA at the end of the B. Tech. programme and will be indicated in the grade card/memo of the 4th year 2nd semester.

7.5 If a student registers for any extra course (other than those listed in the course structure) in the parent department or any other departments/school, the performances in that extra course will not be taken into account while calculating the SGPA and CGPA. For such extra courses, percentage of marks and letter grade alone will be indicated in the grade card as a performance measure, subjected to the completion of attendance and academic requirements.

7.6 A student eligible to appear in the SEE for any course, but absent from it or failed (thereby failing to secure C grade or above) has to reappear for that course in the *supplementary examination* as and when conducted. In such cases, CIE assessed earlier for that course will be carried over, and added to the marks obtained in the supplementary examination for evaluating the performance in that course.

7.7 A student detained in a semester due to shortage of attendance, may be readmitted when the same semester is offered in the subsequent academic years for the fulfillment of academic requirements.

The academic regulations under which student has been re- admitted will be applicable. However, no grade allotments or SGPA/CGPA calculations will be done for the entire semester in which student has been detained.

- 7.8 A student detained due to lack of credits, will be promoted to the next academic year only after acquiring the required credits as per the existing academic regulations in force at the time of readmission.

8.0 Evaluation: Distribution and weightage of marks

- 8.1 Summer Industry Internships 1 and 2 will be organized during vacation after II/2 and III/2 semesters and the evaluation of the same will be carried out during the lab examinations of III/1 and IV/1 semesters. In addition, there will be a project in IV/2 semester, which will be evaluated for 100 marks.

Pattern of CIE for summer industry internships

Sl.No.	Description	Marks
1	Presentation	15
2	Internal Evaluation	15
3	Report	10
	Total	40

Pattern of SEE for summer industry internships

Sl.No.	Description	Marks
1	Presentation	10
2	Demonstration/Technology	40
3	Report	10
	Total	60

Pattern of CIE for project

Sl.No.	Description	Marks
1	Abstract seminar at the end of 3 weeks	5
2	Design Seminar at the end of 8 weeks	5
3	Implementation/Demonstration of Seminar at the end of 12 weeks	10
4	Evaluation of project by internal guide	10
5	Project Report	10
	Total	40

Pattern of SEE for project

Sl.No.	Description	Marks
1	Final project report	10
2	Presentation	20
3	Demonstration/Defense of project	30
	TOTAL	60

8.2 Theory Courses: Pattern of CIE (40 marks)

There will be two mid-examinations in a semester. The first mid-examination will be conducted for Units I, II, and III of the syllabus, and the second mid-examination will be conducted for the remaining 3 units. The total time duration for the mid-examinations will be 2 hours.

Sl.No.	Mid examination pattern	Marks
1.	Part-A: 6 short answer questions of 2 marks each; 2 questions from each unit.	12
2.	Part-B: 4 questions of 6 marks each; 1 question from each unit and the 4 th question will have a choice of (a) or (b) or (c) where (a), (b) and (c) is a question from each unit, respectively. <i>A student has to answer any 3 questions.</i>	18
	Total	30*
	* average of the 2 mid-examinations will be considered for the final evaluation	
	Other evaluations	
a)	Assignment-I (3 marks): Should contain questions from units 1, 2 and 3. A student has to submit all answers before the 1 st mid exam. Assignment-II (3 marks): Should contain questions from units 4, 5 and 6. A student has to submit all answers before the 2 nd mid exam.	6
b)	Attendance *	4
	Grand Total	40

* Marks for attendance will be added to each course based on the attendance obtained in that course. 4 marks are divided as follows

Sl.No.	Attendance Range	Marks
1.	75% and above but less than 85%	2
2.	85% and above	4

If a student is absent in either of the mid exams, or if a student wishes to improve the performance in a course, the student can appear for a **third mid-exam** which will be conducted on the same day of the SEE for 2 hours, covering the full syllabus. The average of the best two of the three mid-exams will then be considered as final (for 30 marks).

Sl.No.	Mid-III exam pattern	Marks
1.	Part-A: 6 short answer questions of 2 marks each; 1 question from each unit.	12
2.	Part-B: 6 questions of 6 marks each; 1 question from each unit. <i>A student has to answer 3 questions.</i>	18
	Total	30

- 8.3 **Pattern for Semester End Examinations (60 marks)** SEE for each theory course will consist of two parts (Part-A & Part-B). The total time duration for the SEE will be 3 hours.

Sl.No.	SEE pattern	Marks
1	Part-A: 6 short answer questions of 2 marks each; 1 question from each unit.	12
2	Part-B: 6 questions of 8 marks each; one question from each unit. Each question will have 2 subunits in the form of (a) or (b).	48
	Total	60

8.4 **Pattern of Evaluation for Labs - (100 marks)**

- 8.4.1 For labs, there will be 40 marks for CIE and 60 marks for SEE. The distribution of 40 marks for CIE is:

Sl. No.	Item	Marks
1.	Write-up (day to day)	10
2.	Viva-voce	10
3.	Internal lab exam	15
4.	Lab record	5
	Total	40

- 8.4.2 The SEE for the labs will be conducted by an internal examiner and an external examiner appointed by the chief superintendent of examinations of the college. The marks are distributed as follows:

Sl. No.	Item	Marks
1.	Write-up	10
2.	Experiment/program	15
3.	Results	15
4.	Lab record	10
5.	Viva-voce	10
	Total	60

- 8.4.3 For oral communication lab and soft skills lab, distribution of marks of CIE and SEE will be as follows:

Sl. No.	CIE	Marks
1.	6 activities (6 x 5marks = 30 marks) including lab records	30
2.	Lab Internal Exam (oral)	10
	Total	40

Sl. No.	SEE	Marks
1.	English Proficiency Evaluation	20
2.	Activity and Viva	20
4.	Lab record	10
5.	Quiz	10
Total		60

- 8.5 For the courses having drawing, the marks distribution will be 40 marks for CIE (20 marks for day-to-day work including drawing and timely submission, 20 marks for average of two internal tests) and 60 marks for the semester end examination which will have the same pattern as other theory papers.

8.6 Technical Seminar

The technical seminar in II year II Semester will be evaluated for 100 marks. The evaluation is purely internal and will be conducted as follows:

Sl. No.	Description	Marks
1	Literature survey, topic and content	20
2	Presentation including PPT	30
3	Seminar Notes	05
5	Final report 3 copies	10
7	Punctuality in giving seminar as per scheduled time and date	10
9	Viva	25
Total		100

A student must secure 40% of the marks to obtain a pass grade.

8.7 Comprehensive viva-voce

A comprehensive viva-voce will be conducted in Semester VI for 100 marks. Two internal exams (oral) of 50 marks each will be considered for CIE.

There will be no SEE for this course. A student must secure 40% of the total marks to obtain a pass grade.

- 8.8 The laboratory marks and the internal theory marks awarded by the college are subject to scrutiny and are scaled down by the departmental committees wherever necessary. In such cases, the internal and laboratory marks awarded by the department will be referred to a committee. The committee will arrive at a scaling factor and the marks will be scaled accordingly. The recommendation of the committee is final and binding.

- 8.9 For all the mandatory courses, a student has to secure 40 marks out of 100 marks in CIE (no SEE for mandatory courses) for passing. There will be NO letter grades, but the Pass/Fail will be mentioned in the grade card/consolidated marks memo. There will be 2 mid examinations of 50 marks each and an optional mid-III (50 marks). The best 2 of 3 mid examinations will be considered for CIE marks.
- 8.10 If the marks secured by a student in CIE is less than 35% (less than 14 marks out of 40 marks) and hence failed the subject (as mentioned in 7.1(a)), the student will be given another chance to re-appear for the mid examination of the same subject.
In such cases, a student must re-register for the failed subject(s) for the 30 marks mid examination within four weeks of the commencement of the class work in next academic year.
In the event of the student taking this chance, the mid examination marks for 30 and the end examination marks for 60 obtained in the previous attempt stand cancelled. However, the marks for assignments and attendance will be carried forward.
The student has to earn 35% of total internal marks (14 out of 40 marks) to pass the subject.

9.0 Grading procedure

- 9.1 Marks will be awarded to indicate the performance of a student in each theory course, laboratory/practical, seminar, project and comprehensive viva-voce. Based on the percentage of marks obtained (CIE plus SEE), a corresponding letter grade will be given.
- 9.2 As a measure of the performance the student, the percentage of marks will be converted to a 10-point grading system using the following letter grades (as per UGC/AICTE guidelines):

% of Marks Secured	Letter Grade	Grade Points
≥ 90%	O (outstanding)	10
< 90% and ≥ 80%	A+ (excellent)	9
< 80% and ≥ 70%	A (very good)	8
< 70% and ≥ 60%	B+ (good)	7
< 60% and ≥ 50%	B (average)	6
< 50% and ≥ 40%	C (pass)	5
< 40%	F (fail)	0
Absent	Ab	0

- 9.3 A student obtaining an F or Ab grade in any course will be deemed to have failed and is required to appear for the supplementary examination, as and when offered. In such cases, internal marks in those courses will remain the same as those obtained earlier.
- 9.4 A letter grade does not indicate any specific percentage of marks secured by the student, but indicates only the range of percentage of marks.

- 9.5 A student earns a grade point (GP) in each course based on the letter grade secured in that course. The corresponding credit point (CP) is computed by multiplying the grade point with credits for that particular course

$$\mathbf{CP = (GP) \times (credits)}$$

- 9.6 A student will pass a course only if the $GP \geq 5$ (i.e., C grade or above).
- 9.7 The Semester Grade Point Average (SGPA) for each semester is calculated by taking the ratio of the sum of CP secured from all the courses and the total number of credits registered during that semester.

$$\mathbf{SGPA = \frac{\sum_{i=1}^N C_i G_i}{\sum_{i=1}^N C_i}}$$

where N is the total number of courses registered by the student, i is the course indicator index, C_i is the number of credits allotted to the i th course and G_i is the grade point awarded for the i th course.

From the 2nd semester onwards, Cumulative Grade Point Average (CGPA) will also be calculated for merit ranking or comparison purposes or any other listing.

$$\mathbf{CGPA = \frac{\sum_{j=1}^M C_j G_j}{\sum_{j=1}^M C_j}}$$

where M is the total number of courses registered by the student from the 1st semester. Both SGPA and CGPA will be rounded off to two decimal places.

Illustration of SGPA and CGPA calculations

course	credits	grade	grade points	credit points
I Year I Sem				
C1	4	A	8	4 x 8 = 32
C2	4	A+	9	4 x 9 = 36
C3	4	B	6	4 x 6 = 24
C4	3	O	10	3 x 10 = 30
C5	3	B+	7	3 x 7 = 21
C6	1	Ab	0	1 x 0 = 0
TOTAL	19			143
SGPA = 143/19 = 7.53				
CGPA = 7.53				
I Year II Sem				
C7	4	B+	7	4 x 7 = 28
C8	4	O	10	4 x 10 = 40
C9	4	A	8	4 x 8 = 32
C10	3	B	6	3 x 6 = 18
C11	2	C	5	2 x 5 = 10
C12	3	F	0	3 x 0 = 0
Total	20			128
SGPA = 128/20 = 6.4				
CGPA = (143+128)/(19+20) = 271/39 = 6.95				

- 9.8 For SGPA and CGPA calculations, performance in failed/absent courses (securing F/Ab grade) will also be taken into account.

After passing the failed/Absent course(s), SGPA and CGPA will be recalculated using the newly secured letter grades.

10.0 Passing standards

- 10.1 A student will be declared successful or passed in a semester, if the student secures a GP ≥ 5 (C grade or above) in every course in that semester (SGPA ≥ 5). Also, a student should obtain a CGPA ≥ 5 for the award of the B.Tech degree.

- 10.2 After the completion of each semester, all students will obtain a grade card (transcript), indicating the letter grades, credits earned, SGPA and CGPA.

11.0 Declaration of results

- 11.1 Final CGPA will be calculated after completing the course requirements

- 11.2 For calculating percentage of marks, the following formula is used;

$$\% \text{ marks} = (\text{final CGPA} - 0.5) \times 10$$

e.g., if the final CGPA after 8 semesters is 9.5,

$$\% \text{ marks} = (9.5 - 0.5) \times 10 = 90\%$$

12.0 Award of degree

- 12.1 A student who passes all the specified courses in the course structure and secures the total number of credits with CGPA ≥ 5.0 within 16 academic semesters (12 academic semesters for lateral entry students) from the date of commencement of the first academic semester, will qualify for the award of the B.Tech. degree in the chosen branch of Engineering.

- 12.2 A student who qualifies for the award of the degree, as listed in item 12.1 will be placed in one of the following classes:

	CLASS	conditions
12.2.1	1 st CLASS WITH DISTINCTION	(a) CGPA ≥ 8 (b) passed all the courses in the first appearance it- self, within the first eight academic semesters (first six academic semesters for lateral entry students), (c) not detained or prevented from writing the end semester examinations in any of the semesters for any reasons.
12.2.2	1 st CLASS	(a) CGPA ≥ 8 , but did not satisfy the conditions (b) and (c) given in 12.2.1. (a) CGPA ≥ 7 but < 8
12.2.3	2 nd CLASS	CGPA ≥ 6 but < 7
12.2.4	PASS CLASS	CGPA ≥ 5 but < 6
12.2.5	FAIL	CGPA < 5

- 12.3 Students fulfilling the conditions listed under item 12.2.1 alone will be eligible for the award of the Department Gold medal (highest CGPA amongst all the sections of the department) and the Institute Gold Medal (highest CGPA amongst all the Departments).

13.0 Withholding of results

- 13.1 If a student has not paid the required fees to the college at any stage, or has dues pending due to any reason what so- ever, or if a case of indiscipline is pending against her/him, the result will be withheld, and the student will not be allowed to go to the next semester. The award or issue of the degree may also be withheld in such cases.

14.0 Transitory Regulations

- 14.1 A student who has discontinued for any reason, or has been detained for want of attendance or lack of required credits as specified, or has failed after having undergone the degree programme, may be considered eligible for readmission to the same courses (or equivalent courses, as the case may be), and the same professional electives/open electives (or from set/category of electives or equivalents suggested, as the case may be) as and when they are offered (within the time- frame of 8 years from the date of commencement of student's first year first semester).

If a student gets readmitted in another academic regulation, the college has to offer substitute/additional courses after comparing the two academic regulations. The details of such changes should be prepared by Dean (Academic) in consultation with the concerned Chairman, BOS. A copy should be sent to the Controller of Examinations and the student. The student will be governed by the academic regulations at the time of readmission.

15.0 Transfer of Students

- 15.1 There shall be no change of departments after the completion of the admission process.
- 15.2 Students seeking transfer to Sreenidhi Institute of Science and Technology (SNIST) from other institutions have to pass the failed courses which are equivalent to the courses of SNIST and also pass the courses of SNIST which the students have not studied at the earlier institution. The details of the extra/substitute to be studied should be prepared by Dean (Academic) in consultation with the concerned Chair- man, BOS. A copy should be sent to the Controller of Examinations and the student.
- Further, though the students have passed some of the courses at the earlier semesters of SNIST, the students have to study substitute courses in SNIST and get sessional marks by attending 3rd mid test and paying requisite fee as per the rules.
- 15.3 The transferred students from other Institutions to SNIST who are on rolls to be provided one chance to write the CIE(internal marks) in the failed courses and/or courses not studied as per the clearance letter issued by the Institution.
- 15.4 The autonomous affiliated colleges have to provide one chance to write the internal examinations in the failed courses and/or courses not studied, to the students transferred from other universities/institutions to SNIST who are on rolls, as per the clearance (equivalence) letter issued by the University.

16.0 Scope

16.1 The academic regulations should be read as a whole, for the purpose of any interpretation.

16.2 In case of any doubt or ambiguity in the interpretation of the regulations, the decision of the academic council is final and binding.

16.3 SNIST reserves the right to change or amend the academic regulations, course structure or syllabi at any time. Such changes or amendments will be applicable to all students with effect from the date notified by SNIST.

**Academic Rules and Regulations (A22) for B.Tech.
LATERAL ENTRY students (w.e.f. AY 2022-23)**

- 1) Eligibility for award of B.Tech. Degree
The Lateral Entry (LE) students after securing admission will pursue a course of study for not less than 6 academic semesters and not more than 12 academic semesters, failing which the student will forfeit the seat.
- 2) The student will register for 123 credits from the II year I semester to the IV year II semester.
- 3) The students, who fail to fulfill the requirement for the award of the degree in 6 academic years from the year of admission, will forfeit their seat in B.Tech. However, the student can take 2 more years to appear for the examinations.
- 4) The attendance requirements of B.Tech. (regular) will be applicable to B.Tech. (LE).
- 5) Promotion rules based on credits

#	Promotion	Conditions to be fulfilled
1	II year I semester to II year II semester	Regular course of study of second year first semester and not detained due to any reason.
2	II year II semester to III year I semester	i. Regular course of study of second year second semester. ii. Must have secured at least 60% of credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3	III year I semester to III year II semester	Regular course of study of third year first semester and not detained due to any reason.
4	III year II semester to IV year I semester	i. Regular course of study of third year second semester. Must have secured at least 60% of credits up to third year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
5	IV year I semester to IV year II semester	ii. Regular course of study of fourth year first semester and not detained due to any reason.
	All other regulations, applicable to B. Tech. regular students, will hold good for B. Tech. Lateral Entry students.	

**MALPRACTICE RULES DISCIPLINARY ACTION FOR MIS-
CONDUCT OF STUDENTS DURING EXAMINATIONS**

1. **Malpractice:** *During examination, possesses any form of material (paper, note book, programmable calculator, cell phone, etc.) concerned with or related to the subject of the examination (theory or practical), but has not made use of the same.*

Punishment: Expulsion from the examination hall and cancellation of the performance in that subject only.

2. **Malpractice:** *Gives assistance/guidance to or receives from any other student orally or by any other methods.*

Punishment: Expulsion from the examination hall and cancellation of the performance in that subject only for all the students involved.

3. **Malpractice:** *Has copied in the examination hall from any form of material relevant to the subject of the examination (theory or practical) in which the student is appearing*

Punishment: Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and UG major project and will not be permitted to appear for the remaining examinations of the subjects of that semester/year. The hall ticket of the student is to be cancelled and sent to the university.

4. **Malpractice:** *Impersonates any other student in connection with the examination.*

Punishment: The student who has impersonated will be expelled from the examination hall. The student is also debarred and forfeits the seat. The performance of the original student who has been impersonated, shall be cancelled in all the subjects of the examination (including practical and UG major project) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that semester. The student is also debarred for two consecutive semesters from class work and all university examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat. If the imposter is an outsider, she/he will be handed over to the police and a case will be registered against her/him.

5. **Malpractice:** *Smuggles in the answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.*

Punishment: Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and UG major project and shall not be permitted for the remaining examinations of the subjects of that semester. The student is also debarred for two consecutive semesters from class work and all university examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat.

6. **Malpractice:** *Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.*

Punishment: Cancellation of the performance in that subject

7. **Malpractice:** *Refuses to obey the orders of any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the officer-in charge or any person on duty in or outside the examination hall or any injury to his person or to any of his relations whether by words, either*

spoken or written or by signs or by visible representation, assaults the officer-in-charge, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of property in the examination hall or any part of the college campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination.

imposter is an outsider, she/he will be handed over to the police and a case will be registered against her/him.

8. **Malpractice:** *Smuggles in the answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.*

Punishment: Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and UG major project and shall not be permitted for the remaining examinations of the subjects of that semester. The student is also de- barred for two consecutive semesters from class work and all university examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat.

9. **Malpractice:** *Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.*

Punishment: Cancellation of the performance in that subject

10. **Malpractice:** *Refuses to obey the orders of any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the officer-in charge or any person on duty in or outside the examination hall or any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the officer-in-charge, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of property in the examination hall or any part of the college campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination.*

Punishment: In case of students of the college, they shall be expelled from the examination hall and cancellation of their performance in that subject and all other subjects the student(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The students also are debarred and forfeit their seats. In case of outsiders, they will be handed over to the police and a police case is registered against them.

11. **Malpractice:** *Leaves the exam hall taking away answer script or intentionally tears of the script or any part thereof inside or outside the examination hall.*

Punishment: Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and UG major project and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all university examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat.

12. **Malpractice:** *Possess any lethal weapon or firearm in the examination hall.*

Punishment: Expulsion from the examination hall and cancellation of the performance in

that subject and all other subjects the student has already appeared including practical examinations and UG major project and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat.

13. **Malpractice:** *If a student of the college, who is not a student for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.*

Punishment: Student of the college expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and UG major project and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat.

14. **Malpractice:** *Comes in a drunken condition to the examination hall.*

Punishment: Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and UG major project and shall not be permitted for the remaining examinations of the subjects of that semester/year.

15. **Malpractice:** *Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.*

Punishment: Cancellation of the performance in that subject and all other subjects the student has appeared including practical examinations and UG major project of that semester/year examinations.

If any malpractice is detected which is not covered in the above clauses 1 to 12, it shall be reported to the university for further action to award suitable punishment.

Malpractices identified by squad or special invigilators

- Punishments to the students as per the above guidelines.
- Punishment for institutions: (if the squad reports that the college is encouraging malpractices)
 - A show cause notice shall be issued to the college.
 - Impose a suitable fine on the college.
 - Shifting the examination centre from the college to another college for a specific period of not less than one year.

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## **THE TELANGANA PROHIBITION OF RAGGING ACT, 1997**

### **Important Points (reproduced from the ACT NO. 26 OF 1997)**

1. Ragging within or outside any educational institution is prohibited.
2. Whoever, with the intention of causing ragging or with the knowledge that he is likely by such act to cause ragging, commits or abets ragging and thereby,-
  - (i) teases or embarrasses or humiliates a student shall be punished with imprisonment for a term which may extend to six months or with fine which may extend to one thousand rupees, or with both; or
  - (ii) assaults or uses criminal force to or criminally intimidates a student shall be punished with imprisonment for a term which may extend to one year or with fine which may extend to two thousand rupees or with both; or
  - (iii) wrongfully restrains or wrongfully confines or causes hurt to a student shall be punished with imprisonment for a term which may extend to two years or with fine which may extend to five thousand rupees or with both; or
  - (iv) causes grievous hurt to or kidnaps or abducts or rapes or commits unnatural offence with a student shall be punished with imprisonment for a term which may extend to five years and with fine which may extend to ten thousand rupees; or
  - (v) causes death or abets suicide shall be punished with imprisonment for life or with imprisonment for a term which may extend to ten years and with a fine which may extend to fifty thousand rupees.
3. **Dismissal of student.**
  - (1) A student convicted, of an offence under section 4 and punished with imprisonment for a term shall be dismissed from the educational institution.
  - (2) A student convicted of an offence under section 4 and punished with imprisonment for a term of more than six months shall not be admitted in any other educational institution.
4. **Suspension of student.**
  - (3) Without prejudice to the foregoing provisions, whenever any student complains of ragging to the head or manager of an educational institution, such head or manager shall inquire into or cause an inquiry to be made into the same forthwith and if the complaint is prima facie found true, shall suspend the student or students complained against for such period as may be deemed necessary.
  - (4) The decision of the head or manager of the educational institution under sub-section (1) shall be final.
5. **Abetment.**
  - (5) If the head or the manager of an educational institution fails or neglects to take action in the manner specified in sub-section (1) of section 6, such person shall be deemed to have abetted the offence and shall be punished with the punishment provided for the offence.
  - (6) If a student commits suicide due to or in consequence of ragging, the person who commits such ragging shall be deemed to have abetted such suicide.

**(for details, check the SNIST website or**

**[https://www.indiacode.nic.in/bitstream/123456789/8518/1/act\\_26\\_of\\_1997.pdf](https://www.indiacode.nic.in/bitstream/123456789/8518/1/act_26_of_1997.pdf)**)

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**  
**(A22 Regulations)**

**B.Tech (CSE) I YEAR I SEM**

| Sl. No. | Course Category | Course Code | Course                               | L                                        | T        | P         | C           | Max. Marks         |            |
|---------|-----------------|-------------|--------------------------------------|------------------------------------------|----------|-----------|-------------|--------------------|------------|
|         |                 |             |                                      |                                          |          |           |             | CIE                | SEE        |
| 1       | BS              | 9HC04       | Engineering Chemistry                | 2                                        | 1        | 0         | 3           | 40                 | 60         |
| 2       | ES              | 9FC01       | Problem Solving using C              | 3                                        | 0        | 0         | 3           | 40                 | 60         |
| 3       | BS              | 9HC11       | Matrix Algebra and Calculus          | 2                                        | 1        | 0         | 3           | 40                 | 60         |
| 4       | HS              | 9HC01       | Essential English Language Skills    | 2                                        | 0        | 0         | 2           | 40                 | 60         |
| 5       | HS              | 9HC61       | Oral Communication Lab – I           | 0                                        | 0        | 2         | 1           | 40                 | 60         |
| 6       | BS              | 9HC64       | Engineering Chemistry Lab            | 0                                        | 0        | 3         | 1.5         | 40                 | 60         |
| 7       | ES              | 9FC61       | Problem Solving using C Lab          | 0                                        | 0        | 3         | 1.5         | 40                 | 60         |
| 8       | ES              | 9BC61       | Workshop/Manufacturing Processes Lab | 0                                        | 1        | 3         | 2.5         | 40                 | 60         |
| 9       | HS              | 9HC18       | Induction Program                    | 2-weeks in the beginning of the semester |          |           |             | Satisfactory / Not |            |
|         |                 |             | <b>Total</b>                         | <b>9</b>                                 | <b>3</b> | <b>11</b> | <b>17.5</b> | <b>320</b>         | <b>480</b> |

**B.Tech (CSE) I YEAR II SEM**

| Sl. No. | Course Category | Course Code | Course                                       | L         | T        | P         | C           | Max. Marks |            |
|---------|-----------------|-------------|----------------------------------------------|-----------|----------|-----------|-------------|------------|------------|
|         |                 |             |                                              |           |          |           |             | CIE        | SEE        |
| 1       | BS              | 9HC07       | Engineering Physics                          | 2         | 1        | 0         | 3           | 40         | 60         |
| 2       | ES              | 9EC01       | Data Structures                              | 3         | 0        | 0         | 3           | 40         | 60         |
| 3       | BS              | 9HC12       | Advanced Calculus                            | 2         | 1        | 0         | 3           | 40         | 60         |
| 4       | ES              | 9AC48       | Basic Electrical and Electronics Engineering | 3         | 0        | 0         | 3           | 40         | 60         |
| 5       | ES              | 9BC01       | Engineering Graphics                         | 1         | 0        | 4         | 3           | 40         | 60         |
| 6       | HS              | 9HC62       | Oral communication Lab – II                  | 0         | 0        | 3         | 1.5         | 40         | 60         |
| 7       | BS              | 9HC66       | Engineering Physics Lab                      | 0         | 0        | 3         | 1.5         | 40         | 60         |
| 8       | ES              | 9EC61       | Data Structures using C Lab                  | 0         | 0        | 3         | 1.5         | 40         | 60         |
|         |                 |             | <b>Total</b>                                 | <b>11</b> | <b>2</b> | <b>13</b> | <b>19.5</b> | <b>320</b> | <b>480</b> |

### B.Tech (CSE) II YEAR I SEM

| Sl.No        | Course Category | Course Code | Dept Course | Course                                           | L        | T        | P/D       | C           | Max. Marks |            |
|--------------|-----------------|-------------|-------------|--------------------------------------------------|----------|----------|-----------|-------------|------------|------------|
|              |                 |             |             |                                                  |          |          |           |             | CIE        | SEE        |
| 1            | PC              | 9EC02       | CSE         | Object Oriented Programming through Java         | 2        | 1        | 0         | 3           | 40         | 60         |
| 2            | ES              | 9CC51       | ECE         | Digital Electronics                              | 3        | 0        | 0         | 3           | 40         | 60         |
| 3            | BS              | 9HC16       | S&H         | Quantitative Aptitude and Logical Reasoning      | 3        | 0        | 0         | 3           | 40         | 60         |
| 4            | BS              | 9F303       | IT          | Discrete Mathematics                             | 2        | 1        | 0         | 3           | 40         | 60         |
| 5            | HS              | 9HC03       | S&H         | Universal Human Values                           | 3        | 0        | 0         | 3           | 40         | 60         |
| 6            | PC              | 9EC62       | CSE         | Object Oriented Programming through Java Lab     | 0        | 0        | 4         | 2           | 40         | 60         |
| 7            | ES              | 9CC82       | ECE         | Digital Electronics Lab                          | 0        | 0        | 3         | 1.5         | 40         | 60         |
| 8            | ES              | 9AC95       | EEE and ECE | Basic electrical and Electronics Engineering Lab | 0        | 0        | 4         | 2           | 40         | 60         |
| <b>Total</b> |                 |             |             |                                                  | <b>1</b> | <b>2</b> | <b>11</b> | <b>20.5</b> | <b>320</b> | <b>480</b> |

### B.Tech (CSE) II YEAR II SEM

| Sl. No       | Course Category | Course Code | Dept Course | Course                                    | L         | T        | P/D       | C           | Max. Marks  |            |
|--------------|-----------------|-------------|-------------|-------------------------------------------|-----------|----------|-----------|-------------|-------------|------------|
|              |                 |             |             |                                           |           |          |           |             | CIE         | SEE        |
| 1            | ES              | 9CC54       | ECE         | Computer Organization                     | 3         | 0        | 0         | 3           | 40          | 60         |
| 2            | PC              | 9FC04       | IT          | Database Management Systems               | 3         | 0        | 0         | 3           | 40          | 60         |
| 3            | PC              | 9FC02       | IT          | Python Programming                        | 2         | 1        | 0         | 3           | 40          | 60         |
| 4            | BS              | 9HC15       | S&H         | Probability and Statistics                | 2         | 1        | 0         | 3           | 40          | 60         |
| 5            | HS              | 9ZC01       | SMS         | Business Economics and Financial Analysis | 3         | 0        | 0         | 3           | 40          | 60         |
| 6            | HS              | 9HC05       | S&H         | Environmental Science                     | 3         | 0        | 0         | -           | Pass / Fail |            |
| 7            | HS              | 9HC63       | S&H         | Soft Skills Lab                           | 0         | 1        | 2         | 2           | 40          | 60         |
| 8            | PS              | 9E484       | CSE         | Technical Seminar                         | 0         | 1        | 0         | 1           | 100         | --         |
| 9            | PC              | 9FC62       | IT          | Python Programming Lab                    | 0         | 0        | 4         | 2           | 40          | 60         |
| 10           | PC              | 9FC63       | IT          | Database Management Systems Lab           | 0         | 0        | 3         | 1.5         | 40          | 60         |
| 11           | ES              | 9FC64       | IT          | IT Workshop and Computer Organization Lab | 0         | 0        | 4         | 2           | 40          | 60         |
| <b>Total</b> |                 |             |             |                                           | <b>16</b> | <b>4</b> | <b>13</b> | <b>23.5</b> | <b>460</b>  | <b>540</b> |

**Note: Summer Industry Internship-I is to be carried out during the summer vacation between 4<sup>th</sup> and 5<sup>th</sup> semesters**

### B.Tech (CSE) III YEAR I SEM

| Sl.No. | Course Category | Course Code | Dept Course | Course                                                      | L         | T        | P/D       | C         | Max. Marks |            |
|--------|-----------------|-------------|-------------|-------------------------------------------------------------|-----------|----------|-----------|-----------|------------|------------|
|        |                 |             |             |                                                             |           |          |           |           | CIE        | SEE        |
| 1      | PE              |             | CSE/IT      | Professional Elective –I                                    | 3         | 0        | 0         | 3         | 40         | 60         |
| 2      | PC              | 9FC05       | IT          | Data Warehousing and Data Mining                            | 2         | 1        | 0         | 3         | 40         | 60         |
| 3      | PC              | 9EC03       | CSE         | Software Engineering                                        | 2         | 0        | 0         | 2         | 40         | 60         |
| 4      | PC              | 9EC04       | CSE         | Design and Analysis of Algorithms                           | 2         | 1        | 0         | 3         | 40         | 60         |
| 5      | PC              | 9EC05       | CSE         | Computer Networks                                           | 3         | 0        | 0         | 3         | 40         | 60         |
| 6      | PC              | 9IC04       | IOT         | Intellectual Property Rights                                | 3         | 0        | 0         | -         | Pass/Fail  |            |
| 7      | PC              | 9EC63       | CSE         | Software Engineering Lab                                    | 0         | 0        | 3         | 1.5       | 40         | 60         |
| 8      | PC              | 9EC64       | CSE         | Computer Networks and Design and Analysis of Algorithms Lab | 0         | 0        | 4         | 2         | 40         | 60         |
| 9      | PC              | 9FC65       | IT          | Data Warehousing and Data Mining lab                        | 0         | 0        | 3         | 1.5       | 40         | 60         |
| 10     | PS              | 9E585       | CSE         | Summer Industry Internship –I                               | 0         | 0        | 0         | 1         | 40         | 60         |
|        |                 |             |             | <b>Total</b>                                                | <b>15</b> | <b>2</b> | <b>10</b> | <b>20</b> | <b>360</b> | <b>540</b> |

### B.Tech (CSE) III YEAR II SEM

| Sl.No | Course Category | Course Code | Dept Course | Course                                     | L         | T        | P/D      | C         | Max. Marks |            |
|-------|-----------------|-------------|-------------|--------------------------------------------|-----------|----------|----------|-----------|------------|------------|
|       |                 |             |             |                                            |           |          |          |           | CIE        | SEE        |
| 1     | OE              |             |             | Open Elective –I                           | 3         | 0        | 0        | 3         | 40         | 60         |
| 2     | PE              |             | CSE/IT      | Professional Elective –II                  | 3         | 0        | 0        | 3         | 40         | 60         |
| 3     | PE              |             | CSE/IT      | Professional Elective -III                 | 3         | 0        | 0        | 3         | 40         | 60         |
| 4     | PC              | 9EC06       | CSE         | Operating Systems                          | 3         | 0        | 0        | 3         | 40         | 60         |
| 5     | PC              | 9LC03       | AIML        | Machine Learning                           | 2         | 1        | 0        | 3         | 40         | 60         |
| 6     | PC              | 9FC06       | IT          | Web Technologies                           | 2         | 1        | 0        | 3         | 40         | 60         |
| 7     | PC              | 9EC65       | CSE         | Operating Systems and Machine Learning Lab | 0         | 0        | 3        | 1.5       | 40         | 60         |
| 8     | PC              | 9FC66       | IT          | Web Technologies Lab                       | 0         | 0        | 3        | 1.        | 40         | 60         |
| 9     | PC              | 9E686       | CSE         | Comprehensive Viva Voce                    | 0         | 0        | 0        | 1         | 100        | -          |
|       |                 |             |             | <b>Total</b>                               | <b>16</b> | <b>2</b> | <b>6</b> | <b>22</b> | <b>420</b> | <b>480</b> |

**Note: Summer Industry Internship-II is to be carried out during the summer vacation between 6<sup>th</sup> and 7<sup>th</sup> semesters.**

### B.Tech (CSE) IV YEAR I SEM

| Sl.No. | Course Category | Course Code | Dept Course | Course                                       | L         | T        | P/D      | C         | Max. Marks |            |
|--------|-----------------|-------------|-------------|----------------------------------------------|-----------|----------|----------|-----------|------------|------------|
|        |                 |             |             |                                              |           |          |          |           | CIE        | SEE        |
| 1      | OE              |             |             | Open Elective –II                            | 3         | 0        | 0        | 3         | 40         | 60         |
| 2      | PE              |             | CSE/IT      | Professional Elective -IV                    | 3         | 0        | 0        | 3         | 40         | 60         |
| 3      | PE              |             | CSE/IT      | Professional Elective -V                     | 3         | 0        | 0        | 3         | 40         | 60         |
| 4      | PC              | 9EC07       | CSE         | Automata Theory and Compiler Design          | 2         | 1        | 0        | 3         | 40         | 60         |
| 5      | PC              | 9EC08       | CSE         | Linux Programming                            | 2         | 1        | 0        | 3         | 40         | 60         |
| 6      | PC              | 9EC09       | CSE         | Cryptography and Network Security            | 3         | 0        | 0        | 3         | 40         | 60         |
| 7      | PC              | 9EC66       | CSE         | Compiler Design and Information Security Lab | 0         | 0        | 3        | 1.5       | 40         | 60         |
| 8      | PC              | 9EC67       | CSE         | Linux Programming Lab                        | 0         | 0        | 3        | 1.5       | 40         | 60         |
| 9      | PS              | 9E787       | CSE         | Summer Industry Internship –II               | 0         | 0        | 0        | 1         | 40         | 60         |
|        |                 |             |             | <b>Total</b>                                 | <b>16</b> | <b>2</b> | <b>6</b> | <b>22</b> | <b>360</b> | <b>540</b> |

### B.Tech (CSE) IV YEAR II SEM

| Sl.No. | Course Category | Course Code | Dept Course | Course                        | L        | T        | P/D       | C         | Max. Marks |            |
|--------|-----------------|-------------|-------------|-------------------------------|----------|----------|-----------|-----------|------------|------------|
|        |                 |             |             |                               |          |          |           |           | CIE        | SEE        |
| 1      | OE              |             |             | Open Elective –III            | 3        | 0        | 0         | 3         | 40         | 60         |
| 2      | PC              | 9FC07       | IT          | Cyber Security and Cyber laws | 2        | 0        | 0         | 2         | 40         | 60         |
| 3      | PS              | 9E888       | CSE         | Major Project                 | 0        | 0        | 20        | 10        | 40         | 60         |
|        |                 |             |             | <b>Total</b>                  | <b>5</b> | <b>0</b> | <b>20</b> | <b>15</b> | <b>120</b> | <b>180</b> |

## Professional Electives

| Code       | PE– I<br>(3-1)                               | Code       | PE – II<br>(3-2)                                  | Code       | PE – III<br>(3-2)                          | Code        | PE – IV<br>(4-1)                    | Code        | PE – V<br>(4-1)                       |
|------------|----------------------------------------------|------------|---------------------------------------------------|------------|--------------------------------------------|-------------|-------------------------------------|-------------|---------------------------------------|
| <b>IT</b>  | Information Retrieval Systems (9FC08)        | <b>IT</b>  | C# & .Net (9FC11)                                 | <b>IT</b>  | Mobile Application Development (9FC14)     | <b>CSE</b>  | Full stack Development (9EC19)      | <b>AIML</b> | Augmented and Virtual Reality (9LC08) |
| <b>IT</b>  | Image Processing (9FC09)                     | <b>IT</b>  | Computer Graphics (9FC12)                         | <b>CSE</b> | Pattern Recognition (9EC21)                | <b>AIML</b> | Natural Language Processing (9LC18) | <b>CSE</b>  | Computer Vision (9EC22)               |
| <b>CSE</b> | Introduction to Data Science (9EC10)         | <b>CSE</b> | Artificial Intelligence and Deep Learning (9EC13) | <b>IT</b>  | Big Data Analytics (9FC15)                 | <b>AIML</b> | DevOps (9LC17)                      | <b>IT</b>   | Cloud Computing (9FC17)               |
| <b>CSE</b> | Advanced Computer Architecture (9EC11)       | <b>CSE</b> | Distributed Databases (9EC14)                     | <b>IOT</b> | Introduction to Internet of Things (9IC45) | <b>CS</b>   | Block Chain Technologies (9JC05)    | <b>CS</b>   | Ethical Hacking (9JC04)               |
| <b>IT</b>  | Human Computer Interaction (9FC10)           | <b>CSE</b> | Semantic Web & Social Networks (9EC15)            | <b>CSE</b> | Advanced Operating Systems (9EC17)         | <b>CSE</b>  | Mobile Computing (9EC20)            | <b>AIML</b> | Quantum Computing (9LC21)             |
| <b>CSE</b> | Software Requirements and Estimation (9EC12) | <b>IT</b>  | Software Project Management (9FC13)               | <b>CSE</b> | Soft Computing (9EC18)                     | <b>IT</b>   | Agile Software Development (9FC16)  | <b>CSE</b>  | Advanced Algorithms (9EC23)           |

## Open Electives

| Open Elective Streams                 | Open Elective (OE) |                                                       |       |                                          |       |                                         |
|---------------------------------------|--------------------|-------------------------------------------------------|-------|------------------------------------------|-------|-----------------------------------------|
|                                       | Code               | OE – I (3-1)                                          | Code  | OE – II (3-2)                            | Code  | OE – III (4-2)                          |
| <b>Entrepreneurship Stream</b>        | 9ZC22              | Basics of Entrepreneurship                            | 9ZC23 | Advanced Entrepreneurship                | 9ZC24 | Product and Services                    |
| <b>Finance Stream</b>                 | 9ZC05              | Banking Operations and Insurance                      | 9ZC15 | Financial Markets and Services           | 9ZC19 | Project and Risk Management             |
| <b>Mechanical</b>                     | 9BC51              | Introduction to Additive Manufacturing Processes      | 9BC52 | Principles of Operations Research        | 9BC53 | Principles of Automation and Robotics   |
| <b>Electrical</b>                     | 9AC44              | Fundamentals of Measurements and Instrumentation      | 9AC45 | Fundamentals of Renewable Energy Sources | 9AC47 | Power Electronic Devices and Converters |
| <b>Electronics</b>                    | 9CC36              | Fundamentals of Digital Circuits and Micro processors | 9CC37 | Fundamentals of Communication            | 9CC38 | Embedded Systems                        |
| <b>Innovation and Design Thinking</b> | 9ZC08              | Design Literacy and Design Thinking                   | 9ZC09 | Co-Creation and Product Design           | 9ZC10 | Entrepreneurship and Business Design    |



**Syllabus for B.Tech CSE I year I Semester**  
**ENGINEERING CHEMISTRY**  
**(Common to all branches)**

**Course code: 9HC04**

**Prerequisite: NIL**

| L | T | P | C |
|---|---|---|---|
| 2 | 1 | 0 | 3 |

**Course Objective:**

Impart knowledge on the fundamental concepts of chemistry involved in application of several important engineering materials that are used in the industry/day-to-day life.

**Course Outcomes:**

After the completion of the course, student will be able to:

1. Explain the fundamental aspects of atomic, molecular orbitals and metal complexes [L2]
2. List general properties of polymers, lubricants and engineering materials [L1]
3. Outline basic properties of water and its usage in domestic and industrial purposes [L2]
4. Summarize electrochemical procedures related to corrosion [L2]
5. Interpret potential applications of chemistry and practical utility of various organic reactions and drug molecules [L5]

**UNIT - I**

**Atomic and molecular structure**

Molecular orbitals of diatomic molecules and plots of the multicenter orbitals, Equations for atomic and molecular orbitals, Energy level diagrams of diatomic ( $F_2$ ,  $Cl_2$ , CO, NO). Pi-molecular orbitals of butadiene and benzene and aromaticity, Crystal field theory and the energy level diagrams for transition metal ions and their magnetic properties, Band structure of solids and the role of doping on band structures.

**UNIT – II**

**Plastics and Lubricants**

**Plastics: Polymerization**-Addition and Condensation polymerization, Plastics – Thermosetting and Thermoplastics, preparation, properties and **engineering applications of plastics**: PVC, Teflon, Bakelite, Fibers: Nylon 6, 6 and Dacron.

Rubbers – natural and artificial rubber, vulcanization of natural rubber, Buna-S, Buna-N and their **engineering applications**. Fabricated Reinforcing Polymers- **engineering applications**.

**Lubricants**: Definition, classification and function of lubricants, Types of lubrication and mechanisms – Thick Film or Hydrodynamic Lubrication, Thin Film or Boundary Lubrication, Extreme Pressure Lubrication. Classification and properties of lubricants – Viscosity, flash and fire point, cloud and pour point and acid value.

**Engineering applications**

**UNIT - III**

**Water Technology**

- (a) **Introduction**: - Hardness of water – types of hardness (temporary and permanent), calculation of hardness- Numerical problems. Estimation of hardness of water by EDTA

Method.

**(b) Water for Industrial purpose:** Food, sugar, textile, paper and pharma industries, water for steam making characteristics of boiler feed water, boiler troubles- scale and sludge & Carry over (priming & foaming), boiler corrosion, caustic embrittlement.

**(c) Water Treatment:** Internal conditioning- phosphate, carbonate & calgon conditioning. External Treatment: Ion-exchange process. Desalination-reverse osmosis. Municipal water treatment-sedimentation, coagulation, filtration, disinfection-chlorination, ozonization. **Engineering applications: Methodology and working of mineral water plant for drinking purpose.**

## UNIT - IV

### Electrochemistry

Conductance – conductors (metallic and electrolytic), types of conductance – specific, equivalent and molar conductance – effect of dilution on conductance.

Free energy and emf, cell potentials, electrode potential (oxidation and reduction). Types of electrodes - redox electrode (quinhydrone electrode), metal – metal insoluble salt electrode and Ion selective electrode. Cell notation and cell reaction – Nernst equation and applications.

#### Engineering Applications.

**Batteries:** Types of batteries

- (a) Primary batteries – Leclanché cell (dry cell), Lithium cell
- (b) Secondary batteries (Accumulators) – Lead acid battery, Lithium-ion battery
- (c) Fuel cells-  $H_2 - O_2$  fuel cell and  $MeOH-O_2$  fuel cell-advantages and applications.

**Engineering applications – future water powered car, Hydrogen production and storage.**

## UNIT - V

### Corrosion and Surface treatment

Corrosion – basic concepts – types of corrosion, chemical, electrochemical corrosion (absorption of  $O_2$  and evolution of  $H_2$ ) -factors affecting the rate of corrosion.

**Cathodic protection** – sacrificial anodic protection and impressed current cathodic protection method.

#### Surface treatment

Mechanical surface treatment and coatings, casehardening and surface coating, thermal spraying, vapour deposition, Ion implantation, Diffusion coating.

Methods of metallic coatings-hot dipping (tinning and galvanizing), metal cladding (Al cladding), electroplating (copper plating) and electroless plating (nickel plating), electroforming, ceramic, organic and diamond coating.

## UNIT-VI

### Organic reactions and drug molecules

Introduction: reactions involving substitution ( $S_N1$ ,  $S_N2$ ) addition to double bond ( $C=C$ ), elimination ( $E^1$  and  $E^2$ ), oxidation (using  $KMnO_4$ ,  $CrO_3$ ), reduction (Hydrogenation by  $Ni/H_2$ ,  $Pd/C$ )

**Drugs:** Definition, classification structure and applications of commonly used drug molecules-

paracetamol, aspirin, ibuprofen and diphenhydramine (Benadryl)

Principles of spectroscopy and selection rules: Electronic spectroscopy. Fluorescence and its applications in medicine. Vibrational and rotational spectroscopy of diatomic molecules-  
**Applications.**

#### **TEXT BOOKS:**

1. Engineering Chemistry: PK Jain & MK Jain, Dhanapathrai Publications(2018)
2. Engineering Chemistry: by Thirumala Chary Laxminarayana & Shashikala, Pearson Publications(2020)

#### **REFERENCE BOOKS:**

1. Textbook of Engineering Chemistry: Jaya Shree Anireddy, Wiley Publications(2019)
2. Engineering Chemistry: by & B. Rama Devi, PrsantaRath & Ch. Venkata Ramana Reddy, Cengage Publications(2018)
3. Engineering Chemistry: Shashi Chawla, Dhanapathrai Publications(2019)
4. Textbook of Engineering Chemistry: SS Dara, SS Umare S. Chand Publications (2004)

**Syllabus for B.Tech CSE I year I Semester**  
**Problem Solving using C**  
**(Common to all branches)**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Code: 9FC01**

**Prerequisite: NIL**

**Course Objective:**

Acquire problem solving and coding skills using C language.

**Course Outcomes:**

After the completion of the course, the student will be able to:

1. Explain the basic computer concepts and Illustrate programming principles of C language. [L2,L4]
2. Develop C programs to solve simple mathematical and engineering problems using decision control , loop statements , arrays and strings.[L3,L4]
3. Apply modular programming using functions to develop reusable code.[L3,L4]
4. Analyze memory oriented concepts using pointers to implement inter function, function communication, pointer arithmetic. [L4]
5. Describe file handling concepts using C. [L2]

## **UNIT I**

**Problem solving Techniques** – Algorithms, pseudo code, flowcharts with examples

**Introduction to Computer Programming Languages** – Machine Languages, Symbolic Languages, High-Level Languages,

**Introduction to C language** – Characteristics of C language, Structure of a C Program. Syntax and semantics.

Data Types, Variables – declarations and initialization, formatting input and output.

## **UNIT – II**

**C Tokens:** Identifiers, Keywords, Constants, variables and operators

**Expressions** – Arithmetic expressions, Precedence and Associativity, evaluating expressions,

**Decision control structures** – if, Two-way selection – if else, nested if, dangling else, Multi-way selection – else if ladder and switch.

**Repetitive control structures** – Pre-test and post-test loops – initialization and updation, while, do while and for loop and nested loops.

**Unconditional statements:** break, continue and goto statements with examples.

## **UNIT III**

**Arrays** – Definition and declaration, initialization, accessing elements of in arrays, storing values in arrays,

1-D arrays, 2-D arrays, character arrays and multidimensional arrays.

**Function and arrays:** passing individual elements to arrays, passing 1-D array, 2-D array to function.

**Applications:** Linear search, matrix addition, subtraction, multiplication and transpose

## UNIT – IV

**Functions** – User – defined functions - Function definition, arguments, return value, prototype, arguments and parameters, inter-function communication. Standard functions – Math functions. Scope – local, global.

**Parameter passing** – Call by value and call by reference.

**Recursive functions** – Definition, examples, advantages and disadvantages.

**Macros** – Definition, examples, comparison with functions.

**Storage Classes** – auto, extern, static and Register

## UNIT V

**Introduction to Pointers** – pointer constants, pointer values, pointer variables, accessing variables through pointers, pointer declaration and definition, declaration versus redirection, initialization of pointer variables, Pointer for inter function communication, pointer to pointers, pointer to function.

**Arrays and pointers** – Pointer arithmetic and arrays, array of pointers

**Strings** – Declaration, Initialization, Input and Output functions, strings and pointer, string handling functions.

## UNIT VI

**Files** – Concept of a file, streams, text and binary files, stream file processing, system created streams, Standard library I/O functions, file open and close, formatting I/O functions, character I/O functions, Binary I/O, command line arguments, file status functions, positioning functions.

Applications: Basic operations on files.

### Text Books:

1. E. Balaguruswamy, Programming in ANSI C, Tata McGraw-Hill
2. Let Us C by Yashavant Kanetkar

### Reference Books:

1. Programming in C (2nd Edition) by Ashok N Kamthane
2. Brian W. Kernighan and Dennis M. Ritchie, The C Programming Language
3. Prentice Hall of India

**Syllabus for B.Tech CSE I year I Semester**  
**MATRIX ALGEBRA AND CALCULUS**  
**(Common to all branches)**

**Code: 9HC11**

| L | T | P | C |
|---|---|---|---|
| 2 | 1 | 0 | 3 |

**Pre Requisites:** Mathematics Knowledge at Pre-University Level

**Course Objective:**

Introduce the fundamental concepts of linear algebra, develop skills in differential calculus for problem-solving and analysis, Teach analytical methods for solving ordinary differential equations.

**Course Outcomes:**

After the course completion the students will be able to:

1. Solve the linear system of equations.[ L3]
2. Determine the Eigen values and Eigenvectors of the matrix and apply Cayley Hamilton Theorem to obtain the higher powers of a matrix. [L3, L5]
3. Identify the nature of the quadratic form and reduction of the quadratic form to its canonical form.[L2]
4. Apply appropriate mean value theorems to obtain the mean values and find the power series expansion of a function [L2, L3]
5. Solve the first order and higher order ordinary differential equations with constant coefficients. [L3]

**UNIT-I**

**System of Linear Equations:** Elementary row/column operations -Echelon form, Rank of a matrix, Inverse of a matrix by Gauss Jordan method, Non-Homogenous and Homogenous system of linear equations- consistency or inconsistency of a system, Gauss Elimination method, Rank method and problems, Symmetric, Skew-symmetric and Orthogonal matrices.

**UNIT-II**

**Eigen values and Eigen vectors:** Definitions and Properties (without proofs). Evaluation of Eigen values and Eigenvectors for a given matrix, Cayley-Hamilton Theorem (without proof) and its applications in finding higher powers & inverse of a matrix, Diagonalization of a matrix, Hermitian, Skew-Hermitian and Unitary matrices.

**UNIT-III**

**Quadratic forms:** Quadratic forms, Nature, rank, index and signature of a quadratic form. Reduction of quadratic form to canonical form

**UNIT-IV**

**Single Variable Calculus:** Rolle's Theorem, Lagrange's and Cauchy's mean value theorems (without proof); Taylor's and Maclaurin's series (without proof) and their application for series expansions of standard functions.

## UNIT-V

**First order ODE:** Exact differential equations, equations reduced to exact, Linear and Bernoulli's equations, Newton's law of cooling, Law of natural Growth/Decay.

## UNIT-VI

**Higher order ODE:** Higher order linear differential equations with constant coefficients- Complementary function, Particular Integral, Method of variation of parameters.

### Text Books:

1. R K Jain and S R K Iyengar Advanced Engineering Mathematics, Narosa Publications.
2. B.S. Grewal, Elementary Engineering Mathematics, Khanna Publishers
3. Alan Jeffery, Advanced Engineering Mathematics, Academic Press
4. Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11th Reprint, 2010.
5. Erwin kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.

**Syllabus for B.Tech CSE I year I Semester**  
**Essential English Language Skills**  
**(Common to all branches)**

| L | T | P/D | C |
|---|---|-----|---|
| 2 | 0 | 0   | 2 |

**Course Code: 9HC01**

**Prerequisite: Nil**

**Course Objective:**

Enhance communicative skills, developing essential soft skills, and building a strong foundation in English language.

**Course Outcomes:**

After completion of the course, the student will be able to:

1. Demonstrate the use of diverse forms of vocabulary in their communication. [L2]
2. Illustrate different grammatical structures and use the appropriate ones in their communication.[L2]
3. Develop effective reading skills by applying strategies to comprehend different types of texts. [L3]

**Units**

**1. Vocabulary-1:**

- 1.1 Root words
- 1.2 Synonyms and Antonyms
- 1.3 Homonyms, Homophones and Homographs
- 1.4 One word substitutes

**2. Vocabulary-2**

- 2.1 Idioms and Phrases
- 2.2 Confusables

**3. Grammar-1**

- 3.1 The Parts of Speech
- 3.2 Use of Articles
- 3.3 Omission of Articles

**4. Grammar-2**

- 4.1 Tenses
- 4.2 Prepositions
- 4.3 Concord

**5. Reading & Writing**

- 5.1 Techniques of Reading, Reading Comprehension
- 5.2 Kinds of Sentences
- 5.3 Punctuation

**6. Writing-2**

- 6.1 Voice – Active voice and Passive Voice
- 6.2 Speech-Direct & Reported Speech
- 6.3 Common errors in English



### **Suggested Reading & References:**

1. Word Power Made Easy by Norman Lewis
2. English Grammar In Use: A Self Study Reference And Practice Book Intermediate Learners Book by Raymond Murphy
3. The Logic of English Words by Logophilia Education
4. English Vocabulary In Use Elementary Book With Ans And Cd-Rom by Felicity Odell (Second Edition)
5. Effective Technical Communicatioin by M. Ashraf Rizvi
6. Intermediate grammar usage and composition; M.L.Tickoo, A.E.Subramanian, P.R.Subramanyam; OBS
7. An Interactive Grammar to Modern English by Shivendra K. Verma and Hemalatha Nagarajan, Frank Bros. & Co.

**Syllabus for B.Tech CSE I year I Semester**  
**Oral Communications Lab-1**  
**(Common to all branches)**

**Course Code: 9HC61**

**Prerequisite: Nil**

| L | T | P/D | C |
|---|---|-----|---|
| 0 | 0 | 2   | 1 |

**Course Objective:**

Develop communication skills that will enable in functioning effectively in civic, professional and interpersonal skills.

**Course Outcomes:**

After completion of the course, the student will be able to:

1. Discover people, objects and situations using simple sentences with proper pronunciation. [L4]
2. Use apt expressions and narrate stories in simple sentences. [L3]

**OC LAB**

**Unit 1: Communication Skills**

Communication basics, essential elements of effective communication, barriers to communication, setting SMART communication goals.

**Activities:**

- Ice-breaking activities
- Personal Communication SWOT Analysis
- Communication Case Studies: The Terrible & The Terrific

**Unit 2: Pronunciation Matters**

Importance of pronunciation, neutralizing mother tongue interference (MTI).

**Activities:**

- Odd Word Out
- Minimal Pairs Masti
- Shadow reading

**Unit 3: Use apt expressions in diverse situations**

Self-introduction, Greetings, apologizing, complimenting, inviting, complaining etc.

**Activity:**

Role play in different contexts using the appropriate expressions

**Unit 4: Mind your Tenses**

Describing present and past habits, states, and events.

Talking about actions in progress, relating past to the present, talking about the future.

Framing questions. (confirmation/information questions)

**Activities:**

- Speaking activity on daily routine, how students spent their recent vacation, speaking about their childhood, speaking about future plans.

- Dumb Charades (Present/Past continuous - Present/ Past perfect)
- Guessing game (10/20 yes or no questions)

### **Unit 5: Hone your Describing skills**

Describing people, objects, and situations

#### **Activities:**

- Picture descriptions.
- Guessing games - listening to the descriptions.
- Narrating memorable incidents from life.
- Describe your ideal world
- Once upon a time.....

### **Unit 6: The Art of Storytelling**

Story telling for career success, the basics of story telling

#### **Activities:**

- Building stories - chain activity.
- Story prompts activity.
- Narrate the story. (all the hints are given except linking words and tenses)

#### **Suggested Reading & References:**

1. "An Interactive Grammar of Modern English" by Shivendra K Verma and HemalathaNagarajan, Frank Bros. & Co.
2. "Skill Sutras" by JayashreeMohanraj, Prism Books Pvt. Ltd.
3. "Better English pronunciation" by J.D. Connor.
4. "Effective Communication" John Adair, Pan Macmillan Ltd.
5. "Body Language", by Allan Pease, Sudha Publications.
6. "Communicative English", by Hariprasad M. and Prakasam V, Neel Kamal Publications.

**Syllabus for B.Tech CSE I year I Semester  
ENGINEERING CHEMISTRY LAB**

**Code: 9HC64**

**Prerequisite: Nil**

| L | T | P/D | C   |
|---|---|-----|-----|
| 0 | 0 | 3   | 1.5 |

**Course Objective:**

Enable the learners to get hands-on experience on the principles discussed in theory sessions and to understand the applications of these concepts in engineering.

**Course Outcomes:**

After completion of the course, the student will be able to:

1. Estimate the hardness and chloride in water. [L5]
2. Determine strength of acid by potentiometric and conductometric methods. [L5]
3. Demonstrate preparation of polymer, aspirin and inorganic compound. [L2]

**List of Experiments**

1. Preparation of coordination complex NiDMG Complex
2. Determination of surface tension
3. Determination of viscosity
4. Saponification/acid value of oil
5. Ion exchange column for removal of hardness of water / Estimation of Hardness of water by EDTA Method
6. Determination of chloride content of water
7. Determination of cell constant and conductance of solutions (HCl Vs NaOH / Mixture of acid Vs Strong base)
8. Potentiometry - determination of redox potential and emf ( $\text{FeSO}_4$  Vs  $\text{KMnO}_4$  / HCl Vs NaOH)
9. Determination of the rate constant of acid catalyzed hydrolysis of methylacetate
10. Synthesis of a polymer- Thiokol rubber / Urea-Formaldehyde resin
11. Synthesis of a drug-Aspirin
12. Estimation of  $\text{Mn}^{+7}$  by Colorimetry method

**Syllabus for B.Tech CSE I year I Semester**  
**Problem Solving using C Lab**

**Code: 9FC61**

**Prerequisite: Nil**

| L | T | P/D | C   |
|---|---|-----|-----|
| 0 | 0 | 3   | 1.5 |

**Course Objective:**

Design and use the code and test a 'C' Program to solve a computational problem.

**Course Outcomes:**

After completion of this course student will be able to:

1. Develop programs to solve simple mathematical and engineering applications using C language [L3]
2. Illustrate various operations on files using C language [L2]

**[The laboratory should be preceded or followed by a tutorial to explain the approach or algorithm to be implemented for the problem given.]**

**1. Unit I (Cycle 1)**

1. Write an algorithm for converting a given Celsius temperature to its equivalent Fahrenheit temperature and draw a flowchart.
2. Write an algorithm to find the largest of three given numbers and draw a flowchart.
3. Write an algorithm and draw a flowchart for finding the roots and nature of roots of a quadratic equation, given its coefficients.
4. Write an algorithm and flowchart for finding the first n Fibonacci numbers, give n.

**2. Unit II (Cycle 2)**

1. Write an algorithm, flowchart, and C program for:
2. Finding the area and circumference of a circle of given radius.
3. Finding the volume of a sphere of given radius.
4. Finding the lateral surface area of a right circular cone of given base radius and height.
5. Finding selling price of an item, given its cost price and profit percent.
6. Finding the interest on a given principal for a given period of time at a given rate of per year.
7. Write a C program to display all the sizes of data types in C.
8. Write a C program to display a given decimal integer into an equivalent octal number and hexadecimal number using %o and %x in printf function.

**3. Unit II (Cycle 3)**

1. Write a C program to find the roots and nature of the roots of a quadratic equation, given its coefficients.
2. Write a C program for finding the largest of three given numbers.
3. A salesman gets a commission of 5% on the sales he makes if his sales is below Rs.5000/- and a commission of 8% on the sales that exceeds Rs.5000/- together with Rs.250/-. Write an algorithm or a flowchart and develop C program for computing the commission of the salesman, given his sales.
4. Write a C Program to demonstrate Marcos.

**4. Unit III (Cycle 4)**

1. Write three C programs to print a multiplication table for a given number using while, do-while, and for loops.
2. Write a C program to compute the sum of:
3.  $1+x+x^2+x^3+\dots+x^n$ , given  $x$  and  $n$ .
4.  $1! + 2! + 3! + \dots + n!$ , given  $n$ .
5.  $1 - x^2/2! + x^4/4! - x^6/6! + x^8/8! - x^{10}/10! + \dots$  to  $n$  terms where the  $n^{\text{th}}$  term becomes less than 0.0001.

#### 5. Unit III (Cycle 5)

1. Write a C program in the menu driven style to perform the operations  $+$ ,  $-$ ,  $*$ ,  $/$ ,  $\%$  between two given integers.
2. Write a C program to find the largest and the least of some numbers given by the user.
3. Write a C program to find the sum of the digits of a positive integer.

#### 6. Unit IV (Cycle 8)

1. Write a program to store the numbers given by the user in an array, and then to find the mean, deviations of the given values from the mean, and variance.
2. Write a C program to initially store user given numbers in an array, display them and then to insert a given number at a given location and to delete a number at a given location.
3. Write a program to store user given numbers in an array and find the locations of minimum and maximum values in the array and swap them and display the resulting array.

#### 7. Unit IV (Cycle 9)

1. Write a C program to implement the operations of matrices – addition, subtraction, multiplication.
2. Write a program to find whether a given matrix is symmetric, lower triangular, upper triangular, diagonal, scalar, or unit matrix.

#### 8. Unit III (Cycle 6)

1. Write C functions for the following:
  - a) A function that takes an integer  $n$  as argument and returns 1 if it is a prime number and 0 otherwise.
  - b) A function that takes a real number  $x$  and a positive integer  $n$  as arguments and returns  $x^n$ .
  - c) A function that takes a positive integer  $n$  as an argument and returns the  $n^{\text{th}}$  Fibonacci number.
2. Using recursion write C functions for the following:
  - a) Factorial of a non-negative integer  $n$ .
  - b) Number of combinations of  $n$  things taken  $r$  at a time.
  - c) Greatest Common Divisor of two integers.
  - d) Least Common Multiple of two integers.

#### 9. Unit III (Cycle 7)

- a) Write a menu driven style program to compute the above functions (cycle 6) on the choice of the function given by the user.
- b) Define macros for the following and use them to find sum of the squares of the minimum and maximum of two given numbers.
  - a) Larger of two numbers.
  - b) Smaller of two numbers.

- c) Sum of the squares of two numbers.
- c) Write a program to generate Pascal's triangle.
- d) Write a program to count the number of letters, words, and lines in a given text.

#### **10. Unit V (Cycle 10)**

1. Write a function to swap two numbers.
2. Write a function to compute area and circumference of a circle, having area and circumference as pointer arguments and radius as an ordinary argument.

#### **11. Unit VI (Cycle 12)**

1. Write a program to:  
Create a file by the name given by the user or by command line argument and add the text given by the user to that file.
  - a) Open the file created above and display the contents of the file.
  - b) Copy a file into some other file, file names given by the user or by command line arguments.
  - c) Append a user mentioned file to another file.
  - d) Reverse the first n characters of a file.

#### **12. (Cycle 13)**

Case study on Electricity Billing, Restaurant Billing System

**Syllabus for B.Tech CSE I year I Semester**  
**WORKSHOP/MANUFACTURING PROCESSES LAB**  
**(Common to all branches)**

**Course Code: 9BC61**

**Prerequisite: NIL**

| L | T | P | C   |
|---|---|---|-----|
| 0 | 1 | 3 | 2.5 |

**Course Objective:**

Impart knowledge and skill to use tools, machines, equipment, and measuring instruments.

**Course Outcomes:**

After completion of the course, the student will be able to:

1. Demonstrate and make use of the workshop tools for Fitting, Carpentry, Welding, Casting, Smithy, Moulding, Glass cutting and Electric connections [L2]
2. Design and Fabricate jobs with wood, MS flat, GI Sheet material [L3]

**LIST OF EXPERIMENTS**

| S.No | Trades                           | Experiment name                                                                                                                                    |
|------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Fitting Shop                     | 1. Preparation of T-Shape Work piece<br>2. Preparation of U-Shape Work piece which contains: Filing, Sawing, Drilling, Grinding.                   |
| 2    | Carpentry                        | 3. Cross Half Lap joint<br>4. Half Lap Dovetail joint                                                                                              |
| 3    | Electrical & Electronics         | 5. One lamp one switch<br>6. Stair case wiring                                                                                                     |
| 4    | Welding ( Arc & Gas) & Soldering | 7. Practice of Lap and Butt joint by Arc welding Demonstration: Gas welding, Resistance welding & Soldering                                        |
| 5    | Casting                          | 8. Preparation of mould cavity using solid pattern<br>9. Preparation of mould cavity using split pattern<br>Demonstration: pouring of molten metal |
| 6    | Tin Smithy                       | 10. Preparation of Rectangular Tray<br>11. Preparation of Square box                                                                               |
| 7    | Machine Shop                     | Turning, Drilling and grinding operations on Lathe, Drilling and grinding machines                                                                 |
| 8    | Plastic moulding & Glass Cutting | 12 a) Injection Moulding<br>b) Glass Cutting with hand tools                                                                                       |
| 9    | Domestic Appliances              | Study of internal components & circuit of appliances such as Fans, Mixers, Air blower, Iron box, Rice cooker, Emergency light etc.,                |



**Syllabus for B.Tech CSE I year II Semester  
ENGINEERING PHYSICS**

| L | T | P | C |
|---|---|---|---|
| 2 | 1 | 0 | 3 |

**Code: 9HC07**

**Prerequisite: Nil**

**Course Objective:**

Study and understand the concepts of physics involved in application of several important engineering components/equipment that are used in the industry/day-to-day life.

**Course Outcomes:**

After completion of the course, the student will be able to:

1. Interpret the concept of quantum mechanics, Schrodinger wave equation and its application for one dimensional potential box. [L2]
2. Explain the principle, construction and working of lasers and fiber optics along with their applications. [L2]
3. Summarize the phenomenon of magnetism and superconductivity. [L2]
4. Outline the concepts of dielectrics, polarization and apply the same for Piezo, Ferro and Pyro-electricity. [L2,L3]
5. Identify the nature of semiconductors and demonstrate the semiconductor devices. [L2,L3]
6. Characterize the nano and bulk materials for various applications. [L3]

**Unit: 1**

**Wave nature of particles, Schrodinger equation and its application**

Waves and Particles, de-Broglie Hypothesis, Matter waves, Davisson and Germer's Experiment, G.P. Thomson Experiment, Heisenberg's Uncertainty Principle, Schrodinger's Time Independent Wave Equation – Physical Significance of the Wave Function–Application of Schrodinger wave equation - Particle in One Dimensional Potential Box.

**Unit: 2**

**Lasers and Fiber Optics**

**Lasers:**

Characteristics of LASER, Spontaneous and Stimulated Emission of Radiation, Einstein's Coefficients and their significance. Meta-stable State, Pumping, Population Inversion and Optical resonator. Ruby Laser, Helium-Neon Laser, Semiconductor Diode Laser, Applications of Lasers.

**Fiber optics:**

Principle and construction of optical fiber, Acceptance Angle and Numerical Aperture. Based on refractive index profile classification of Optical Fibers: Single mode & Multimode mode Step index fibers, Single mode & Multimode mode graded index fibers. Attenuation in Optical Fibers (scattering, absorption and bending losses), optical Fiber communication system, Fiber Optic Sensors-Temperature sensor, Pressure sensor and Medical Endoscopy.

**Unit: 3**

**Magnetism and Superconductivity**

**Magnetic Materials:**

Origin of Magnetic Moment-Bohr Magnetron. Ferromagnetic domain, Magnetization process by using domain, B-H curve explanation based on Domain theory and important outcomes of the curve. Hard and Soft Magnetic Materials.

#### **Super conducting Materials:**

General properties of Superconductors. Effect of Magnetic Field, Critical current density, Meissner effect, Penetration depth. Type-I and Type-II superconductors, BCS theory, Magnetic levitation.

#### **Unit: 4 Dielectric materials**

Electric Dipole, Dipole Moment, Dielectric Constant, Electric Susceptibility. Electronic, Ionic polarizability (Quantitative) and Orientation Polarization (Qualitative). Internal fields in Solids, Clausius-Mossotti equation, Frequency and temperature effect on Dielectrics (Qualitative), Applications - Piezo-electricity, Pyro-electricity and Ferro-electricity.

#### **Unit: 5 Semiconductors and Semiconductor devices**

##### **Semiconductors:**

Fermi Level in Intrinsic and Extrinsic Semiconductors. Carrier concentration of Intrinsic and Extrinsic Semiconductor (qualitative). Direct & Indirect Band Gap Semiconductors, Hall Effect in semiconductors.

##### **Semiconductor devices:**

Formation of a PN Junction and working of a PN Junction diode, Energy band Diagram of open circuited PN Diode, I-V Characteristics of PN Junction diode. Applications: LED, Solar Cell and Photo diode.

#### **Unit: 6**

**Nanomaterials and their fabrication:** Origin of Nanotechnology, Nano Scale, Surface to Volume Ratio, Quantum Confinement, Bottom-up Fabrication, Sol-gel method, Chemical vapor Deposition technique (CVD); Top-down Fabrication; Ball Milling, Characterization of Nano materials (XRD & TEM), carbon Nano tubes (CNTs), Applications of Nano Materials.

#### **Text Books:**

1. B.K. Pandey & S. Chaturvedi Engineering Physics, Cengage Learning

#### **Reference Books:**

1. P K Palanisamy, Engineering Physics, Sitech Publications
2. Charles Kittel, Introduction to Solid State Physics, John Wiley Publisher
3. A.S. Vasudeva , Modern engineering Physics, S. Chand
4. Dekker, Solid State Physics
5. Dr. M.N. Avadhanulu, Engineering Physics, S. Chand
6. Dekker, Solid State Physics
7. Halliday and Resnick, Physics
8. S.O. Pillai, Solid State Physics
9. A. Ghatak - Optics

**Syllabus for B.Tech CSE I year II Semester**  
**DATA STRUCTURES**  
**(Common to all Branches)**

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Prerequisites:** Problem Solving using C

**Course:** 9EC01

**Course Objective:**

Develop skills to apply appropriate data structures in solving societal/industrial problems.

**Course Outcomes:**

After completion of this course student will be able to:

1. Explain different data structure and select the appropriate data structure to develop applications. [L2,L3]
2. Illustrate various Linear, Non Linear data structure in developing applications. [L2,L3]
3. Demonstrate various hashing and collision resolution techniques for enhancing the performance of algorithms.[L3]
4. Asses the performance of various searching and sorting techniques. [L5]

**UNIT I:**

**Structures:** Introduction, types, initialization and accessing, Array of Structures, Nested Structures, Self-referential structures. Unions, enum, typedef, Dynamic Memory allocation.

**UNIT II:**

**Introduction to data structures:** Abstract data type (ADT), Stacks, Queues and Circular queues and their implementation with arrays.

Applications of Stack: infix to post fix conversion, postfix expression evaluation.

Applications of Queues.

**UNIT III:**

**Linked list:** introduction, advantages of Linked list over Arrays.

**Single linked list:** creation, insertion, deletion and display operations

**Double linked list:** creation, insertion, deletion and display operations

**Circular linked list:** creation, insertion, deletion and display operations,

Implementation of Stacks and Queues with singly linked list.

**UNIT IV:**

**Trees:** Terminology, Binary Tree: types, representation and traversals (in-order, pre-order, post-order).

**Binary Search Tree:** introduction, operations (insertion, deletion, display)

**AVL Trees:** Definition, examples, and operations (insertion, deletion and searching).

**UNIT V:**

**Graphs:** terminology, representation, traversals (DFS and BFS).

**Heaps:** Introduction, Min Heap, Max Heap, Operations on Heaps, Heap Sort.

**Hashing:** Hash Table, Hash functions.

**Collision resolution techniques:** separate chaining, open addressing-linear probing, quadratic probing, double hashing.

**UNIT VI:**

**Searching:** linear and binary search methods.

**Sorting:** Bubble Sort, Insertion Sort, Selection Sort, Quick sort, Merge sort  
Performance analysis of Searching and Sorting Algorithms.

**TEXT BOOKS:**

1. Data Structures Using C second edition by [Reema Thareja](#) Oxford university press
2. Data Structure through C by Yashavant Kanetkar.

**REFERENCES:**

1. Alfred V. Aho, Jeffrey D. Ullman, John E. Hopcroft. Data Structures and Algorithms. Addison Wesley, 1983 .
2. Data Structures using C Aaron M. Tenenbaum, Yedidyah Langsam, Moshe J. Augenstein.
3. Introduction to Data Structures in C By Kamtane
4. Data Structures, A pseudocode Approach with C by Richard F. Gilberg and Behrouz A. Forouzan.

**Syllabus for B.Tech CSE I year II Semester**  
**ADVANCED CALCULUS**  
**(Common to all branches)**

| L | T | P | C |
|---|---|---|---|
| 2 | 1 | 0 | 3 |

**Code: 9HC12**

**Pre Requisites:** Mathematics Knowledge at Pre-University Level

**Course Objective:**

Introduce the fundamental concepts of multivariable calculus, including partial derivatives, multiple integrals, and their applications. Develop an understanding of Fourier series and their role in representing periodic functions. Teach the principles of vector calculus, including vector integrals and applications of vector integral theorems.

**Course Outcomes:**

After the course completion the students will be able to

1. Compute the Jacobian transformation, the extreme values of a multivariable function and solve the first order linear and nonlinear PDEs. [L2, L3]
2. Evaluate double integrals using change of order of integration and change of variables, triple integrals [L3, L5]
3. Determine Fourier series expansion of a function over the interval.[L5]
4. Find directional derivative and solve the problems on line, surface and volume integrals. [L1, L3]

**UNIT-I: Functions of several variables:** Limits, Continuity and partial derivative, total derivative, Jacobian, Maxima and minima of two variable functions (without constraints).

**UNIT-II: Multiple Integrals:** Double integrals, change of order of integration, change of variables (Cartesian to polar), Triple integrals (Cartesian form).

**UNIT-III: Partial Differential Equations:** Formation of partial differential equations, solutions to first order linear and non-linear partial differential equations – standard Forms.

**UNIT-IV: Fourier series:** Dirichlet conditions, Fourier series of functions over the intervals of length  $2l$  &  $2\pi$ . Half range sine and cosine series, Problems on Parseval's theorem (without proof).

**UNIT-V: Vector Differentiation:** Vector and scalar point functions, gradient, directional derivatives; divergence and curl of a vector point function and problems.

**UNIT-VI: Vector Integration:** Line integrals, surface integrals, volume integrals, Green, Gauss divergence and Stokes theorems (without proofs) and problems.

**Text Books:**

1. R K Jain and S R K Iyengar Advanced Engineering Mathematics, Narosa Publications.
2. B.S. Grewal, Elementary Engineering Mathematics, Khanna Publishers
3. Alan Jeffery, Advanced Engineering Mathematics, Academic Press
4. Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11th

- Reprint,2010.
5. Erwin kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons,2006.

**Syllabus for B.Tech CSE I year II Semester**  
**BASIC ELECTRICAL AND ELECTRONICS ENGINEERING**

**Code: 9AC48**

**Prerequisite: Nil**

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Course Objective:**

Understand the concepts of Electrical engineering and their applications.

**Course Outcomes:**

After studying this course, the student will be able to

1. Apply the principles of electrical circuits and DC generation with basic equations.[L3]
2. Illustrate the working principles of DC and three phase AC motors.[L2]
3. Analyze the construction and working principles of diode, various transistors with applications.[L4]
4. Use numbering systems to solve Boolean expressions.[L3]

**Unit – I: Fundamentals of Electrical Engineering and DC Machines:**

Ohm's Law, Kirchhoff's Laws, types of sources, passive elements. Series parallel circuits, mesh and nodal analysis. Superposition, Reciprocity theorem.

**DC Machines:** Principle of operation of D.C generators, types, E.M.F equation. Principle of operation of D.C motors, Types motors, Torque equation, Losses and efficiency, simple problems on D.C Generators and motors.

**Unit – II: Fundamentals of AC circuits:**

AC voltage wave form and basic definitions: Peak Value, R.M.S. value, Average values, Form factor and Peak factor, 'j' operator, Analysis of single phase AC circuits series and parallel (Simple circuits). Three phase circuits – Star - delta connection, Relation between line and phase voltages / currents in a 3-phase Star-Delta balanced system.

**Unit – III: Induction Motors and Instruments:**

Concept of Faraday's laws, 3- phase induction motor working principle, operation and construction details.

**Instruments:** Introduction, classification of instruments, operating principles, essential features of measuring instruments, permanent magnet moving coil (PMMC) instruments, moving iron (MI) instruments.

**UNIT IV-DIODE:** Overview of Semiconductors, PN junction diode and Zener diode –Diode circuits: rectifiers (bridge type only), filters, clippers and clampers.

**UNIT V- TRANSISTOR:** BJT construction, operation, characteristics (CB, CE and CC configurations) and uses – JFET and MOSFET construction, operation, characteristics (CS configuration) and uses.

**UNIT VI-DIGITAL ELECTRONICS:** Number systems – binary codes –binary arithmetic - Boolean algebra, laws & theorems - simplification of Boolean expression using K maps -

logic gates - implementation of Boolean expressions is using logic gates - standard forms of Boolean expression.

**Text Books:**

1. Basic Electrical Engineering –T.K. Nagesarkar and M.S. Sukhja, Oxford University Press. 2<sup>nd</sup> edition.
2. Basic electrical Engineering – M.S. Naidu and S. Kamakshiah – TataMcGraw-Hill, 2005 edition.
3. Principles of Electronics - V.K.Mehta, S.Chand Publications, 2nd edition.

**References:**

1. Theory and problems of Basic electrical Engineering- D.P.Kotahari & I.J.Nagrath PHI.
- Electronic Devices and Circuits, Millman & Halkias, TMH publications.



**Syllabus for B.Tech CSE I year II Semester**  
**ENGINEERING GRAPHICS**  
**(Common to All Branches)**

**Code: 9BC01**

**Pre Requisites:** Nil

| L | T | P | C |
|---|---|---|---|
| 1 | 0 | 4 | 3 |

**Course objective:**

Develop the skills to communicate ideas graphically and to create engineering drawings.

**Course outcomes:**

After completion of this course, the student will able to:

1. Outline the basics of the Engineering Graphics, Curves and Auto Cad [L2]
2. Interpret the concepts of orthographic and isometric projections of lines, planes and solids. [L2]
3. Draw the sections of solids and development of surfaces. [L1]
4. Develop orthographic views from isometric views and vice versa [L3]

**UNIT – I**

**Introduction to Engineering Drawing:** Drawing Instruments and their uses, types of lines, Lettering, Dimensioning-Terms & notations, placing of dimensions, general rules of dimensioning **Scales**(concepts): RF, Reducing, Enlarging and Full Scales

**Curves:** Conic Sections including Rectangular Hyperbola - General method, Cycloid and Involute of circle.

**UNIT – II**

**Orthographic Projection:** Principles of Projection – Methods of projection, First angle and third angle projections.

**Projections:** Projections of Points, Projections of straight lines –line inclined to one plane and line inclined to both reference planes.

**UNIT –III**

**Projections of regular Planes:** types of planes, plane inclined to one reference plane, Oblique planes

**Projections of regular Solids:** types of solids, Projections of: Prisms, Cylinders, Pyramids, Cones – simple position and axis inclined to one plane only

**UNIT –IV**

**Sections and sectional views of Solids:** Sections and Sectional views of Right Regular Solids – Prism, Cylinder, Pyramid – Auxiliary views.

**Development of Surfaces:** Methods of development, Development of lateral Surfaces of Right Regular Solids – Prisms, Cylinders, Pyramids, Cones and their sections.

**UNIT – V**

**Isometric Projection:** Introduction, Isometric axes, lines and planes, Isometric Scale – Isometric drawing or View – Isometric drawing of planes and simple solids such as prisms, pyramids, cylinder, cone.

## **UNIT –VI**

Conversion of isometric views to orthographic views of simple objects.

**Introduction to CAD:** Benefits of CAD, Graphic input and output devices - Function performed by CAD Software, AUTOCAD-Drawing Entities, Editing commands.

### **Text Books:**

Bhatt N.D., Panchal V.M. & Ingle P.R., (2014), Engineering Drawing, Charotar Publishing House (In First-angle Projection Method)

### **Reference Books:**

1. Shah, M.B. & Rana B.C. (2008), Engineering Drawing and Computer Graphics, Pearson Education
2. Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMH Publication
3. AUTOCAD Software Theory and User Manuals

**Syllabus for B.Tech CSE I year II Semester**  
**ORAL COMMUNICATION LAB – II**  
**(Common for all Branches)**

**Code: 9HC62**

**Prerequisite: English**

| L | T | P | C   |
|---|---|---|-----|
| 0 | 0 | 3 | 1.5 |

**Course Objective:**

Enhance the skills and knowledge necessary to communicate effectively in various personal and professional situations.

**Course Outcomes:**

After completion of the course, the student will be able to:

1. Use appropriate language in varied real-world scenarios [L3]
2. Develop a winning presentation and present themselves with ease in various competitive situations.[L3]

| Unit   | OC Lab (2 hrs. per week)                                                                                                                                                                                                                                                                                                                                   | No. of periods (32 hrs.) |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Unit 1 | <b>Small talk and conversational techniques</b><br>Tips on enhancing conversation skills.<br>Conversation starters, small talk questions, how to talk to strangers and practice activities on initiating informal conversations.<br><ul style="list-style-type: none"> <li>• Talk about your favourite things.</li> <li>• Interview each other.</li> </ul> | 6                        |
| Unit 2 | <b>Role Play/skit/one act play</b><br><ul style="list-style-type: none"> <li>• Role play assuming fictional characters and non-fictional characters.</li> <li>• One Act plays</li> <li>• Ad' Venture: Advertisement creation and enacting.</li> </ul>                                                                                                      | 4                        |
| Unit 3 | <b>Just a minute (JAM)</b><br>One-minute speaking activity on topics of students' choice and Extempore.                                                                                                                                                                                                                                                    | 6                        |
| Unit 4 | <b>Presentation skills</b><br>Introduction to structural talk. Techniques of making effective presentations.<br><ul style="list-style-type: none"> <li>• Five minute PowerPoint presentations.</li> </ul>                                                                                                                                                  | 6                        |
| Unit 5 | <b>Group Discussions</b><br>Tips on Dos and Don'ts of Group Discussion (GD). Discussion on evaluation pattern during GD.<br><ul style="list-style-type: none"> <li>• Practice sessions: GDs on different topics.</li> </ul>                                                                                                                                | 4                        |
| Unit 6 | <b>Facing questions: Mock Interviews</b>                                                                                                                                                                                                                                                                                                                   | 6                        |

|  |                                                                                                                                                       |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | Strategies of handling Question and Answer sessions after Presentations/seminars.<br><br>• Question Toss: Practice on asking and answering questions. |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------|--|

### **Suggested Reading:**

1. “Effective Technical Communication” by M. Ashraf Rizvi, McGraw Hill.
2. “Skill Sutras” by Jayashree Mohanraj, Prism Books Pvt. Ltd.
3. “Technical Communication: Principles and Practice” by Meenakshi Raman, OUP.
4. “Effective Communication” John Adair, Pan Macmillan Ltd.
5. “Body Language”, by Allan Pease, Sudha Publications.
6. “Business Communication: From Principles to Practice” MM Monippally, TataMcGraw Hill.

**Syllabus for B.Tech CSE I year II Semester**  
**ENGINEERING PHYSICS LAB**

| L | T | P | C   |
|---|---|---|-----|
| 0 | 0 | 3 | 1.5 |

**Code:** 9HC66

**Prerequisite:** Engineering Physics

**Course Objective:**

Provide practical experience in the field of physics.

**Course Outcomes:**

After completing the experiments, students are able to

1. Demonstrate the concepts of photo electric effect, total internal reflection, diffraction and dispersion of light. [L2]
2. Demonstrate the concepts of rigidity modulus, periodicity and oscillations. [L2]
3. Compare and contrast the Biot-Savart law with Oersted law and explain the concept of Magnetostriction. [L3,L4]
4. Analyze the electrical resonance, time constant, band gap and forward resistance of a semiconductor diode. [L4]

**List of Experiments**

**1. Photo voltaic cell:**

-Determination of Planck's constant by using photo voltaic cell.

**2. Dispersive power:**

-Calculation of dispersive power of a given material of prism by using Spectrometer.

**3. Diffraction Grating:**

-Determination of wavelength of a given laser source of light by using diffraction grating.

**4. Numerical Aperture:**

-Determination of a Numerical Aperture (NA) of an optical fiber.

**5. Stewart-Gee's Experiment:**

-Determination of magnetic induction flux density along the axis of a current carrying circular coil using Stewart and Gee's experiment.

**6. Sonometer:**

-Calculating the frequency of AC supply by using the Sonometer.

**7. LCR Circuit:**

-Study of series and parallel resonance of an LCR circuit.

**8. Torsional pendulum:**

-Determination of rigidity modulus of a given wire material using the Torsional pendulum.

**9. Energy Gap:**

-Determination of the energy gap ( $E_g$ ) of a given semiconductor.

**10. Light Emitting Diode:**

-Studying the LED characteristics and calculating the forward resistance of it.

**11. RC Circuit:**

-Determination of time constant of an RC-circuit.

**NOTE:** Any **TEN** of the above experiments are to be conducted.

**Syllabus for B.Tech CSE I year II Semester**  
**DATA STRUCTURES Lab Using C**  
**(Common to all Branches)**

| L | T | P/D | C   |
|---|---|-----|-----|
| 0 | 0 | 3   | 1.5 |

**Code: 9EC61**

**Prerequisites:** Problem Solving using C Lab

**Course objective:**

Develop the skills to choose appropriate data structures, and to apply them to real-world problems.

**Course Outcomes:**

After completion of the course, the student will be able to:

1. Develop programs to illustrate various linear and nonlinear data structures using C language [L3]
2. Develop programs to assess the performance of various searching and sorting techniques using C language [L3,L5]

Note: Lab Projects will be allocated to the students at the beginning of the semester.

**Cycle 1:**

1. Define a structure for complex number. Write functions on complex numbers (addition, subtraction, absolute value, multiplication, division, complex conjugate) and implement them in a menu driven style.
2. Define a structure student having members roll no., name, class, section, marks. Create an array of 10 students give the data and find the average marks, section-wise.

**Cycle 2:**

3. Write a C program that implement stack and its operations using arrays
4. Write a C program that implement Queue and its operations using arrays.
5. Write a C program that implement Circular Queue and its operations using arrays.

**Cycle 3:**

6. Write a C program that uses Stack operations to perform the following:
  - i) Converting infix expression into postfix expression
  - ii) Evaluating the postfix expression

**Cycle 4:**

7. Write a C program that uses functions to perform the following operations on Single linked list:
  - i) Creation ii) Insertion iii) Deletion iv) Traversal

**Cycle 5:**

8. Write a C program that uses functions to perform the following operations on Doubly linked list:
  - i) Creation ii) Insertion iii) Deletion iv) Traversal in both ways

**Cycle 6:**

9. Write a C program using functions to perform the following operations on Circular linked list:  
i) Creation ii) Insertion iii) Deletion iv) Traversal

**Cycle 7:**

10. Write a C program to implement operations on the following Data Structures Using Singly linked list:  
i) Stack ii) Queue

**Cycle 8:**

11. Write a C program that uses functions to perform the following:  
i) Creating a Binary Search Tree.  
ii) Traversing the above binary tree in pre-order, in-order and post-order.

**Cycle 9:**

12. Write C programs that use both recursive and non-recursive functions to perform the following searching operations for a Key value in a given list of integers:  
i) Linear Search ii) Binary Search

**Cycle 10:**

13. Write C programs that implement the following sorting methods to sort a given list of integers in ascending order:  
i) Bubble Sort ii) Insertion Sort iii) Selection Sort

**Cycle 11:**

14. Write C programs that implement the following sorting methods to sort a given list of integers in ascending order:  
i) Quick sort ii) Merge sort iii) Heap Sort

**Cycle 12:**

15. Lab Projects- Design and Develop Case Studies such as, Graph Traversal Techniques, Collision Resolution Techniques

**Syllabus for B.Tech CSE II year I Semester**  
**OBJECT ORIENTED PROGRAMMING THROUGH JAVA**  
**(Common to CSE, IT and ECM)**

| L | T | P/D | C |
|---|---|-----|---|
| 2 | 1 | 0   | 3 |

**Code: 9EC02**

**Prerequisite:** Problem Solving Using C

**Course Objective:**

Provide a versatile, platform-independent, and object oriented programming language for developing a wide range of applications.

**Course Outcomes:**

After completion of the course, the student will be able to:

1. Illustrate the concepts of Object-Oriented Programming in Java [L2] – U1, U2
2. Develop real time applications using I/O streams, packages and collections framework [L3] – U3
3. Utilize multithreading and exception handling concepts to build applications [L3] – U4
4. Apply AWT/SWING controls and applets to develop applications [L3] – U5
5. Build network-based applications [L3] - U6

**UNIT I**

History of Java, Java buzzwords, datatypes, variables, simple java program, scope and life time of variables, operators, expressions, control statements, type conversion and casting, arrays,, classes and objects – concepts of classes, objects, constructors, methods, access control, this keyword, garbage collection, overloading methods and constructors, recursion, string handling, StringTokenizer.

Applications: Basic operations on the bank account of a customer.

**UNIT II**

Inheritance –Definition, single inheritance, benefits of inheritance, Member access rules, super class, polymorphism- method overriding, Dynamic method dispatch, using final with inheritance, abstract classes, Base class object.

Interfaces: definition, variables and methods in interfaces, differences between classes and interfaces, usage of implements and extends keyword, interfaces, uses of interfaces, packages Applications: Extending the banking operations to the loan applicants.

**UNIT III**

Packages: Definition, types of packages, Creating and importing a user defined package. Introduction to I/O programming: DataInputStream, DataOutputStream, FileInputStream, FileOutputStream, BufferedReader.

Collections: interfaces, Implementation classes, and Algorithms (such as sorting and searching).

Applications: Searching for a string in the text. PNR status check, students' result sorting.

**UNIT IV**

Exception handling -exception definition, benefits of exception handling, exception hierarchy, usage of try, catch, throw, throws and finally, built in exceptions, creating own



exception sub classes.

Multi-Threading: Thread definition, types of multitasking, uses of multitasking, thread life cycle, creating threads using Thread class and Runnable interface, synchronizing threads, daemon thread.

Applications: Illegal entry handling in the registration form. (Example: entering incorrect intermediate hall-ticket number in EAMCET Registration form)

## **UNIT V**

Advantages of GUI over CUI ,The AWT class hierarchy, Introduction to Swings, Swings Elements:- JComponent, JFrame, user interface components- JLabels, JButton, JScrollbar, text components, check box, check box groups, choices, lists panels – scrollpane, menubar, graphics, layout, managers –boarder, grid, flow, card and grid bag.

Event handling: Delegation event model, closing a Frame, mouse and keyboard events, Adapter classes.

Applications: developing calculator, developing feedback form, developing bio data.

## **UNIT VI**

Applets – Concepts of Applets, differences between applets and applications, life cycle of an applet, types of applets, creating applets, passing parameters to applets.

Applications: Developing of simple advertisements.

Networking – Basics of network programming, addresses, ports, sockets, simple client server program, multiple clients, sending file from server to client.

Applications: One to one Chat application

## **TEXT BOOKS:**

1. Java; the complete reference, 6th edition, Herbert Schildt,TMH.
2. Introduction to Java programming 6th edition, Y. Daniel Liang, Pearsoneducation.

## **REFERENCES:**

1. Core Java 2, Vol 1, Fundamentals, Cay. S. Horstmann and Gary Cornell, seventh Edition, Pearson Education.
2. Core Java 2, Vol 2, Advanced Features, Cay. S. Horstmann and Gary Cornell, Seventh Edition, Pearson Education

**Syllabus for B.Tech CSE II year I Semester**  
**Digital Electronics**

**Code: 9CC51**

**Prerequisite: Nil**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course objective:**

Learn the concepts of various number systems, design of Combinational and Sequential Circuits using Logic gates and PLDs.

**Course outcomes:**

After completing of the course, the students will be able to:

1. Explain number systems, and apply Boolean theorem, postulates and K-maps to minimize switching expressions.[L2, L3-U1,U2]
2. Design the various combinational and sequential logic circuits using logic gates and flip flops.[L6-U3, U4]
3. Analyze and Design the sequential machines such as counters, registers etc.[L4, L6-U4, U5]
4. Develop the combinational logic circuits using PLD's and interpret the memory architectures. [L2, L3- U6]

**UNIT – I**

**Fundamentals of Digital Systems and logic families**

Digital signals, digital circuits, AND, OR, NOT, NAND, NOR and Exclusive-OR operations, signed binary, octal, hexadecimal number, number conversion. Characteristics of digital ICs, Error detecting and correcting codes (7 bit Hamming Code).

**UNIT – II**

**Boolean algebra**

Boolean algebra, Postulates and theorems, Standard representation for logic functions, K-map representation, simplification of logic functions using K-map, Boolean function minimization using Quine-Mcclusky method.

**UNIT - III**

**Design of combinational circuits** Design of Multiplexer, De-Multiplexer/Decoders, Adders, Subtractors, Parallel adder/Subtractor, BCD arithmetic, digital comparator, parity checker /generator, code converters.

**UNIT - IV**

**Sequential circuits-I**

A 1-bit memory, the circuit properties of Bistable latch, the clocked SR flip flop, J- K, T and D-types flipflops, triggering mechanism of flip-flops, flip-flop conversion.

**UNIT - V**

**Sequential circuits-II**

Applications of flip-flops: Ripple (Asynchronous) counters, synchronous counters, counters design using flip flops, asynchronous sequential counters, shift registers, applications of shift registers, serial to parallel converter, parallel to serial converter, ring counter.

## **UNIT - VI**

### **Semiconductor memories and Programmable logic devices**

Memory organization and operation, Memory hierarchy and expanding memory size, classification and characteristics of memories, sequential memory, read only memory (ROM), read and write memory(RAM), Programmable logic devices: PROM, PLD and PAL.

#### **Text Books:**

1. R. P. Jain, "Modern Digital Electronics", McGraw Hill Education, 2009.
2. M. M. Mano, "Digital logic and Computer design", Pearson Education India, 2016.
3. A. Kumar, "Fundamentals of Digital Circuits", Prentice Hall India, 2016.

#### **References:**

1. ZviKohavi and Niraj K Jha -Switching & Finite Automata theory – Cambridge, 3rd Edition.
2. SubrataGhoshal, Digital Ele ctronics,2012, Cengage Learning Fletcher -An Engineering Approach to Digital Design – PHI.

**Syllabus for B.Tech CSE II year I Semester  
Quantitative Aptitude and Logical Reasoning**

**Code: 9HC16**

**Pre Requisites:** Nil

| L | T | P /D | C |
|---|---|------|---|
| 3 | 0 | 0    | 3 |

**Course objective:**

Solve general problems in his everyday life within in short time and improve the certain skills such as numerical and logical ability, mental capacity.

**Course outcomes:**

After completion of the course, the student will be able to:

1. Solve logical thinking and analytical related problems and thereby develop their employability skills to face the real world scenario. [L3]

**Unit I:**

Number System: Test for Divisibility, Test of prime number, Division and Remainders – HCF and LCM of Numbers–Fractions and Decimals -Average-Problems on Ages- Problems on Numbers- Ratio and Proportion.

**Unit II:**

Percentage – Profit, Loss and Discount – Partnership and Share-Simple Interest - Compound Interest. Time and Work- Pipes and Cisterns-Time and Distance- Problems on Trains- Boats and Streams.

**Unit III:**

Allegation or Mixtures, Clocks & Calendar, Mensuration: Area of Plane Figures, Volume and Surface Area of Solid Figures. Data Interpretation: Tabulation, Bar Graphs, Pie Charts, Line Graphs.

**Unit–IV:**

Series Completion: Number Series, Alphabet Series, Alpha – Numeric Series.

Analogy: Completing the Analogous Pair, Simple Analogy, Choosing the Analogous pair, Double Analogy, Word Analogy, and Number Analogy.

Classification: Word Classification, Number Classification and Letter Classification.

Coding & Decoding: Letter Coding, Number Coding, Matrix Coding, Substitution, Deciphering Message Word Codes, Jumbled Coding. Crypt arithmetic-Inequalities-Input Output Tracing

**Unit–V:**

Blood Relations– Direction sense test- Number, Ranking & Time Sequence Test – Mathematical Operations-Arithmetical Reasoning. Puzzle Test: Classification Type Questions, Seating Arrangements, Comparison Type Questions, Sequential Order of Things, Selection Based on Given Conditions, Family Based Puzzles, Jumbled Problems.

**Unit –VI:**

Logical Venn Diagrams –Cubes and Dice – Analytical Reasoning-Assertions and Reason– Logical Deductions-Syllogism -Statement and Arguments-Statement and Conclusions- -Data Sufficiency.

**Text Books:**

1. Quantitative Aptitude by R.S.Agarwal
2. Verbal and Non Verbal Reasoning by R.S.Agarwal.

**Syllabus for B.Tech CSE II year I Semester**  
**Discrete Mathematics**

| L | T | P /D | C |
|---|---|------|---|
| 2 | 1 | 0    | 3 |

**Code: 9F303**

**Prerequisites:** Mathematics- I and II

**Course Objective:**

Learn and apply mathematical facts in solving real-world problems.

**Course Outcomes:**

After completion of the course, the student will be able to:

1. Interpret mathematical arguments and identify fallacious reasoning, predicates and effectively use automatic theorem proving [L2]
2. Demonstrate operations on sets, functions, relations. [L2]
3. Solve problems that involve: computing permutations and combinations, Binomial and multinomial theorems [L3]
4. Analyze and deduce problems involving recurrence relations and generating functions [L4]
5. Apply graph theory Spanning trees, planar graphs, Isomorphism and connectivity [L3]

**UNIT – I**

**Propositional Logic:** Statement and notations, Connectives, Well-formed Formulas, Truth Tables, Tautology, Equivalence, Implication, Rules of inference, Arguments, Proof by contradiction, Conditional Proof Normal forms, and Automatic theorem proving.

**UNIT-II**

**First order logic:** Predicates, Quantifiers, Free and Bound variables, Rules of inference, Consistency, Automatic Theorem Proving.

**UNIT – III**

**Relations:** Properties of Binary Relations, Equivalence, transitive closure, Compatibility & Partial Ordering Relations, Hasse Diagrams, Lattice and its properties.

**Algebraic structures:** Algebraic systems, Examples and general properties, Semi groups and Monoids. Groups, Subgroups, Ring, Homomorphism, Isomorphism.

**UNIT –IV**

**Elementary Combinatorics:**

Basics of counting, Combinations & Permutations with and without repetitions, Constrained repetitions. Binomial coefficients, Binomial and Multinomial theorems, Euler function, Derangements, Principle of inclusion and exclusion, Pigeon hole principle and its applications.

**UNIT V**

**Recurrence relations:** Generating functions. Function of sequences. Recurrence relations, Solving recurrence relations by substitution and generating functions. Characteristic roots. Solution of Inhomogeneous recurrence relations.

## **Unit VI**

**Graph Theory:** Basic concepts, Representation of Graph, DFS, BFS, Spanning trees, Planar graphs, coloring, Isomorphism and subgraphs, Multi graphs and connected graphs, Euler Circuits, Hamiltonian graphs, Chromatic numbers,.

### **TEXT BOOKS:**

1. Discrete Mathematics for Computer Scientists & Mathematicians, J.L. Mott, A. Kandel, T.P. Baker, PHI.
2. Discrete mathematics with applications to computer science, J.P.Tremblay and R.Manohar, TMH
3. Elements of Discrete mathematics – A computer Oriented Approach- C L Liu, D P Mohapatra. Third Edition, Tata MacGraw Hill.

### **REFERENCES:**

1. Discrete and Combinational Mathematics- An Applied Introduction-5<sup>th</sup> Edition – Ralph. P.Grimaldi.Pearson Education.
2. Discrete Mathematics and its Applications, Kenneth H. Rosen, Fifth Edition.TMH.
3. Discrete Mathematical structures Theory and application-Malik & Sen, Cengage.

**Syllabus for B.Tech CSE II year I Semester**  
**Universal Human Values**  
**(Common to All Branches)**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Code: 9HC03**

**Prerequisite: Nil**

**Human Values Courses:** This course also discusses their role in their family. It, very briefly, touches issues related to their role in the society and the nature, which needs to be discussed at length in one more semester for which the foundation course named as “H-102 Universal Human Values 2: Understanding Harmony” is designed which may be covered in their III or IV semester. During the Induction Program, students would get an initial exposure to human values through Universal Human Values – I. This exposure is to be augmented by this compulsory full semester foundation course.

**Course Objective:**

Strengthening the values and morals to lead an ethical life.

**Course Outcomes:**

After completion of the course, the student will be able to:

1. Interpret an understanding of holistic perspective using self-exploration, natural acceptance and basic human aspirations.[L2]
2. Observe and practice harmony about self, family, society, nature and existence.[L1,L6]
3. Operate themselves on the above holistic understanding and professional ethics.[L3]

**Unit 1:**

**Course Introduction** - Need, Basic Guidelines, Content and Process for Value Education

1. Purpose and motivation for the course, recapitulation from Universal Human Values-I
2. Self-Exploration-what is it? - Its content and process; ‘Natural Acceptance’ and Experiential Validation- as the process for self-exploration
3. Continuous Happiness and Prosperity- A look at basic Human Aspirations
4. Right understanding, Relationship and Physical Facility- the basic requirements for fulfillment of aspirations of every human being with their correct priority
5. Understanding Happiness and Prosperity correctly- A critical appraisal of the current scenario
6. Method to fulfill the above human aspirations: understanding and living in harmony at various levels.

Include practice sessions to discuss natural acceptance in human being as the innate acceptance for living with responsibility (living in relationship, harmony and co-existence) rather than as arbitrariness in choice based on liking-disliking.

**Unit 2:**

**Understanding Harmony in the Human Being** - Harmony in Myself!

7. Understanding human being as a co-existence of the sentient ‘I’ and the material ‘Body’
8. Understanding the needs of Self (‘I’) and ‘Body’ - happiness and physical facility
9. Understanding the Body as an instrument of ‘I’ (I being the doer, seer and enjoyer)
10. Understanding the characteristics and activities of ‘I’ and harmony in ‘I’
11. Understanding the harmony of I with the Body: Sanyam and Health; correct appraisal of Physical needs, meaning of Prosperity in detail

12. Programs to ensure Sanyam and Health. Include practice sessions to discuss the role others have played in making material goods available to me. Identifying from one's own life. Differentiate between prosperity and accumulation. Discuss program for ensuring health vs dealing with disease.

### **Unit 3:**

**Understanding Harmony in the Family and Society-** Harmony in Human-Human Relationship

13. Understanding values in human-human relationship; meaning of Justice (nine universal values in relationships) and program for its fulfillment to ensure mutual happiness; Trust and Respect as the foundational values of relationship

14. Understanding the meaning of Trust; Difference between intention and competence

15. Understanding the meaning of Respect, Difference between respect and differentiation; the other salient values in relationship

16. Understanding the harmony in the society (society being an extension of family): Resolution, Prosperity, fearlessness (trust) and co-existence as comprehensive Human Goals

17. Visualizing a universal harmonious order in society- Undivided Society, Universal Order- from family to world family.

Include practice sessions to reflect on relationships in family, hostel and institute as extended family, real life examples, teacher-student relationship, goal of education etc. Gratitude as a universal value in relationships. Discuss with scenarios. Elicit examples from students' lives.

### **Unit 4:**

**Understanding Harmony in the Nature and Existence -** Whole existence as Coexistence

18. Understanding the harmony in the Nature

19. Interconnectedness and mutual fulfillment among the four orders of nature- recyclability and self-regulation in nature

20. Understanding Existence as Co-existence of mutually interacting units in all-pervasive space

21. Holistic perception of harmony at all levels of existence.

Include practice sessions to discuss human being as cause of imbalance in nature (film "Home" can be used), pollution, depletion of resources and role of technology etc.

### **Unit 5:**

**Implications of the above Holistic Understanding**

22. Natural acceptance of human values

23. Definitiveness of Ethical Human Conduct

24. Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order

### **Unit 6:**

**Harmony on Professional Ethics**

25. Competence in professional ethics:

a. Ability to utilize the professional competence for augmenting universal human order

b. Ability to identify the scope and characteristics of people-friendly and eco-friendly production systems,

c. Ability to identify and develop appropriate technologies and management patterns for above production systems.

26. Case studies of typical holistic technologies, management models and production systems

27. Strategy for transition from the present state to Universal Human Order:

a. At the level of individual: as socially and ecologically responsible engineers, technologists and managers



b. At the level of society: as mutually enriching institutions and organizations

28. Sum up.

Include practice Exercises and Case Studies will be taken up in Practice (tutorial) Sessions

eg. to discuss the conduct as an engineer or scientist etc.

### **Text Book**

1. Human Values and Professional Ethics by R R Gaur, R Sangal, G P Bagaria, Excel Books, New Delhi, 2010 3.

### **Reference Books**

1. Jeevan Vidya: Ek Parichaya, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.

2. Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.

3. The Story of Stuff (Book).

4. The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi

5. Small is Beautiful - E. F Schumacher.

6. Slow is Beautiful - Cecile Andrews

7. Economy of Permanence - J C Kumarappa

8. Bharat Mein Angreji Raj – Pandit Sunderlal

9. Rediscovering India - by Dharampal

10. Hind Swaraj or Indian Home Rule - by Mohandas K. Gandhi

11. India Wins Freedom - Maulana Abdul Kalam Azad

12. Vivekananda - Romain Rolland (English)

13 .Gandhi - Romain Rolland (English)

**Syllabus for B.Tech CSE II year I Semester**  
**OBJECT ORIENTED PROGRAMMING THROUGH JAVA LAB**  
**(Common to CSE, IT and ECM)**

| L | T | P/D | C |
|---|---|-----|---|
| 0 | 0 | 4   | 2 |

**Code:** 9EC62

**Prerequisite:** Problem Solving Using C

**Course Objective:**

Understand, design and execute the programs involving concepts of Java and object-oriented programming principles.

**Course Outcomes:**

After completion of the course, the student will be able to:

1. Develop programs by applying the concepts of object oriented programming to solve mathematical and engineering problems [L2, L3]
2. Build robust applications using multithreading, exception handling and collections framework and GUI network programming [L2, L3]

**List of Programs:**

1. A) Write a program to print prime numbers up to a given number.  
B) Write a program to print roots of a quadratic equation  $ax^2+bx+c=0$ .  
C) Write a program to print Fibonacci sequence up to a given number.
2. A) Define a class to represent a bank account and include the following members  
Instance variables:  
(i) Name of depositor (ii) Account No (iii) Type of account  
(iv) Balance amount in the account

Instance Methods:

To assign instance variables (Constructors-Zero argument and parameterized)

1. To deposit an amount
  2. To withdraw amount after checking the balance  
To display name and address  
Define Execute Account classes in which define main method to test above class.
- B) In the above account class, maintain the total no. of account holders present in the bank and also define a method to display it. Change the main method appropriately.
- C) In main method of Execute Account class, define an array to handle five accounts.
- D) In Account class constructor, demonstrate the use of “this” keyword.
- E) Modify the constructor to read data from keyboard.
- F) Overload the method deposit() method (one with argument and another without argument)
- G) In Account class, define set and get methods for each instance variable.

**Example:**

For account no variable, define the methods get Account No() and set Account No (int accno) In each and every method of Account class, reading data from and writing data to instance variables should be done through these variables.

3. A) Define Resister class in which we define the following members: Instance

variables: resistance Instance Methods: give Data():To assign data to the resistance variable display Data(): To display data in the resistance variable constructors

Define subclasses for the Resistor class called Series Circuit and Parallel Circuit in which define methods: calculate Series Resistance ( ) and calculate Parallel Resistance ( ) respectively. Both the methods should take two Resistor objects as arguments and return Resistor object as result. In main method, define another class called Resistor Execute to test the above class.

B) Modify the above two methods which should accept array of Resistor objects as argument and return Resistor object as result.

4. A) Write a program to demonstrate method overriding.  
B) Write a program to demonstrate the uses of “super” keyword (three uses)  
C) Write a program to demonstrate dynamic method dispatch (i.e. Dynamic polymorphism).

5. A) Write a program to check whether the given string is palindrome or not.  
B) Write a program for sorting a given list of names in ascending order.  
C) Write a program to count the no. of words in a given text.

6. A) Define an interface “Geometric Shape” with methods area( ) and perimeter( ) (Both method’s return type and parameter list should be void and empty respectively).

Define classes like Triangle, Rectangle and Circle implementing the “Geometric Shape” interface and also define “Execute Main” class in which include main method to test the above class

- B) Define a package with name “sortapp” in which declare an interface “SortInterface” with method sort( ) whose return type and parameter list should be void and empty. Define “subsortapp” as subpackage of “sortapp” package in which define class “SortImpl” implementing “SortInterface” in which sort() method should print a message linear sort is used. Define a package “searchingapp” in which declare an interface “SearchInterface” with search( ) method whose return type and parameter list should be void and empty respectively. Define “searchingimpl” package in which define a “Search Impl” class implementing “Search Interface” defined in “searching app” package in which define a search( ) method which should print a message linear search issued.

Define a class Execute Package with main method using the above packages (classes and its methods).

Use Array List class of Collections Framework to and use algorithms to search and sort the element of an array.

- 7) Modify the withdraw() method of Account class such that this method should throw “Insufficient Fund Exception” if the account holder tries to withdraw an amount that leads to condition where current balance becomes less than minimum balance otherwise allow the account holder to withdraw and update the balance accordingly.

- 8.A) Define two threads such that one thread should print even numbers and another thread should print odd numbers.
- B) Define more than one thread to print tables using synchronization concept.
- C) Define two threads such that one thread should read a line of text from text file and another thread should write that line of text to another file. (Thread communication example).

- 9) Design the user screen as follows and handle the events appropriately.

|               |                                              |
|---------------|----------------------------------------------|
| First Number  | <input type="text"/>                         |
| Second Number | <input type="text"/>                         |
| Result        | <input type="text"/>                         |
|               | <div><div>ADD</div><div>SUBTRACT</div></div> |

- 10) Write a Java program for handling mouse events and key events.
- 11) a) Write a program for handling window events.  
b) Develop an applet that displays a simple message.
- 12) Develop a client that sends data to the server and also develop a server that sends data to the client (two-way communication)
- 13, 14) Develop Lab projects (such as banking application, simple gaming application, scientific calculator, Client-Server Communication in incorporating file handling mechanisms, etc.)

**Syllabus for B.Tech CSE II year I Semester**  
**Digital Electronics Lab**

**Code: 9CC82**

**Prerequisite: Nil**

| L | T | P/D | C   |
|---|---|-----|-----|
| 0 | 0 | 3   | 1.5 |

**Course Objective:**

Learn the various numbering systems, Boolean functions and design of Combinational and Sequential Circuits.

**Course Outcomes:**

After completing this course, the students will be able to:

1. Design combinational circuits using logic gates[L6]
2. Demonstrate the applications of sequential circuits. [L2]

**Lab experiments:**

Verify the operations of the Digital ICs (Hardware) in the Laboratory

1. Realization of A-O-I Gates using Universal gates
2. Verification of Demorgans Theorem
3. Implementation of 4-Bit binary to Gray code converter
4. Verification of 4-bit Binary Adder using IC 74x283
5. Implementation of 4-bit parity generator
6. Realization of 8x1 Multiplexer using IC 74x151
7. Verification of 3x8 Decoder using IC 74x138
8. Verification of Priority encoder using 74x148
9. Verification of D Flip-Flop IC 74x74
10. Conversion of JK-Flipflop to D-Flipflop
11. Verification of Decade counter using IC74x90
12. Verification of Universal Shift Register

**Syllabus for B.Tech CSE II year I Semester  
Basic Electrical and Electronics Engineering Lab**

**Code: 9AC95**

**Prerequisite: NIL**

| L | T | P/D | C |
|---|---|-----|---|
| 0 | 0 | 4   | 2 |

**Course Objective:**

Teach students how to analyse and construct electrical and electronic circuits.

**Course Outcomes:**

After completing this course, the students will be able to:

1. Examine the performance of DC machines and AC machines.[L4]
2. Illustrate the principles circuit theorems.[L2]
3. Examine the principle of operation of diode and its applications.[L4]
4. Analyze the principle of operation of transistor.[L4]
5. Develop the principles of verification of logic gates.[L3]

**Electrical Experiments**

1. Brake test on 3-phase induction motor (performance characteristics).
2. Speed control of DC shunt motor by
  - a) Armature Voltage Control .
  - b) Field flux control method.
3. Brake test on DC shunt motor.
4. Swinburne's test on DC shunt machine.
5. OCC characteristics of DC shunt generator.
6. Verification of superposition and Reciprocity Theorems.

**Electronics Experiments**

1. PN Junction diode characteristics A. Forward bias B. Reverse bias.
2. Zener diode characteristics
3. Half wave Rectifier with and without filters.
4. Full wave Rectifier (Centre tapped and Bridge)with and without filters
5. Transistor CE characteristics (Input and Output)
6. Verification of Logic gates

## Syllabus for B.Tech CSE II year II Semester

### Computer Organization

**Code:** 9CC54

**Prerequisite:** Digital Electronics

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

#### Course Objective:

Learn about basic structure of computer, different data representations and Instruction sets; 8086 architecture, addressing modes and instruction set also write efficient programs to interface devices with 8086 processor.

#### Course Outcomes:

After completing this course, student should be able to

1. Interpret the number system and the structure of computer along with the instruction set and Register transfer language. [L2]
2. Distinguish micro-programmed and hard-wired control units. [L4]
3. Explain 8086 microprocessor architecture and develop programming. [L2]
4. Develop the various interfacing circuits with 8086 microprocessor. [L3]

#### UNIT-I

**Basic Structure of Computer:** Computer Types, Functional units, Basic operational concepts, Bus structures, Software, Performance, multiprocessors and multi computers.

**Data Representation:** Fixed Point and Floating – Point Representation

#### UNIT-II

**Register Transfer Language and Micro-operations:** Register Transfer language. Arithmetic Micro-operations, logic micro-operations, shift micro operations, Arithmetic logic shift unit. Instruction codes, Computer instructions – Instruction cycle. Memory – Reference instructions. Input – Output and Interrupt; STACK organization; Instruction formats.

#### UNIT-III

**Control Unit Design:** Control memory, Address sequencing, micro-program example, design of control unit-Hard wired control, Micro-programmed control.

**Computer Arithmetic Operations:** Addition and subtraction, multiplication Algorithms, Division Algorithms, Fixed point Arithmetic operations.

#### UNIT-IV

Architecture of 8086 Microprocessor. Special functions of General purpose registers. 8086 flag register and function of 8086 Flags, Addressing modes of 8086, Instruction set of 8086.

#### UNIT-V

Assembler directives, simple programs, procedures, and macros. Assembly language programs involving logical, Branch & Call instructions, sorting, evaluation of arithmetic expressions, string manipulation.

#### UNIT-VI

Pin diagram of 8086-Minimum mode and maximum mode of operation.Timing diagram. Memory interfacing to 8086 (Static RAM&EPROM).8255 PPI-Various modes of operation and interfacing to 8086. Stepper motor Interface to 8086. Interrupt structure of 8086. Vector interrupt table. Interrupt service routines.

**TEXT BOOKS:**

1. Computer Systems Architecture – M.Moris Mano, IIIrd Edition, Pearson
2. Microprocessors and interfacing – Douglas V.Hall, TMH, 2<sup>nd</sup> Edition, 1999.

**REFERENCES:**

1. Computer Organization and Architecture – William Stallings Sixth Edition, Pearson
2. Microcomputer systems, The 8086/8088 Family Architecture, Programming and Design – Y.Liu and G.A. Gibson, PHI, 2<sup>nd</sup> Edition.
3. Advanced microprocessor and Peripherals – A.K.Ray and K.M.Bhurchandi, TMH, 2000.



**Syllabus for B.Tech CSE II year II Semester**  
**DATABASE MANAGEMENT SYSTEMS**  
(Common to CSE, IT & ECM)

**Code: 9FC04**

**Prerequisite: NIL**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Understand the fundamental concepts of database management systems.

**Course Outcomes:**

After completion of this course student will be able to:

1. Illustrate the basic concepts of DBMS, ER model and ER diagrams. [L3]
2. Implementing queries using SQL and Applying the Relational algebra and Calculus to define expressions for queries in Databases. [L3]
3. Apply normalization techniques to improve the database design. [L3]
4. Utilise principles of transaction management to ensure data consistency and Concurrency Control in multi-user database environments. [L3]
5. Analyze External Storage Organization mechanisms and apply Indexing in databases for query optimization to enhance system performance. [L4]

**UNIT I**

**Data Base Systems:** Data Vs Information, Data base System Applications, data base System VS file System – View of Data – Data Abstraction –Instances and Schemas – data Models – the ER Model – Relational Model – Other Models – Database Languages – DDL – DML – database Access for applications Programs – data base Users and Administrator – Transaction Management – data base System Structure – Storage Manager – the Query Processor.

Data base design and ER diagrams – Entities, Attributes and Entity sets – Relationships and Relationship sets – Additional features of ER Model – Concept Design with the ER Model – Data Modeling checklist.

Application- ER diagram for a college

**UNIT II**

Introduction to the Relational Model – Integrity Constraint Over relations – Enforcing Integrity constraints – Querying relational data – Logical data base Design – Introduction to Views – Destroying /altering Tables and Views.

Relational Algebra – Selection and projection set operations – renaming – Joins – Division – Examples of Algebra overviews – Relational calculus – Tuple relational Calculus – Domain relational calculus.

Application - Student database design.

**UNIT III**

Form of Basic SQL Query – Examples of Basic SQL Queries – Introduction to Nested Queries – Correlated Nested Queries Set – Comparison Operators – Aggregative Operators – NULL values – Comparison using Null values – Logical connectivity's – AND, OR and NOT – Impact on SQL Constructs – Outer Joins – Disallowing NULL values – Complex Integrity Constraints in SQL Triggers, Embedded SQL.

Application - working with Aviation company database.

#### **UNIT IV**

Schema refinement – Problems Caused by redundancy – Decompositions – Problem related to decomposition – reasoning about FDS – FIRST, SECOND, THIRD Normal forms – BCNF – Lossless join Decomposition – Dependency preserving Decomposition – Multi valued Dependencies – FORTH NormalForm.

Application - Faculty Evaluation Report.

#### **UNIT V**

Transaction Concept- Transaction State- Implementation of Atomicity and Durability – Concurrent – Executions – Serializability- Recoverability – Implementation of Isolation – Testing for serializability- Lock –Based Protocols – Timestamp Based Protocols- Validation- Based Protocols – Multiple Granularity, Recovery and Atomicity – Log – Based Recovery – Recovery with Concurrent Transactions – Buffer Management – Failure with loss of nonvolatile storage.

Application - Production Management System.

#### **UNIT VI**

Data on External Storage – File Organization and Indexing – Cluster Indexes, Primary and Secondary Indexes – Index data Structures – Hash Based Indexing – Tree base Indexing – Comparison of File Organizations – Indexed Sequential Access Methods (ISAM) – B+ Trees: A Dynamic Index Structure.

Application – Creating B+ tree on Instructor File.

#### **TEXT BOOKS:**

1. Data base System Concepts, Silberschatz, Korth, McGraw hill, Vedition.
2. Data base Management Systems, Raghurama Krishnan, Johannes Gehrke, TATAMcGrawHill, 3rdEdition
3. Database Management Systems, Peter Rob, A.AnandaRao, Carlos Coronel ,CENGAGE Learning

#### **REFERENCES:**

1. Data base Systems design, Implementation, and Management, Peter Rob and Carlos Coronel 7thEdition.
2. Fundamentals of Database Systems, ElmasriNavratePearsonEducation
3. Introduction to Database Systems, C.J.DatePearsonEducation

**Syllabus for B.Tech CSE II year II Semester**  
**Python Programming**

**Code: 9FC02**

**Prerequisite:** Problem Solving using C, Object-Oriented Programming

| L | T | P/D | C |
|---|---|-----|---|
| 2 | 1 | 0   | 3 |

**Course Objective:**

Develop structured as well as object-oriented programming skills using Python.

**Course Outcomes:**

After completion of this course student will be able to:

1. Outline the basic concepts of the python, illustrate syntax and semantics of python. [L1&L2]
2. Infer components of function, demonstrate functionalities of string, List, Tuples and Dictionary to build application.[L2&L3]
3. Make use of files, built - in modules and packages by exploring its functions in building specific application. [L3]
4. Examine Python-OOP concepts and identify Motives or causes of different exceptions and handling them. [L3]
5. Develop the skills for data processing and GUI designing in Python. [L6]

**Unit -I:**

**Introduction to Python:** History, Features, Modes of Execution, Setting up path, working with Python Basic Syntax, Variable and Data Types, Operators. Conditional Statements (If, If- else, Nested if-else) Looping (for, While Nested loops) Control Statements (Break, Continue, Pass).

**Input-Output:** Printing on screen, Reading data from keyboard, Opening and closing file

**Unit-II:**

**Functions:** Defining a function, calling a function, Types of functions, Function Arguments, Anonymous functions, Global and local variables

**String Manipulation:** Accessing Strings, Basic Operations, String slices, Function and Methods

**Lists:** Accessing list, Operations, Working with lists Function and Methods

**Tuple:** Accessing tuples, Operations, Working.

**Dictionaries:** Accessing values in dictionaries, working with dictionaries, Properties Functions and Methods.

**Unit-III:**

**Modules:** Importing module, Math module, Random module, Packages

**Exception Handling:** Exception, Exception Handling, Except clause, Try? Finally clause User Defined Exceptions

**Unit-IV:**

**Python- OOPs concept:** Class and object, Attributes, Inheritance, Overloading Overriding, Data hiding.

**Regular expressions:** Match function, Search function, Matching VS Searching, Modifiers Patterns.

**Unit -V:**

Introduction to Files, File Handling, Working with File Structure, Directories, Handling Directories

**Unit -VI:**

Case Study with NumPy/PlotPy/SciPy/GUI Programming, Introduction, Tkinter programming, Tkinter widgets

**TEXT BOOK:**

1. [Apress]-Beginning Python. From Novice to Professional, 2nd ed. - [Hetland] (2008)

**Reference books:**

1. Introduction to Computation and Programming using Python, Revised and Expanded Edition, John V. Guttag, The MIT Press.
2. Programming Python, Fourth Edition by Mark Lutz, O'Reilly
3. Python Programming using problem solving approach, ReemaThareja, Oxford Higher Education.

**Syllabus for B.Tech CSE II year II Semester**  
**Probability and Statistics**

**Code: 9HC15**

**Pre Requisites:** Mathematics Knowledge at Pre-University Level

| L | T | P/D | C |
|---|---|-----|---|
| 2 | 1 | 0   | 3 |

**Course Objective:**

Introduce the fundamental concepts of various probability distributions, including discrete and continuous distributions; provide an understanding of sampling distributions and their importance in statistical inference. Teach inferential statistics, including estimation, hypothesis testing, and decision-making based on data. Also explore the concepts of correlation and regression to analyze relationships between variables and predict outcomes.

**Course Outcomes:**

After the course completion the students will able to:

1. Make Use of the probability distributions to obtain mean and Standard deviation of a random variable. [L3].
2. Compute the Sampling distribution of a statistic and estimate the population parameter. [L2, L3]
3. Test the hypothesis based on the large and small samples. [L4]
4. Calculate Correlation coefficient using Karl Pearson's formula and Spearman's rank correlation. [L3]
5. Construct a curve by least squares method and find the regression lines. [L3]

**UNIT-I:**

**Random Variables and Probability Distributions:** Random variables – Discrete and Continuous, probability mass and density functions, expectation and variance. Binomial, Poisson and Normal Distributions.

**UNIT-II:**

**Sampling Distributions and Estimation:** Sampling distribution of the mean ( $\sigma$  - known and unknown), sums and differences, Central limit theorem. Estimation: Point estimation and Interval estimation concerning means for large samples.

**UNIT-III:**

**Tests of Hypothesis for Large Samples:** Tests of Hypothesis, Type-I and Type-II Errors, Hypothesis testing concerning one mean and two means and test of hypothesis concerning to one Proportion and difference of proportions.

**UNIT-IV:**

**Tests of Hypothesis for Small Samples:** Student t-test, Hypothesis testing concerning one mean and two Means, F-test and  $\chi^2$  test-Goodness of fit, Independence of Attributes.

**UNIT-V:**

**Correlation:** Types of correlation, coefficient of correlation, Properties. Methods of finding the coefficient of correlation, Scatter diagram, direct method - Karl Pearson's formula Spearman's rank correlation,

**UNIT-VI:**

**Curve fitting and Regression:** Curve fitting by the method of least squares- fitting of straight lines, second degree parabolas and more general curves. Types of Regression- linear regression, multiple regressions.

**Suggested Readings:**

1. Ramana B.V., Higher Engineering Mathematics, Tata McGraw Hill New Delhi, 11<sup>th</sup> Reprint, 2010.
2. Miller and Freund's, Probability and Statistics for Engineers, 8th Edition, Pearson Educations
3. S C Gupta and V K Kapoor, Fundamentals of Mathematical statistics, Khanna publications.

**Syllabus for B.Tech CSE II year II Semester  
Business Economics and Financial Analysis**

**Code: 9ZC01**

**Prerequisite: Nil**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Learn the concepts of managerial economics and the market dynamics namely demand elasticity of demand and pricing in different market structures.

**Course Outcomes:**

After the course completion the students will able to:

1. Explain the Basics of Economics and its relation to Business Economics [L2]
2. Analyze the production function in terms of cost and revenue [L4]
3. Examine the basic market structures and their relevance to business [L4]
4. Outline the fundamentals of financial accounting and prepare financial statements [L2]
5. Apply financial ratios to analyze the financial performance of the company [L3]

**UNIT I**

**INTRODUCTION TO BUSINESS ECONOMICS:**

Definition, Nature and Scope of Business Economics, Micro and Macro Economics concepts- National Income, Gross domestic product (GDP), Per capita income, Demand Analysis: Demand Determinants, Law of Demand and its exceptions, Elasticity of Demand, Types of Elasticity of Demand and Demand Forecasting–Statistical and Non-Statistical techniques.

**UNIT II**

**THEORY OF PRODUCTION AND COST ANALYSIS:**

Production Function – Isoquants and Isocosts, Internal and External Economies of Scale, Law of Returns Cost Analysis: Cost concepts, different types of costs, Break-even Analysis (BEA)-Determination of Break-Even Point (simple problems).

**UNIT III**

**INTRODUCTION TO MARKETS**

Market structures: Types of competition, Features of Perfect competition, Monopoly and Monopolistic Competition, Pricing Methods and strategies.

**UNIT IV**

**FINANCIAL ACCOUNTING-I:**

Accounting concepts and Conventions, Double-Entry system of Accounting, Accounting Cycle, Rules for maintaining Books of Accounts, Journal, Posting to Ledger, Preparation of Trial Balance.

**UNIT V**

**FINANCIAL ACCOUNTING –II:**

Introduction to Final accounts, Revenue and Capital Expenditure, elements of Financial Statements, Preparation of Final Accounts with simple adjustments (simple problems)

## **UNIT-VI**

### **FINANCIAL ANALYSIS THROUGH RATIOS:**

Concept of Ratio Analysis, Various Types of Ratios: Liquidity Ratios (short term solvency ratios), Leverage Ratios (long term solvency ratios), Turnover Ratios and Profitability Ratios (simple problems).

### **TEXTBOOKS:**

1. Aryasri: Managerial Economics and Financial Analysis, 2/e, TMH, 2005.

### **REFERENCES:**

1. Ambrish Gupta, Financial Accounting for Management, Pearson Education, New Delhi.
2. H. Craig Peterson & W. Cris Lewis, Managerial Economics, PHI, 4th Ed.
3. Suma Damodaran, Managerial Economics, Oxford University Press.



**Syllabus for B.Tech CSE II year II Semester**  
**Environmental Science**

**Code: 9HC05**

**Prerequisite:** Engineering Chemistry

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | - |

**Course Objective:**

Understand and disseminate knowledge about environmental problems at local, regional and global scale.

**Course Outcomes:**

After completion of the course, the student will be able to:

1. Demonstrate ecological principles and natural resources. [L2]
2. Interpret biodiversity and Environmental Pollution. [L2]
3. Develop technologies on the basis of and environmental regulations which in turn help in sustainable development. [L6]

**UNIT-I Ecosystems:** Definition, Scope, and Importance of ecosystem. Classification, structure, and function of an ecosystem, Food chains, food webs, and ecological pyramids. Flow of energy, Biogeochemical cycles, Bioaccumulation, Biomagnification, ecosystem value, services and carrying capacity.

**UNIT-II Natural Resources:** Classification of Resources: Living and Non-Living resources, water resources: use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems. Mineral resources: use and exploitation, environmental effects of extracting and using mineral resources, Land Energy resources: growing energy needs, renewable and non-renewable energy sources, use of alternate energy source.

**UNIT-III Biodiversity and Biotic Resources:** Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity; consumptive use, productive use, social, ethical, aesthetic and optional values. India as a mega diversity nation, Hot spots of biodiversity. Field visit. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts; conservation of biodiversity: In-Situ and Ex-situ conservation.

**UNIT-IV Environmental Pollution and Control Technologies:** Environmental Pollution: Classification of pollution, Air Pollution: Primary and secondary pollutants. Acid rain-Threshold limit values of chemicals present in environment, Global warming, Ozone layer depletion, Water pollution: Sources and types of pollution. Soil Pollution: Sources and types, Impacts of modern agriculture, degradation of soil. Noise Pollution: Sources and Health hazards, standards, Solid waste: Municipal Solid Waste management, composition and characteristics of e-Waste and its management. Pollution control technologies: Sewage water Treatment, Kyoto protocol, and Montréal Protocol.

**UNIT-V Sustainable development and Green Technology:** Concept of sustainable development, threats to sustainability population and its explosion, Crazy consumerism, over-exploitation of resources, strategies for achieving sustainable development environmental education, conservation of resources, urban sprawl sustainable cities and sustainable communities, human health , role of IT in Environment, Environmental Ethics, Environmental Economic – Concept of Green Building, Clean Development Mechanism ( CDM ).

**UNIT-VI Environmental Policy, Legislation & Environment Impact Assessment:** Environmental Protection act, Legal aspects Air Act- 1981, Water Act, Forest Act, Wild life Act, Municipal solid waste management and handling rules, biomedical waste management and handling rules, hazardous waste management and handling rules. EIA: EIA structure, methods of baseline data acquisition. Overview on Impacts of air, water, biological and Socio-economical aspects. Strategies for risk assessment, Concepts of Environmental Management Plan (EMP).

**TEXT BOOKS:**

1. Perspectives in *Environmental Studies*: Kaushik A. and Kaushik, C.P. New Age International (P) Ltd. (2008)

**REFERENCE BOOKS:**

1. Environmental Studies by Erach Bharucha, 2005 University Press.
2. Environmental Science: towards a sustainable future by Richard T. Wright. 2008 PHL Learning Private Ltd. New Delhi.
3. Environmental Engineering and science by Gilbert M. Masters and Wendell P. Ela. 2008 PHI Learning Pvt. Ltd.
4. Environmental Science by Daniel B. Botkin & Edward A. Keller, Wiley INDIA edition.
5. Environmental Studies by Anubha Kaushik, 4th Edition, New age international publishers.
6. Text book of Environmental Science and Technology - Dr. M. Anji Reddy 2007, BS Publications.

**Syllabus for B.Tech CSE II year II Semester  
Soft Skills Lab**

| L | T | P/D | C |
|---|---|-----|---|
| 0 | 1 | 2   | 2 |

**Code:** 9HC63

**Prerequisite:** Universal Human Values, English

**Course objective:**

Encourage the all-round development of students by focusing on soft skills.

**Course Outcomes:**

After completion of the course, the student will be able to:

1. Assess themselves and develop soft skills for employability and life-long learning.  
[L3,L5]
2. Demonstrate values and etiquette in their personal and professional lives.[L2]

**Tutorial (1 per week)**

• **Unit-1**

- 1.1 Introduction to soft skills
- 1.2 SWOT / SWOC Analysis
- 1.3 SWOT / SWOC Grid
- 1.4 Johari window

**Unit-2**

- 2.1 Emotional intelligence
- 2.2 Time management
- 2.3 Goal Setting

**Unit-3**

- 3.1 Attitude
- 3.2 Professional etiquette & Grooming

**Unit-4**

- 4.1 Styles of Communication
- 4.2 **Inter-personal Skills**
- 4.3 Team work, Team building
- 4.4 Leadership Skills

**Unit-5**

- 5.1 Problem Solving & Decision making
- 5.2 Critical & Creative thinking

**Unit-6**

- 6.1 Values: Personal, Social & Cultural

**Lab (2 per week)**

**Unit-1**

- Activities based on Soft skills
- Self-Analysis
- Questionnaire,
- SWOT Practice

## **Unit-2**

Activities:

- big picture challenge
- Goal setting charts

## **Unit-3**

Practice activities on

- Attitude
- Professional etiquette & Grooming

## **Unit-4**

- Activities on social skills
- Role Plays
- Team building activities

## **Unit-5**

Practice activities on

- Problem solving situations
- Games and puzzles
- Case Studies and Group Discussions on decision making and problem solving, creativity and innovation.

## **Unit-6**

Practice activities

- Role Plays

## **Text Book:**

SOFT SKILLS – Dr. K. Alex, S. Chand publications

## **Suggested Readings:**

1. SOFT SKILLS – MeenakshiRaman ; \* Step Ahead with Soft Skills - Oxford University Press ; \* Skill Sutras- JayashreeMohanraj \* The Power of Soft Skills – Robert A. Johnson ; \* Soft Skills for Everyone – Jeff Butterfield

**Syllabus for B.Tech CSE II year II Semester**  
**Technical Seminar**

| L | T | P/D | C |
|---|---|-----|---|
| 0 | 1 | 0   | 1 |

**Code: 9E484**

**Prerequisite: NIL**

**Course objective:**

Develop an ability to understand and present the latest technological developments in the relevant field to enhance oratory and interview-facing skills.

**Course Outcomes:**

After completion of this course, students will be able to:

1. Identify topics related to relevant domains or disruptive technologies. [L3]
2. Survey and organize technical content in presentable form. [L4 & L3]
3. Interpret effectively in technological know-how. [L5]

**Procedure:**

1. Seminar In-charge shall highlight the significance of the Technical Seminar in the first two sessions and enlighten the students on the utility of these seminars.
2. The slots and titles shall be decided upfront and the seminar In-charge shall take signatures from students.
3. The same sheet shall be affixed to the respective classrooms and seminar register.
4. If any student fails to present his/her seminar in the given slot, for genuine reasons, they may be asked to present in the subsequent slot/week.
5. Progress of the seminars needs to be reviewed by the concerned HOD once in 15 days.
6. The evaluation for Technical Seminars has to be informed to students and displayed in the classrooms.
7. The report and presentation must contain a topic, introduction, explanation, diagrams, tables, applications, and conclusions.

There shall be a technical seminar evaluated for 100 marks. The evaluation is purely internal and will be as follows:

| S. No | Description                                          | Marks |
|-------|------------------------------------------------------|-------|
| 1     | Literature survey, topic, and content                | 10    |
| 2     | Presentation including PPT                           | 10    |
| 3     | Seminar Notes                                        | 05    |
| 4     | Interaction with the audience after the presentation | 05    |
| 5     | Final Report 3 copies                                | 10    |

|   |                                                                                |           |
|---|--------------------------------------------------------------------------------|-----------|
| 6 | Classroom participation                                                        | 05        |
| 7 | Punctuality in giving a seminar as per the Scheduled time and date             | 10        |
| 8 | Mid Semester Viva (on the seminar topics completed up to the end of the 9 week | 15        |
| 9 | End Semester Viva                                                              | 30        |
|   | Total                                                                          | 100 Marks |

Student must secure 40% i.e. 40 marks to be successful in sum total (Hundred Marks) in Technical Seminar.

## Syllabus for B.Tech CSE II year II Semester

### Python Programming Lab

**Code: 9FC62**

**Prerequisite:** Problem Solving using C Lab

| L | T | P/D | C |
|---|---|-----|---|
| 0 | 0 | 4   | 2 |

#### Course Objective:

Develop applications using various modules on GUI designing, AI, machine learning, numerical computing, mobile applications, and game development.

#### Course Outcomes:

At completion of this course, the student will be able to:

1. Develop the skills in python basics, OOPS concepts and build applications to demonstrate the core concepts of python. [L1 & L6]
2. Explain python modules, packages, files, data processing and GUI design to develop python applications. [L2 & L3]

#### Week -1:

1. Use a web browser to go to the Python website <http://python.org>. This page contains information about Python and links to Python-related pages, and it gives you the ability to search the Python documentation.
2. Start the Python interpreter and type help() to start the online help utility.
3. Start Python interpreter and use it as Calculator.

#### Week -2:

4. If you run a 10 kilometer race in 43 minutes 30 seconds, what is your average time per mile? What is your average speed in miles per hour? (Hint: there are 1.61 kilometers in a mile).
5. The volume of a sphere with radius r is  $\frac{4}{3}\pi r^3$ ? (Use Sphere volume formula)
6. Suppose the cover price of a book is \$24.95, but bookstores get a 40% discount. Shipping costs \$3 for the first copy and 75 cents for each additional copy. What is

the total wholesale cost for 60 copies?

#### Week -3:

7. A function object is a value you can assign to a variable or pass as an argument. For example, do\_twice is a function that takes a function object as an argument and calls it twice:

```
def do_twice(f):
```

```
    f()
```

```
    f()
```

Here's an example that uses do\_twice to call a function named print\_spam twice.

```
def print_spam():
```

```
print'spam'
```

```
do_twice(print_spam)
```

- a. Type this example into a script and test it.
- b. Modify `do_twice` so that it takes two arguments, a function object and a value, and calls the function twice, passing the value as an argument.
- c. Write a more general version of `print_spam`, called `print_twice`, that takes a string as a parameter and prints it twice.
- d. Use the modified version of `do_twice` to call `print_twice` twice, passing 'spam' as an argument.

8. Write a function that draws a grid like the following:

```
+ - - - + - - - +  
|       |       |  
|       |       |  
|       |       |  
+ - - - + - - - +  
|       |       |  
|       |       |  
|       |       |  
|       |       |  
+ - - - + - - - +
```

Hint: to print more than one value on a line, you can print a comma-separated sequence.

9. Write a function called `gcd` that takes parameters `a` and `b` and returns their greatest common divisor.
10. Write a function called `is_palindrome` that takes a string argument and returns `True` if it is a palindrome and `False` otherwise. Remember that you can use the built-in function `len` to check the length of a string.

#### Week-4:

11. Write a function called `is_sorted` that takes a list as a parameter and returns `True` if the list is sorted in ascending order and `False` otherwise.
12. Write a function called `has_duplicates` that takes a list and returns `True` if there is any element that appears more than once. It should not modify the original list.



### **Week-5:**

12. Write a function called `remove_duplicates` that takes a list and returns a new list with only the unique elements from the original. Hint: they don't have to be in the same order.
13. The wordlist I provided, `words.txt`, doesn't contain single letter words. So you might want to add "I", "a", and the empty string.
14. Write a python code to read a dictionary values from the user. Construct a function to invert its content. i.e., keys should be values and values should be keys.
15. If there are 23 students in your class, what are the chances that two of you have the same birthday? You can estimate this probability by generating random samples of 23 birthdays and checking for matches.  
Hint: you can generate random birthdays with the `randint` function in the `random` module
16. How does a module source code file become a module object?

### **Week-6:**

17. Why might you have to set your `PYTHONPATH` environment variable?
18. What is a namespace, and what does a module's namespace contain?
19. How do you make a module? Give an example of construction of a module using different geometrical shapes and operations on them as its functions.
20. What is the purpose of a `__init__.py` file in a module package directory? Explain with a suitable example.
21. Use the structure of exception handling all general purpose exceptions.

### **Week-7:**

22. a. Write a function called `draw_rectangle` that takes a `Canvas` and a `Rectangle` as arguments and draws a representation of the `Rectangle` on the `Canvas`.  
b. Add an attribute named `color` to your `Rectangle` objects and modify `draw_rectangleso` that it uses the `color` attribute as the fill color.  
c. Write a function called `draw_point` that takes a `Canvas` and a `Point` as arguments and draws a representation of the `Point` on the `Canvas`.  
d. Define a new class called `Circle` with appropriate attributes and instantiate a few `Circle` objects. Write a function called `draw_circle` that draws circles on the canvas.

### **Week-8:**

23. Write a Python program to demonstrate the usage of MRO in multiple levels of Inheritances.
24. Write a python code to read a phone number and email-id from the user and validate it for correctness.

### **Week-9:**

25. Write a Python code to merge two given file contents into third file.
26. Write a Python code to open a given file and construct a function to check for given words present in it and display on found.

**Week-10:**

27. Import numpy, Plotpy and Scipy and explore their functionalities.
28. Write a GUI program to create a window wizard having two text labels, two text fields and two buttons as Submit and Reset.

**Week -11:** Case Study on Mario Game, Hangman game

**Syllabus for B.Tech CSE II year II Semester**  
**DATABASE MANAGEMENT SYSTEMS LAB**

**Code: 9FC63**

**Prerequisite: NIL**

| L | T | P/D | C   |
|---|---|-----|-----|
| 0 | 0 | 3   | 1.5 |

**Course objective:**

To design the optimal queries using structured and unstructured query languages like SQL and PL/SQL by making use of control structures, cursors, triggers and functions/procedures.

**Course Outcomes:**

After completion of this course, the student will be able to:

1. Infer database language commands to create simple database and analyze the database using queries to retrieve records. [L3]
2. Applying PL/SQL for processing database. [L3]

**Exercises:**

1. Creation, altering and dropping of tables and inserting rows into a table (use constraints while creating tables) examples using SELECT command.
2. Queries (along with sub Queries) using ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSECT, Constraints.  
Example: - Select the roll number and name of the student who secured fourth rank in the class.
3. Queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.
4. Queries using Conversion functions (to\_char, to\_number and to\_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next\_day, add\_months, last\_day, months\_between, least, greatest, trunc, round, to\_char, to\_date)
5. i) Creation of simple PL/SQL program which includes declaration section, executable section and exception – Handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found)  
ii) Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block.
6. Develop a program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions.
7. Program development using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT-IN Exceptions, USER defined Exceptions, RAISE- APPLICATION ERROR.
8. Programs development using creation of procedures, passing parameters IN and OUT of PROCEDURES.
9. Program development using creation of stored functions, invoke functions in SQL Statement and write complex functions.
10. Program development using creation of package specification, package bodies, private objects, package variables and cursors and calling stored packages.
11. Develop programs using features parameters in a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables.

12. Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers.
13. Queries using SQL-INJECTION: AND/OR Attack, Comments Attack, String Concatenation Attack, UNION Injection Attack

**TEXT BOOKS:**

- 1) ORACLE PL/SQL by example. Benjamin Rosenzweig, Elena Silvestrova, Pearson Education 3Edition
- 2) ORACLE DATA BASE LOG PL/SQL Programming SCOTT URMAN, Tata Mc-Graw Hill.
- 3) SQL and PL/SQL for Oracle 10g, Black Book, Dr. P. S.Deshpande.

**Syllabus for B.Tech CSE II year II Semester**  
**IT Workshop and Computer Organization Lab**

**Code: 9FC64**

**Prerequisite:** Problem Solving using C Lab

| L | T | P/D | C |
|---|---|-----|---|
| 0 | 0 | 4   | 2 |

**Course Objective:**

Learn basic knowledge of computer hardware, software, installation of system software and ability to implement assembly language program.

**Course outcomes:**

After completion of the course, the student will be able to:

1. Apply knowledge for computer assembling and software installation and ability to solve the trouble shooting problems along with computing various works using MS-Office such as Ms Power Point Presentation, Ms Word and Ms Excel (by using wide range of function in Excel). [L3]
2. Demonstrate the architecture of 8086 processor, assembly language programming and interfacing with modules and determine Arithmetic operations of binary numbers system[L2]

## **IT Workshop**

### **Week 1:**

**Introduction to Computer:** Identify the peripherals of a computer, components/peripherals in a CPU & its functions. Introduction to the types of Operating System, Assembling and disassembling demonstration.

### **Week 2:**

Install computer with dual boot operating system (Windows, Linux with PowerPoint presentation). Comparison of types of OS in different platform

### **Week3:**

Introduction to S/W's, difference b/w hardware and software. Introduction to MS-Office and its importance.

### **Ms Word**

### **Ms Power Point Presentation**

### **Week4:**

Introduction to Excel

**Features:** Accessing, Overview at toolbars, saving excel files, Gridlines, Format cells, Summation, Auto fill, formatting text.

Formula in excel – Average, Standard Deviation, Charts, Roaming & Inserting worksheets, Hyper linking, count function, lookup / Vlookup, sorting, Conditional formatting.

**PART – A**

**Introduction to MASM/TASM Assembler**

**Familiarization with 8086 Kit**

**Experiment I, II**

**Write ALP and execute the program to**

1. Add two 8-bit numbers
2. Add two 16-bit numbers
3. Add two 32-bit numbers
4. Subtract two 8-bit numbers
5. Subtract two 16-bit numbers
6. Subtract two 32-bit numbers
7. Multiply two 8-bit numbers
8. Multiply two 16-bit numbers
9. Perform 8-bit division
10. Perform 16-bit division
11. Find square of a number
12. Find cube of a number
13. Exchange two numbers
14. Find factorial of a given number

**Experiment III**

**Write ALP and execute the program to**

15. Add a given series of numbers
16. Find average of a given series of numbers
17. Find sum of squares of a given series of numbers
18. Find sum of cubes of a given series of numbers

**Experiment IV**

**Write ALP and execute the program to**

19. Find largest number from a given series of numbers
20. Find smallest number from a given series of numbers
21. Sort a series of given numbers in ascending order
22. Sort a series of given numbers in descending order

**Experiment V**

**Write ALP and execute the program to**

23. Display Fibonacci series
24. Move a string of data bytes from one location to another
25. Concatenate two strings
26. Reverse a given string

## **Experiment V1**

**Write ALP and execute the program to**

27. Compare two strings
28. Find length of a given string
29. Find whether the given byte is in the string or not

## **PART-B**

Write ALP and interface with 8086

1. Interface a stepper motor
2. Generate a triangular wave, square wave and saw tooth waves, Interface keyboard

## **TEXT BOOKS:**

1. Computer Systems Architecture – M.Moris Mano, IIIrd Edition, Pearson
2. Microprocessors and interfacing – Douglas V.Hall, TMH, 2<sup>nd</sup> Edition, 1999.

## **REFERENCES:**

1. Computer Organization and Architecture – William Stallings Sixth Edition, Pearson
2. Microcomputer systems, The 8086/8088 Family Architecture, Programming and Design – Y.Liu and G.A. Gibson, PHI, 2<sup>nd</sup> Edition.
3. Advanced microprocessor and Peripherals – A.K.Ray and K.M.Bhurchandi, TMH, 2000.

**Syllabus for B.Tech (CSE) III YEAR I SEM**  
**INFORMATION RETRIEVAL SYSTEMS**  
**(Professional Elective-I)**

**Code:** 9FC08

**Prerequisites:** Data Structures

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Learn the important concepts and algorithms in IRS and also understand the data/file structures that are necessary to design, and implement information retrieval (IR) systems.

**Course Outcomes:**

After completion of this course student will be able to:

1. Identify and Outline IR principles, capabilities of IR systems, and relate associations with DBMS, Data warehousing.[L1]-U1
2. Infer functions of cataloguing and Indexing, discuss behaviour of various data structures[L2]-U2,U3
3. Analyze different clustering algorithms.[L4]-U3
4. Choose various searching methods, searching process metrics to implement search and visualization applications.[L3]-U4,U5
5. Discover the components of Multimedia Information Retrieval and illustrate a distributed information retrieval system.[L4]-U6

**UNIT - I**

Introduction to Information Retrieval Systems: Definition of Information Retrieval System, Objectives of Information Retrieval Systems, Functional Overview, Relationship to Database Management Systems, Digital Libraries and Data Warehouses Information Retrieval System Capabilities: Search Capabilities, Browse Capabilities, Miscellaneous Capabilities

**UNIT - II**

Cataloguing and Indexing: History and Objectives of Indexing, Indexing Process, Automatic Indexing, Information Extraction

Data Structure: Introduction to Data Structure, Stemming Algorithms, Inverted File Structure, N-Gram Data Structures, PAT Data Structure, Signature File Structure, Hypertext and XML Data Structures, Hidden Markov Models

**UNIT - III**

Automatic Indexing: Classes of Automatic Indexing, Statistical Indexing, Natural Language, Concept Indexing, Hypertext Linkages Document and Term Clustering: Introduction to Clustering, Thesaurus Generation, Item Clustering, Hierarchy of Clusters

**UNIT - IV**

User Search Techniques: Search Statements and Binding, Similarity Measures and Ranking, Relevance Feedback, Selective Dissemination of Information Search, Weighted Searches of Boolean Systems, Searching the INTERNET and Hypertext Information Visualization: Introduction to Information Visualization, Cognition and Perception, Information Visualization Technologies



## **UNIT V**

**Text Search Algorithms:** Introduction, Software text search algorithms, Hardware text search systems. Integrating Structured Data and Text Historical progression, Information retrieval as a relational application, Semi structured search using a relational schema.

## **UNIT VI**

**Multimedia Information Retrieval** – Models and Languages – Data Modeling, Query Languages, Indexing and Searching.

Distributed Information Retrieval: A Theoretical model of distributed retrieval, web search

### **TEXT BOOK:**

1. Information Storage and Retrieval Systems – Theory and Implementation, Second Edition, Gerald J. Kowalski, Mark T. Maybury, Springer

### **REFERENCE BOOKS:**

1. Frakes, W.B., Ricardo Baeza-Yates: Information Retrieval Data Structures and Algorithms, Prentice Hall, 1992.
2. Information Storage & Retrieval By Robert Korfhage – John Wiley & Sons.
3. Modern Information Retrieval By Yates and Neto Pearson Education.

**Syllabus for B.Tech (CSE) III YEAR I SEM**  
**IMAGE PROCESSING**  
**Professional Elective - I**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Code: 9FC09**

**Prerequisite :** Computer Graphics

**Course Objective:**

Make decisions from image data, online inspection and face recognition.

**Course Outcomes:**

After completion of this course student will be able to:

1. Identify Image Processing terminology & interpret the components of image processing system and its applications. [L3]-U1
2. Examine various methods for Image Enhancement approaches in spatial domain. [L3]-U2
3. Make use of the filtering techniques and Fourier, Geometric transformations and filtering techniques for image restoration. [L3]-U3
4. Analyze Colour image Processing using various approaches for transformation, segmentation, smoothing and sharpening of colour images. [L4]-U4
5. Illustrate and examine Image compression and Image morphological processing methods. [L4]-U5
6. Apply and evaluate image segmentation and recognition methods. [L3, L5]-U6

**UNIT – I**

**Introduction:** Examples of fields that use digital image processing, fundamental steps in digital image processing, components of image processing system.. Digital Image Fundamentals: A simple image formation model, image sampling and quantization, basic relationships between pixels

**UNIT – II**

**Image enhancement** in the spatial domain: Basic gray-level transformation, histogram processing, enhancement using arithmetic and logic operators, basic spatial filtering, smoothing and sharpening spatial filters, combining the spatial enhancement methods

**UNIT – III**

**Image restoration:** A model of the image degradation/restoration process, noise models, restoration in the presence of noise—only spatial filtering, Weiner filtering, constrained least squares filtering, geometric transforms; Introduction to the Fourier transform and the frequency domain, estimating the degradation function

**UNIT– IV**

**Color Image Processing:** Color fundamentals, color models, pseudo color image processing, basics of full-color image processing, color transforms, smoothing and sharpening, color segmentation.

**UNIT – V**

**Image Compression and Morphology:** Fundamentals, image compression models, error-free compression, lossy predictive coding, image compression standards, Morphological

Image Processing: Preliminaries, dilation, erosion, open and closing, hit or miss transformation

## **UNIT – VI**

**Image Segmentation and Recognition:** Detection of discontinuous, edge linking and boundary detection, thresholding, region-based segmentation, Patterns and patterns classes, recognition based on decision-theoretic methods, matching, optimum statistical classifiers

### **TEXT BOOKS:**

1. Digital Image Processing, RafealC.Gonzalez, Richard E.Woods, Third Edition, Pearson Education/PHI.

### **REFERENCE BOOKS:**

1. Image Processing, Analysis, and Machine Vision, Milan Sonka, Vaclav Hlavac and Roger Boyle, Second Edition, Thomson Learning.
2. Introduction to Digital Image Processing with Matlab, Alasdair McAndrew, Thomson Course Technology
3. Computer Vision and Image Processing, Adrian Low, Second Edition, B.S.Publications
4. Digital Image Processing, William K. Prat, Wily Third Edition
5. Digital Image Processing and Analysis, B. Chanda, D. Datta Majumder, Prentice Hall of India, 2003

**Syllabus for B.Tech (CSE) III YEAR I SEM**  
**INTRODUCTION TO DATA SCIENCE**  
**(Professional Elective- I)**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Code: 9EC10**

**Prerequisite:** Database Management Systems, Python Programming, Probability and Statistics

**Course Objective:**

Learn concepts, algorithms, techniques and tools to analyze the data, data modelling and visualization of data.

**Course Outcomes:**

After completion of the course, the student should be able to:

1. Explain different levels of Data, various Datasets and demonstrate mathematical models for representation. [L2,L3]-U1,U2
2. Summarize and apply the statistical techniques to test the sampling distributions and their goodness. [L2,L3]-U2,U3
3. Apply Array handling, Data Manipulation techniques on various datasets and analyze using Visualization techniques. [L3, L4]-U4
4. Explain different error handling techniques on data.[L2]-U5
5. Classify and explain different Machine Learning Techniques. [L2]-U6

**UNIT 1**

Data science: Basic terminology - Example – Sigma Technologies

Types of Data: Structured versus unstructured data - Quantitative versus qualitative data -The four levels of data – Nominal – Ordinal – Interval – Ratio

**UNIT 2**

Basic Mathematics - Basic symbols and terminology - Vectors and matrices - Arithmetic symbols - Summation - Proportional - Dot product - Graphs - Logarithms/exponents - Set theory - Linear algebra - Matrix multiplication

Basic Statistics: Obtaining data - Observational - Experimental - Sampling data - Probability sampling - Random sampling - Unequal probability sampling - Measuring statistics - Measures of center - Measures of variation - Measures of relative standing - The Empirical rule

**UNIT 3**

Advanced Statistics: Point estimates - Sampling distributions - Confidence intervals - Hypothesis tests - Conducting a hypothesis test - One sample t-tests - Example of a one sample t-tests - Assumptions of the one sample t-tests - Type I and type II errors - Hypothesis test for categorical variables - Chi-square goodness of fit test - Chi-square test for association/independence

**UNIT 4**

The Basics of NumPy Arrays - The Basics of NumPy Arrays - Aggregations: Min, Max, and Everything in Between - Computation on Arrays: Broadcasting - Fancy Indexing Data Manipulation with Pandas - Introducing Pandas Objects - Data Indexing and Selection - Handling Missing Data - Combining Datasets: Concat and Append - Aggregation

and Grouping. Visualization with Matplotlib - Importing matplotlib - Simple Line Plots - Simple Scatter Plots - Histograms, Binnings, and Density – case study: Effect of Holidays on US Births

## **UNIT 5**

The bias variance tradeoff, Error due to bias, Error due to variance, Two extreme cases of bias/variance tradeoff, Underfitting, Overfitting, How bias/variance play into error functions ,K folds cross-validation, Grid searching, Visualizing training error versus cross-validation error

## **UNIT-6**

Basic Machine Learning: Introduction to machine learning, types of machine learning: Supervised and unsupervised

Types of Supervised learning: Classification and regression, Classifiers – Decision tree and k-Nearest Neighbors (k-NN),

Types of unsupervised: Clustering k-means, Association Rule mining

## **TEXT BOOKS:**

1. Principles of Data Science, Sinan Ozdemir, Packt, (2016)
2. Python Data Science handbook, Essential Tools for Working with Data, Jake Vanderplas, O'Reilly, (2017)

## **REFERENCES:**

1. Pang-Ning Tan, Michael Steinbach, Vipin Kumar, Introduction to Data Mining, Pearson
2. Laura Igual and Santi Seguí, Introduction to Data Science, Springer

**Syllabus for B.Tech (CSE) III YEAR I SEM**  
**ADVANCED COMPUTER ARCHITECTURE**  
**(Professional Elective - I)**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Code:** 9EC11

**Prerequisites:** Computer Organization

**Course Objective:**

Learn the concepts and techniques of parallel and advanced computer architectures to design modern computer systems.

**Course Outcomes:**

After completion of this course student will be able to:

1. Classify different Computational models and Computer Architectures.[L2]-U1,U2
2. Illustrate the Concepts of parallel and Scalable Architectures.[L3]-U2,U3,U4
3. Summarize the concepts of Pipelining, Superscalar processors, multiprocessors, cache and shared memory.[L2]-U3,U4,U5,U6

**UNIT - I**

Theory of Parallelism, Parallel computer models, The State of Computing, Multiprocessors and Multicomputers, Multivector and SIMD Computers, PRAM and VLSI models, Architectural development tracks, Program and network properties, Conditions of parallelism, Program partitioning and Scheduling, Program flow Mechanisms, System interconnect Architectures.

**UNIT - II**

Principals of Scalable performance, Performance metrics and measures, Parallel Processing applications, Speed up performance laws, Scalability Analysis and Approaches, Hardware Technologies, Processes and Memory Hierarchy, Advanced Processor Technology, Superscalar and Vector Processors, Memory Hierarchy Technology, Virtual Memory Technology

**UNIT - III**

Bus Cache and Shared memory, Backplane bus systems, Cache Memory organizations, Shared-Memory Organizations, Sequential and weak consistency models, Pipelining and superscalar techniques, Linear Pipeline Processors, Non-Linear Pipeline Processors, Instruction Pipeline design, Arithmetic pipeline design, superscalar pipeline design.

**UNIT - IV**

Parallel and Scalable Architectures, Multiprocessors and Multicomputers, Multiprocessor system interconnects, cache coherence and synchronization mechanism, Three Generations of Multicomputers.

**UNIT – V**

Message-passing Mechanisms, Multivector and SIMD computers, Vector Processing Principals, Multivector Multiprocessors, Compound Vector processing, SIMD computer Organizations, The connection machine CM-5

**UNIT – VI**

Scalable, Multithreaded and Dataflow Architectures, Latency-hiding techniques, Principals of

Multithreading, Fine-Grain Multicomputers, Scalable and multithreaded Architectures, Dataflow and hybrid Architectures.

**TEXT BOOK:**

1. Advanced Computer Architecture Second Edition, Kai Hwang, Tata McGraw Hill Publishers.

**REFERENCES:**

1. Computer Architecture, Fourth edition, J. L. Hennessy and D.A. Patterson. ELSEVIER.

**Syllabus for B.Tech (CSE) III YEAR I SEM**  
**HUMAN COMPUTER INTERACTION**  
**(Professional Elective - I)**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Code: 9FC10**

**Prerequisite:** IT Workshop, Web Technologies

**Course Objective:**

Gain an overview of Human-Computer Interaction (HCI), with an understanding of user interface design in general, and alternatives to traditional computing.

**Course Outcomes:**

After completion of this course student will be able to:

1. Explain basics and benefits of good Graphical User Interface.[L2]-U1,U2
2. Use appropriate GUI components to design an interface.[L3]-U3,U4,U5
3. Identify hardware and software tools for different contextual scenarios.[L3]-U6

**UNIT-I**

Introduction: Importance of user Interface definition, importance of good design. Benefits of good design. A brief history of Screen design.

**UNIT-II**

The graphical user interface – popularity of graphics, the concept of direct manipulation, graphical system, Characteristics, Web user – Interface popularity, characteristics-Principles of user interface.

**UNIT-III**

Design process – Human interaction with computers, importance of human characteristics human consideration in Design, Human interaction speeds, understanding business functions.

**UNIT-IV**

Screen Designing: Design goals – Screen planning and purpose, organizing screen elements, ordering of screen data and content – screen navigation and flow – Visually pleasing composition – amount of information – focus and emphasis – presentation information simply and meaningfully – information retrieval on web – statistical graphics – Technological consideration in interface design.

**UNIT-V**

Windows – New and Navigation schemes selection of window, selection of devices based and screen based controls.

Components – text and messages, Icons and increases – Multimedia, colors, uses problems, choosing colors.

**UNIT-VI**

Software tools – Specification methods, interface – Building Tools.



Interaction Devices – Keyboard and function keys – pointing devices-speech recognition digitization and generation – image and video displays – Printers.

**TEXTBOOKS:**

1. The essential guide to user interface design, Wilbert O Galitz, Wiley Dreama Tech.
2. Designing the user interface, 3<sup>rd</sup> Edition Ben Shneidermann, Pearson Education Asia.

**REFERENCES:**

1. Human – Computer Interaction. ALANDIX, JANETFINCAY, GREGORYD, ABOWD, RUSSELLBEALG, PEARSON. Interaction Design PRECE, ROGERS, SHARPS. Wiley Dreamtech,
2. User Interface Design, Soren Lauesen, Pearson Education

**Syllabus for B.Tech (CSE) III YEAR I SEM**  
**SOFTWARE REQUIREMENTS AND ESTIMATION**  
**(Professional Elective – I)**

**Code:** 9EC12

**Prerequisite:** Software Engineering

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Understand proper contents of a Software Requirements Specification document and estimation techniques.

**Course Outcomes:**

After completion of this course student will be able to:

1. Describe the software requirements and use elicitation techniques to gather requirements. [L1,L3]-U1,U2
2. Analyze various software design models. [L2]-U3
3. Identify the software requirement management tools.[L3]-U6
4. Apply various Estimation methods and use appropriate tools to evaluate the cost of the project.[L3,L5]-U4,U5,U6

**UNIT- I**

**Software Requirements: What and Why** Essential Software requirement, Good practices for requirements engineering, Improving requirements processes, Software requirements and risk management

**UNIT-II**

**Software Requirements Engineering** Requirements elicitation, elicitation techniques, requirements analysis, documentation, review, Software quality attributes, risk reduction through prototyping, setting requirements priorities, verifying requirements quality.

**UNIT – III**

**Software Requirements Modelling** Analysis Models, Use Case Modelling, Dataflow diagram, state transition diagram, class diagrams.

**Software Requirements Management** Requirements management Principles and practices, Requirements attributes, Change Management Process, Requirements Traceability Matrix, Links in requirements chain.

**UNIT IV**

**Software Estimation** Components of Software Estimations, Estimation methods, Problems associated with estimation, Key project factors that influence estimation Size **Estimation** Two views of sizing, Function Point Analysis, Mark II FPA, Full Function Points, LOC Estimation, and Conversion between size measures.

**UNIT - V**

**Effort, Schedule and Cost Estimation** What is Productivity? Estimation Factors, Approaches to Effort and Schedule Estimation, COCOMO II, Putnam Estimation Model, Cost Estimation.

## **UNIT – VI**

**Tools for Requirements Management and Estimation** Requirements Management Tools: Benefits of using a requirements management tool, commercial requirements management tool, Rational Requisite pro, Caliber – RM, implementing requirements management automation.

**Software Estimation Tools:** Desirable features in software estimation tools, IFPUG, USC's COCOMO II, and SLIM (Software Life Cycle Management) Tools.

### **TEXT BOOK:**

1. Software Requirements and Estimation by Rajesh Naik and Swapna Kishore, Tata Mc Graw Hill.

### **REFERENCES:**

1. Software Requirements by Karl E. Weigers, Microsoft Press.
2. Managing Software Requirements, Dean Leffingwell & Don Widrig, Pearson Education, 2003.
3. Mastering the requirements process, second edition, Suzanne Robertson & James Robertson, Pearson Education, 2006.
4. Estimating Software Costs, Second edition, Capers Jones, TMH, 2007.
5. Practical Software Estimation, M.A. Parthasarathy, Pearson Education, 2007.
6. Measuring the software process, William A. Florac & Anita D. Carleton, Pearson Education, 1999.

## Syllabus for B.Tech (CSE) III YEAR I SEM

### DATA WAREHOUSING AND DATA MINING

**Code:** 9FC05

**Prerequisite:** Database Management Systems

| L | T | P/D | C |
|---|---|-----|---|
| 2 | 1 | 0   | 3 |

**Course Objective:**

Understand Data warehouse architecture and Data Mining techniques.

**Course Outcomes:**

After completion of this course student will be able to:

1. Illustrate the functionality of the various data mining functions.[L3]-U1
2. Apply pre-processing techniques on various datasets.[L3]-U2
3. Create a Data Warehouse and demonstrate the business analysis with OLAP tools.[L5,L3]-U3
4. Categorize the kinds of patterns that can be discovered by association rule mining.[L4]-U4
5. Compare and contrast different classification and clustering algorithms.[L4]-U5,U6

**UNIT – I**

**Introduction:** Fundamentals of data mining, KDD process, Architectures of Data Mining Systems ,Data Mining Functionalities, Classification of Data Mining systems, Data Mining Task primitives, Integration of a Data mining System with a Database or a Data warehouse systems, Major issues in Data Mining.

**UNIT – II**

**Data Preprocessing:** Needs for Preprocessing the Data, Descriptive Data Summarization, Data Cleaning, Data Integration and Transformation, Data Reduction, Discretization and Concept Hierarchy Generation.

**UNIT – III**

**Data Warehouse and OLAP Technology for Data Mining:** Data Warehouse, Multidimensional Data Model, Data Warehouse Architecture, Data Warehouse Implementation, From Data Warehousing to Data Mining.

**UNIT – IV**

**Mining Frequent, Associations and Correlations:** Basic concepts, Frequent Item set mining methods, Mining multilevel association rules from Transaction Databases, Mining Multidimensional association rules from Relational databases and Data Warehouses, From Association Mining to Correlation Analysis, Constraint-Based Association Mining.

**UNIT – V**

**Classification and Prediction:** Issues regarding Classification and Prediction, Classification by Decision Tree Induction, Bayesian Classification, Classification by Back propagation, Support Vector Machines (SVMs),k-nearest neighbour classifier, Other Classification Methods. Prediction, Classifier Accuracy

## **UNIT – VI**

**Cluster Analysis Introduction:** Types of Data in Cluster Analysis, Major Clustering methods, Partitioning Methods, Density-Based methods, Grid-Based methods, Model-Based Clustering methods, Outlier Analysis.

**Advanced Concepts:** Text Mining, Web Mining.

### **TEXT BOOKS:**

1. Data mining: Concepts and Techniques, Jiawei Han and Micheline Kamber, 2nd Edition, Elsevier, 2006.
2. Data Mining Techniques – ARUN K PUJARI, University Press.

### **REFERENCES:**

1. Data Mining Introductory and advanced topics –MARGARET H DUNHAM, PEARSON EDUCATION
2. Data Mining Techniques – ARUN K PUJARI, University Press.
3. Data Warehousing in the Real World – SAM ANAHORY & DENNIS MURRAY. Pearson Edn Asia.
4. Data Warehousing Fundamentals – PAULRAJ PONNAIAH WILEY STUDENT EDITION.
5. The Data Warehouse Lifecycle Toolkit – RALPH KIMBALL WILEY STUDENT EDITION
6. Introduction to Data Mining - First Edition, by Pang-Ning Tan, Michael Steinbach and Vipin Kumar, ISBN-13: 978-0321321367

## Syllabus for B.Tech (CSE) III YEAR I SEM Software Engineering

**Code: 9EC03**

**Prerequisite: Nil**

| L | T | P/D | C |
|---|---|-----|---|
| 2 | 0 | 0   | 2 |

### **Course Objective:**

Understand the software engineering lifecycle models in the development of software.

### **Course Outcomes:**

After completion of this course student will be able to:

1. Explain the concepts of software engineering and distinguish software development lifecycle models.[L2]-U1,U2
2. Analyze different customer requirements and design models. [L4]-U2,U3
3. Identify and use the testing techniques and tools to improve the quality of the product.[L2]-U4,U5,U6

### **UNIT I**

**Introduction to Software Engineering:** The evolving role of software, Changing Nature of Software, Characteristics of Software, Software myths.

**A Generic view of Process:** Software engineering- A layered technology, a Process framework. The Capability Maturity Model – Integrated (CMM-I)

### **UNIT II**

**Software Requirements:** Functional and Non-functional Requirements, User Requirements, System Requirements, the Software Requirements Document.

**Requirements engineering process:** Feasibility studies, requirements elicitation and analysis, requirements validation, Requirements management, RTM

**Process Models:** Software Development Life Cycle, The waterfall model, Incremental Process models, RAD Model, Unified Process Model, Evolutionary Process models (Prototype Model, Spiral Model, Concurrent Development Model, V-Model, Agile Model)

### **UNIT III**

**Introduction to UML:** Importance of Modelling, Principles of Modelling, Conceptual model of the UML, Architecture.

**Diagrams:** Class Diagrams, Forward and Reverse engineering for class Diagram. Use case Diagrams, Sequence Diagram.

### **UNIT IV**

**Testing:** What is Testing, Characteristics of Test Engineers, Software Testing Life Cycle, Levels of Testing, Testing Approaches, Test Cases: Format for Writing Test Case, Test plan: Format to prepare Test plan

Purpose of testing, Dichotomies, Consequences of bugs

## UNIT V

**Flow graphs and Path testing:** Basics concepts of path testing

**Data Flow Testing:** Basics of Data flow Testing

**Logic Based Testing:** Decision Tables

**Software Metrics:** Software Quality, Metrics for Analysis Model, Metrics for Design Model, Metrics for source code, Metrics for testing, Metrics for maintenance

## UNIT VI

**Software Testing Techniques:** Black-box Testing, White box testing, Gray box Testing, Boundary value Analysis, Regression Testing, Cause Effect graphing, CRUD, Database, Mutation Testing , Random Testing, Performance testing, Stress Testing, Spike Testing, Accepting Testing, Alpha and Beta Testing , V& V

**Automated Testing Tools:** Need for Automated testing tools, Taxonomy of Testing Tools, Exposure to Software Testing Tools: Selenium, UFT

### TEXT BOOKS:

1. Software Engineering, A Practitioner's Approach- Roger S. Pressman, 6th edition. McGrawHill International Edition. (Unit 1,2,5)
2. Grady Booch, James Rumbaugh, Ivar Jacobson: The Unified Modeling Language User Guide, Pearson Education.(Unit 3)
3. Software testing techniques – Boris Beizer, Dreamtech, second edition.(Unit 4,5,6, Refer Internet)

### REFERENCES:

1. Software Testing and continuous Quality Improvement, by William E.Lewis,Gunasekaran,2<sup>nd</sup> Edition Auerbach publications (Unit 6,Refer Internet)
2. Software testing tools – by Dr. K.V.K.K Prasad Dreamtech (Unit 4)
3. Software Engineering- Sommerville, 7th edition, Pearson education.
4. Software Engineering- K.K. Agarwal & Yogesh Singh, New Age International Publishers
5. Software Engineering, an Engineering approach- James F. Peters, WitoldPedrycz, John Wiely.
6. Systems Analysis and Design- Shely Cashman Rosenblatt,Thomson Publications.
7. Software Engineering principles and practice- Waman S Jawadekar, The McGraw-Hill Companies
8. Meilir Page-Jones: Fundamentals of Object Oriented Design in UML, Pearson Education.
9. Pascal Roques: Modeling Software Systems Using UML2, WILEY-Dreamtech India Pvt. Ltd.
10. AtulKahate: Object Oriented Analysis & Design, The McGraw-Hill Companies.
11. Mark Priestley: Practical Object-Oriented Design with UML,TATA McGrawHill
12. Craig Larman Applying UML and Patterns: An introduction to Object – Oriented Analysis and Design and Unified Process, Pearson Education
13. Hans-Erik Eriksson, Magnus Penker, Brian Lyons, David Fado: UML 2 Toolkit, WILEY-Dreamtech India Pvt. Ltd

## Syllabus for B.Tech (CSE) III YEAR I SEM

### Design and Analysis of Algorithms

| L | T | P/D | C |
|---|---|-----|---|
| 2 | 1 | 0   | 3 |

**Code:** 9EC04

**Prerequisite:** Data Structures

#### **Course Objective:**

Understand various algorithmic techniques to design different applications.

#### **Course Outcomes:**

After completion of this course student will be able to:

1. Outline the running times of algorithms using asymptotic analysis. [L1]-U1
2. Assess the performance of applications solved using divide and conquer method. [L5]-U2
3. Analyze greedy method, dynamic programming and their applications. [L4]-U3,U4
4. Analyze the Backtracking, Branch and Bound methods, their applications and explain the concept of P and NP Problems. [L4]-U5,U6

#### **UNIT I**

**Introduction:** Algorithm, Pseudo code for expressing algorithms, Performance Analysis-Space complexity, Time complexity, **Asymptotic Notation-** Big oh notation, Omega notation, Theta notation and Little oh notation, Probabilistic Analysis, Connected and Bi-connected Components..

Applications: Designing optimal solution with respect to time for a problem.

#### **UNIT II**

**Divide and conquer:** General method, **Applications**-Binary search, Quick sort, Merge sort, Strassen's matrix multiplication.

Applications: PNR number Search, sorting the Google search results.

#### **UNIT III**

**Greedy method:** General method, **Applications**-Job sequencing with deadlines, 0/1 knapsack problem, Minimum cost spanning trees, Single source shortest path problem.

Applications: Allocation of funds/resources based on the priority in the computer systems.

#### **UNIT IV**

**Dynamic Programming:** General method, **Applications**-Matrix chain multiplication, Optimal binary search trees, 0/1 knapsack problem, All pairs shortest path problem, Travelling sales person problem.

Applications: Routing Algorithms in the computer networking

#### **UNIT V**

**Backtracking:** General method, **Applications**-n-queen's problem, sum of subsets problem, graph colouring, Hamiltonian cycles, Maze generation Problem

Applications: Undo in MS-Word, Games



## **UNIT VI**

**Branch and Bound:** General method, **Applications** - Travelling sales person problem, 0/1 knapsack problem- LC Branch and Bound solution, FIFO Branch and Bound solution.

**Introduction to NP-Hard and NP-Complete problems:** Basic concepts of non deterministic algorithms, Definitions of NP-Hard and NP-Complete classes. Modular Arithmetic

Applications: Performance evaluation in the dynamic systems.

### **TEXT BOOKS:**

1. Fundamentals of Computer Algorithms, Ellis Horowitz, Satraj Sahni and Rajasekharam, Galgotia publications pvt. Ltd.
2. Algorithm Design: Foundations, Analysis and Internet examples, M.T.Goodrich and R.Tomassia, Johnwiley and sons.

## Syllabus for B.Tech (CSE) III YEAR I SEM COMPUTER NETWORKS

**Code : 9EC05**

**Prerequisite:** NIL

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

### **Course Objective:**

Learn about computer network organization and implementation, obtaining a theoretical understanding of data communication and networking.

### **Course Outcomes:**

After completion of this course student will be able to:

1. Describe the different types of network topologies and protocols useful for real time applications and transmission media.[L1]-U1
2. Identify design issues of data link layer and solve problems on Checksum and flow control.[L2]-U2
3. Analyze MAC layer protocols and LAN technologies.[L4]-U3,U4
4. Use the concepts of subnetting, routing mechanisms and congestion control to design congestion free network.[L3]-U4,U5
5. Generalize the concepts, services and protocols of Transport and Application layers. [L2]-U6

### **UNIT I**

**Introduction:** Uses of Computer Networks, Types of networks: WAN, LAN, MAN, Network Topologies, Reference models: OSI, TCP/IP.

**Physical Layer:** Transmission media: magnetic media, twisted pair, coaxial cable, fiber optics, wireless transmission.

### **UNIT II:**

**Data link layer:** Design issues in data link layer: framing, flow control, error control, Error Detection and Correction: Parity, CRC checksum, Hamming code, Flow Control: Sliding Window Protocols, Applications: Data link layer protocols HDLC, PPP.

### **UNIT III:**

**Medium Access sub layer:** Channel allocation problem, MAC Protocols: ALOHA, CSMA, CSMA/CD, MAC addresses, IEEE 802.X, Standard Ethernet, Wireless LANS. Bridges, Types of Bridges.

### **UNIT IV:**

**Network Layer:** Design issues in Network Layer, Virtual circuit and Datagram subnets-Routing algorithm: Shortest path routing, Flooding, distance vector routing, Link state routing, Hierarchical routing, Broad casting, Multi casting, Routing for mobile hosts.

**Internetworking:** Concatenated Virtual Circuits, Connectionless internetworking, Tunnelling, Internetwork routing, Fragmentation

## **UNIT V:**

**Network layer in internet:** IPv4, IP addresses, Sub netting, Super netting, NAT. Internet control protocols: ICMP, ARP, RARP, DHCP.

**Congestion Control:** Principles of Congestion, Congestion Prevention Policies.

Congestion Control in datagram Subnet: Choke packet, load shedding, jitter control.

Quality of Service: Leaky Bucket algorithm and token bucket algorithm.

## **UNIT VI:**

**Transport Layer:** Transport Services, Connection establishment, Connection release and TCP and UDP protocols.

**Application Layer:** Domain name system, FTP, HTTP, SMTP, WWW.

## **TEXT BOOKS:**

1. Computer Networks — Andrew S Tanenbaum, 4th Edition. Pearson Education/PHI
2. Data Communications and Networking – Behrouz A. Forouzan. Third Edition TMH.
3. Data Communication and Networks - Bhushan Trivedi-OXFORD Publications.

## **REFERENCES:**

1. An Engineering Approach to Computer Networks-S.Keshav, 2nd Edition, Pearson Education
2. Understanding communications and Networks, 3rd Edition, W.A. Shay, Thomson

## Syllabus for B.Tech (CSE) III YEAR I SEM INTELLECTUAL PROPERTY RIGHTS

| L | T | P | C   |
|---|---|---|-----|
| 3 | 0 | 0 | P/F |

**Code: 9IC04**

**Prerequisite:** Universal Human Values

### **Course Objective:**

Impart awareness on intellectual property rights and various regulatory issues related to IPR.

### **Course Outcomes:**

After completion of this course student will be able to:

1. Demonstrate knowledge of Intellectual property and rationale for the legal protection of intellectual property.[L3]-U1
2. Summarize Patents, copyright & GI: Searching, filling, drafting, legislation.[L2]-U2,U3
3. Distinguish Trade Mark & Trade Secret and protection of Integrated Circuit and Industrial Design in Searching, filing, drafting, legislation.[L4]-U4,U5
4. Adapt different national and international conventions and treaties governing the IPRs to meet international norms.[L6]-U6

### **UNIT I**

**Introduction to IPR:** Discovery, Invention, Creativity, Innovation, History & Significance of IPR, Overview of IPR -Patent, Copyright, Trade Mark, Trade Secret , GI, Industrial Design & Integrated Circuit, Non-patentable criteria

### **UNIT II**

**Patents:** Patents- Patentability Criteria, Types of Patents-Process, Product & Utility Models, Software Patenting and protection, Patent infringement-Case studies-Apple Vs Samsung, Enfish LLC Vs Microsoft, Overview of Patent search-Types of Searching, Public & Private Searching Databases, Basics of Patent Filing & Drafting, Indian Patents Law

### **UNIT III**

**Copyrights and Geographical Indications:** Types of Copyrights, Procedure for filing, copyright infringement, Copyright Law, Geographical Indications – Tirupati Laddu, Darjeeling Tea, Basmati rice

### **UNIT IV**

**Trademark and Trade secrets:** Trade Marks –Commercial importance, protection, registration, Case Studies- Sabena and Subena, Castrol Vs Pentagon, Trade Secrets- Case Studies – Kentucky Fried Chicken (KFC), Coca-Cola

### **UNIT V**

**Protection of Industrial Designs & Integrated Circuits:** Industrial Designs–Scope, protection, filing, infringement; Integrated Circuits & Layout design, Semiconductors, Unfair competition, Designs Act.

## **UNIT VI**

**International Conventions & Treaties:** Overview of WTO, GATT, TRIPS, WIPO, Berne Convention, Rome convention, Paris Convention, Patent Cooperation Treaty(PCT), Madrid Protocol, Budapest Treaty, Hague agreement

### **TEXTBOOKS:**

1. DeborahE. Bouchoux, Intellectual Property for Paralegals – The law of Trademarks, Copyrights, Patents & Trade secrets, 3<sup>rd</sup> Edition, Cengage learning, 2012
2. N.S.Gopala krishnan & T.G.Agitha, Principles of Intellectual Property, Eastern Book Company, Lucknow, 2009.

### **REFERENCES:**

1. M.M.S.Karki, Intellectual Property Rights: Basic Concepts, Atlantic Publishers, 2009
2. Neeraj Pandey & Khushdeep Dharni, Intellectual Property Rights, Phi Learning Pvt. Ltd
3. Ajit Parul ekarand Sarita D'Souza, Indian Patents Law–Legal & Business Implications; Macmillan India ltd, 2006.
4. B.L.Wadehra. Law Relating to Patents, Trade Marks, Copyright, Designs & Geographical Indications; Universal law Publishing Pvt.Ltd., India 2000.
5. P.Narayanan; Law of Copyright and Industrial Designs; Eastern law House, Delhi, 2010

**Syllabus for B.Tech (CSE) III YEAR I SEM**  
**Computer Science and Engineering**  
**Software Engineering Lab**

**Code: 9EC63**

**Prerequisite: Nil**

| L | T | P | C   |
|---|---|---|-----|
| 0 | 0 | 3 | 1.5 |

**Course Objective:**

Design and test different models for development of real-time applications.

**Course Outcomes:**

After completion of this course student will be able to:

1. Design various software models to develop a software.[L4]
2. Demonstrate the testing techniques and automation tools to enhance the product quality. [L3, L5]

**Exercises:**

1. The student should take up the case study of Unified Library application which is mentioned in the theory, and Model it in different views i.e. Use case view, logical view, component view, Deployment view, Database design, forward and Reverse Engineering, and Generation of documentation of the project.
2. The student takes up the case studies mentioned below, and model it in different views i.e. Use case view, logical view, component view, Deployment view, Database design, forward and Reverse Engineering, and Generation of documentation of the project.
3. Prepare Test Plan document and write Test Cases for Small scale Project (Like for their B.Tech IV Year Project or Post-Graduate Projects), they are learn how to Analyze SRS document in order to prepare Test Plan Document.
4. Demonstrate skills to use modern software testing tools (EX: UFT, Bugzilla, Selenium) and test application (web, Window application) by using the tools.
5. Demonstrate the ability to differentiate between different Testing tools present in the market (like functional testing tools, Test Management Tools, Bug Tracking Tools and Performance Testing Tools) and prepare Test Plan document and write Test Cases for Small scale Project (Like for their B.Tech IV Year Project or Post-Graduate Projects).

**CASE TOOLS LAB:**

**Case Studies**

1. Automatic Teller Machine
2. Library Management System
3. Railway Reservation System
4. Online Book Shopping System
5. Student Admission System

## **Week 1**

### **1. Basic Structural Modeling:**

- i) Class Diagrams.      ii) Object Diagrams

## **Week 2**

### **2. Basic Behavioral Modeling: i) Usecase Diagrams**

## **Week 3**

### **3. Basic Behavioral Modeling:**

- i) Interaction Diagrams (Collaboration and Sequence Diagram)

## **Week 4**

### **4. i) Basic Behavioral Modeling : Activity Diagram**

- ii) Architectural Modeling : Deployment Diagram

## **Software Testing Lab:**

## **Week 5**

### **5. a) Write programs in 'C' Language to demonstrate the working of the following constructs:**

- i) do...while    ii) while...do    iii) if ...else    iv) switch    v) for
- b) A program written in 'C' language for matrix multiplication fails” Introspect the causes for its failure and write down the possible reasons for its failure.

## **Week 6**

### **6. a) Take any system (e.g. ATM system) and study its system specifications and report the various bugs.**

- i) Open Source Bug Tracking Tool: Bugzilla
- b) Write the test cases for any known application (Ex: Banking application)

## **Week 7**

### **7.a) Create a test plan document for any application (Ex: Internet Banking Application)**

- b) Study of Selenium IDE (open source testing tool)

## **Week 8-10**

### **8. Study of any Functional and Regression Testing Tools: i) Licensed Tool: UFT 12.01**

## Syllabus for B.Tech (CSE) III YEAR I SEM

### Computer Networks and Design and Analysis of Algorithms Lab

**Code: 9EC64**

**Prerequisite:** Data Structures, Problem Solving Using C /Object Oriented Programming

| L | T | P/D | C |
|---|---|-----|---|
| 0 | 0 | 4   | 2 |

### Computer Networks Lab

#### Course Objective:

Understanding the design concepts of Data link, Network, Transport layer and demonstration of NS-2 tool.

#### Course Outcomes:

After completion of this course student will be able to:

1. Analyze different Data link layer concepts such as Framing methods, Error Detection and Error Correction techniques.[L4]
2. Examine different Routing Algorithms.[L3]
3. Demonstrate Wire shark and NS-2 tools and CISCO Packet Tracer.[L3]

#### Computer Networks Lab Exercises:

1. Implement the data link layer framing methods such as
  - a) Character / Byte stuffing
  - b) Bit stuffing.
2. Implement on a data set of characters the three CRC polynomials
  - a) CRC 12
  - b) CRC 16
  - c) CRC CCITT.
3. Implement Hamming code for error detection and error correction
4. Implement Dijkstra's algorithm to compute the shortest path through a graph.
5. Simulate Distance-Vector Routing using CISCO Packet Tracer.
6. Implement Congestion control using Leaky-Bucket Algorithm
7. Execute the basic Networking Commands

|                        |                  |
|------------------------|------------------|
| Arp                    | Hostname         |
| ipconfig               | ipconfig/all     |
| Ipconfig/renew         | Ipconfig/release |
| vii. Ipconfig/flushdns | viii. Pathping   |
| ix. Ping               | x. Route         |
| xi. tracert            |                  |

8. Demonstration of NS-2.
9. Simulate Network Topologies using CISCO Packet tracer.



## **Design and Analysis of Algorithms Lab**

**Prerequisite:** Data Structures (C/C++) Lab

### **Course Objective:**

Solve real-time applications using appropriate methods.

### **Course Outcomes:**

After completion of this course student will be able to:

1. Develop sorting algorithms by applying divide and conquer approach. [L5]
2. Identify the running time analysis of greedy technique, dynamic programming, backtracking, branch and bound approaches. [L2]

### **List of Programs for Lab**

1. Write a C program to implement Merge sort algorithm for sorting a list of integers in ascending order.
2. Write a C program to implement Character sorting.
3. Write a C program to implement Dijkstra's algorithm for the single source shortest path problem.
4. Write a C program that implements Prim's algorithm to generate minimum cost spanning tree.
5. Write a C program to implement greedy algorithm for job sequencing with deadlines.
6. Write a C program to implement Dynamic Programming algorithm for the 0/1 Knapsack problem.
7. Write a C program to implement Dynamic programming algorithm for the Optimal Binary search Tree problem.
8. Write a C program to implement backtracking algorithm for n-queens problems.

### **Other Practice Programs:**

1. Write a C program to implement Quick Sort algorithm for sorting a list of integers in ascending order.
2. Write a C program to implement the DFS algorithm for a graph.
3. Write a C program to implement the BFS algorithm for a graph.
4. Write a C program that implements Kruskal's algorithm to generate minimum cost spanning tree.
5. Write a C program to implement Floyd's algorithm for all pairs shortest path problem.
6. Write a C program to implement the backtracking algorithm for the Hamiltonian circuit's problem.
7. Write a C program to implement backtracking algorithm for the sum of subsets problem.

## Syllabus for B.Tech (CSE) III YEAR I SEM DATA WAREHOUSING AND DATA MINING LAB

**Code: 9FC65**

**Prerequisite:** Database Management Systems

| L | T | P/D | C   |
|---|---|-----|-----|
| 0 | 0 | 3   | 1.5 |

### Course Objective:

Obtain Practical Experience to build data warehouse and perform mining techniques on real data sets.

### Course Outcomes:

After completion of this course student will be able to:

1. Apply transformation techniques by using ETL tool.[L3]
2. Demonstrate the classification, clustering techniques on the data sets.[L3]
3. Analyse the Clustering, Association and Classification techniques applied on the data sets with varied input parameters.[L4]
4. Apply mining techniques for realistic data.[L3]

### Exercises

1. Perform filter transformation for the employee database.
2. Add the commission of 1000 Rs in the Salary field of Employee table using Expression Transformation.
3. Use Aggregator transformation to display the average salary of employees in each department.
4. Use Joiner transformation to display the Sailor\_Name from Sailors table and Boat\_Name from Boats table in a new table.
5. Perform steps to load top 2 salaries for each department without using Rank Transformation and SQL queries in Source Qualifier.
6. Implement the following Multidimensional Data Models
7. i. Star Schema
8. ii. Snowflake Schema
9. iii. Fact Constellation.
10. Compare the GRI and Apriori usage (Prepare a sample data set in Spread Sheet).
11. Determine the Drugs importance w.r.t. Age, Cholestrol and BP using C 5.0.
12. Predict the accuracy of the test data set using Neural Net model using a Case Study of Botanical data set.
13. Compare the C 5.0 and Neural Net using the sample data.
14. Using the BASKETS1n dataset, select the data as given below.
  - a) Customer age < 35 and count the customers who buy dairy and VEG products
  - b) Find the AVG income of customers who buy at least 5 products
15. Using the BASKETS1n dataset, select the data as given below.
  - a) Derive the field whose home own is 'YES' and Age > 30 and sort data w.r.t. income in ascending order, and output only the item fields.
  - b) Find the mean value of salary w.r.t age = {Young, Middle, Senior}.
16. Demonstrate the Weka machine learning toolkit and perform the following exercises

- Determine the numbers of instances of each class are present in the data
- Load the weather dataset and remove all instances where the attribute ‘humidity’ has the value ‘high’?
- Load the iris dataset and view the Visualizer panel.

## Syllabus for B.Tech (CSE) III YEAR I SEM SUMMER INDUSTRY INTERNSHIP – I

**Code: 9E585**

**Prerequisite:** Nil

| L | T | P | C |
|---|---|---|---|
| 0 | 0 | 0 | 1 |

### **Course Objective:**

Develop skills and knowledge in their respective domain to apply and solve engineering problems.

### **Course Outcomes:**

After completion of this course, students will be able to:

1. Use the basic concepts learned in the courses, so far, in conceptualizing, designing, and executing the modules of the projects. [L3]
2. Apply technologies and use modern tools to develop software/hardware. [L3]
3. Inculcate an enthusiasm to use creative ideas to build innovative projects and prototypes that meet the current needs of the market and society as a whole. [L4]
4. Improve their communicative skills and team skills. [L6]

A summer industry internship project shall be carried out by a group of students consisting of 1 or 2 to 3 in number during the summer second-year semester at industries or institutions. This work shall be carried out under the guidance of the faculty assigned as an internal guide as well as an external guide at the industry where students are carrying out summer industry internship projects. The project shall consist of the design, fabrication, software development, or building of a prototype or application app. This can also be of an interdisciplinary nature.

There will be 100 marks in total with 40 marks for internal evaluation and 60 marks for external

The internal evaluation shall consist of:

|                      |           |
|----------------------|-----------|
| Presentation:        | 15 marks  |
| Internal Evaluation: | 15 marks  |
| Report:              | 10 marks  |
|                      | -----     |
|                      | 40 marks  |
|                      | -----     |
| End examination:     | 60 Marks. |

External Evaluation of the project (viva voce) shall be conducted by a committee appointed by the Chief Superintendent. The end examination will be carried out by a committee consisting of an external examiner, the head of the department, a senior faculty member, and the internal guide.

|                            |          |
|----------------------------|----------|
| Presentation:              | 10 marks |
| Demonstration/ Technology: | 40 marks |
| Report:                    | 10 marks |
|                            | -----    |
|                            | 60 marks |

# Syllabus for B.Tech (CSE) III YEAR II SEM

## Basics of Entrepreneurship

### OPEN ELECTIVE – I

**Code: 9ZC22**

**Prerequisite:** Business Economics and Financial Analysis

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

#### Course Objective:

Understand the nature of Entrepreneurship and its importance to business to the engineering students, which will allow them to get the required intuition and interest in starting their own start-up's.

#### Course Outcomes:

After completion of this course student will be able to:

1. Demonstrate the basic knowledge and skill set required for entrepreneurship.[L2]-U1
2. Distinguish business models and their validation in entrepreneurship.[L4]-U2,U3
3. Examine cost and financial structures and decide suitable pricing strategies.[L4]-U3
4. Relate team building and project management styles to project management and entrepreneurship.[L2]-U4,U5
5. Identify different marketing strategies and understand business regulations for start-ups.[L3]-U6

#### UNIT I

**Introduction to Entrepreneurship & Self Discovery:** - Define Entrepreneurship, Entrepreneurship as a Career option, Find your Flow, Stock of Your Means, Characteristics, Qualities and Skills of Entrepreneurship, Effectuation, Principles of Effectuation, Life as an Entrepreneur, Stories of Successful Entrepreneurs.

#### UNIT II

**Opportunity & Customer Analysis:** - Identify your Entrepreneurial Style, Methods of finding and understanding Customer Problems, Run Problem Interview, Process of Design Thinking, Identify Potential Problems worth Solving, Customer Segmentation, Niche Marketing and Targeting, Craft your Values Proportions, Customer-driven Innovation.

#### UNIT III

**Business Model & Validation:** - Introduction to Business Models, Lean approach to Business Model Canvas, Blue and Red Ocean Strategies, the Problem-Solution Fit, Build your Solution Demo, Solution Interview Method, Identify Minimum Viable Product (MVP), Product-Market fit test.

#### UNIT IV

**Economics & Financial Analysis:** - Revenue Analysis, Identify different Revenue Streams and Costs Analysis – Startup Cost, Fixed Cost and Variable Cost, Break Even Analysis, Profit Analysis, Introduction to Pricing, different Pricing Strategies, Sources of Finance, Bootstrapping and Initial Financing, Practice pitching to Investors and Corporate.

#### UNIT V

**Team Building & Project Management:** - Leadership Styles, Shared Leadership Model, Team Building in Venture, Roles and Responsibilities of team in venture, Explore collaboration tools and

techniques, Brainstorming, Introduction to Project Management, Project Life Cycle, Create a Project Plan.

## **UNIT VI**

**Marketing & Business Regulations:** - Positioning, Positioning Strategies, Branding, Branding Strategies, Selecting and Measuring Channels , Customer Acquisition, Selling Process, Selling Skills, Sales Plans. Business regulations – List of Required Registrations, Compliance Check List, Business Structures and Legal Entities.

### **TEXT BOOKS:**

1. Robert D Hisrich, Michael P Peters, Dean A Shepherd, Entrepreneurship, Sixth Edition, New Delhi, 2006.
2. Thomas W. Zimmerer, Norman M. Scarborough, Essentials of Entrepreneurship And Small Business Management, Fourth Edition, Pearson, New Delhi, 2006
3. Alfred E. Osborne, Entrepreneur's Toolkit, Harvard Business Essentials, HBS Press, USA, 2005.
4. Madhurima Lall, ShikhaSahai, Entrepreneurship, Excel Books, First Edition, New Delhi, 2006.

### **REFERENCES:**

1. S.S. Khanka, Entrepreneurial Development, S. Chand and Company Limited, New Delhi, 2007.
2. H. Nandan, Fundamentals of Entrepreneurship, Prentice Hall of India, First Edition, New Delhi, 2007.
3. S.R. Bhowmik, M. Bhowmik, Entrepreneurship-A tool for Economic Growth And A key to Business Success, New Age International Publishers, First Edition, (formerly Wiley Eastern Limited), New Delhi, 2007.
4. <https://www.wfglobal.org/>
5. <https://www.learnwise.org/#/IN/en/home/login>

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**Banking Operations and Insurance**  
**OPEN ELECTIVE – I**

**Code: 9ZC05**

**Prerequisite:** Business Economics and Financial Analysis

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Course Objective:**

Understand the concepts and principles of Indian Banking and Insurance Business and the role of RBI in regulating the Indian Financial System.

**Course Outcomes:**

1. Describe the Indian Banking System and banking regulation.[L1]-U1,U2
2. Explain the role of RBI as a regulator of Indian Banking.[L2]-U3
3. Describe the new dimensions and products served by the banking system in INDIA.[L1]-U4
4. Examine the role of Insurance industry, its regulatory and legal frame work of IRDA.[L3]-U5,U6

**UNIT I**

**INTRODUCTION TO BANKING BUSINESS:** Introduction to banking services - History of banking business in India, Structure of Indian banking system: Types of accounts, advances and deposits in a bank, KYC norms.

**UNIT II**

**BANKING REGULATIONS:** Banking Sector Reforms with special reference to Prudential Norms, Capital Adequacy Norms, Classification of Assets and NPA's, Functions of RBI, Role of RBI in regulating Indian Banking, Banking Ombudsman scheme

**UNIT III**

**CREDIT CONTROL BY RBI:** Definition, Objectives of Credit Control, Quantitative methods of Credit Control by RBI: Bank Rate Policy, Open Market Operations, Variation of Reserve Ratio, Qualitative methods of Credit Control by RBI: Fixation of Margin Requirements, Regulation of consumer Credit, Rationing of Credit, Direct Action, Moral Suasion and Publicity

**UNIT IV**

**NEW DIMENSIONS IN BANKING:** Financial Inclusion – Micro finance, E-Banking: Mobile-Banking, Net Banking, Digital Banking, Artificial Intelligence in Banking, CIBIL Score, Negotiable Instruments: Cheque Truncation system.

**UNIT V**

**INTRODUCTION TO INSURANCE:** Introduction to insurance, Need and importance of Insurance, principles of Insurance, characteristics of insurance contract, types of insurance: Life insurance and its products, General Insurance and its variants.

## **UNIT VI**

**INSURANCE BUSINESS ENVIRONMENT:** Procedure for issuing an insurance policy – Nomination - Surrender Value - Policy Loans – Assignment - Revivals and Claim Settlement; Insurance as a tax mitigation tool, Role of IRDA in Insurance Regulation.

### **TEXT BOOKS:**

1. Varshney, P.N., Banking Law and Practice, Sultan Chand & Sons, New Delhi.
2. General Principles of Insurance Harding and Eantly
3. Mark S. Dorfman: Risk Management and Insurance, Pearson, 2009.

### **REFERENCES:**

1. Scott E. Harringam Gregory R. Nichanus: Risk Management & Insurance, TMH, 2009.
2. Geroge E. Rejda: Principles of risk Management & Insurance, 9/e, pearson Education. 2009.
3. G. Koteswar: Risk Management Insurance and Derivatives, Himalaya, 2008.



**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**INTRODUCTION TO ADDITIVE MANUFACTURING PROCESSES**  
**OPEN ELECTIVE – I**

**Code: 9BC51**

**Prerequisites:** Metal Cutting

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Course Objective:**

Learn the fundamental concepts of additive manufacturing processes, techniques involved and various applications of these technologies in relevant areas such as mechanical, Bio-medical, Aerospace, Electronics etc.

**Course Outcomes:**

After the completion of the course students will be able to:

1. Illustrate the additive manufacturing processes and their relationship with subtractive manufacturing.[L2]-U1
2. Demonstrate liquid and solid based rapid prototype processes with appropriate applications.[L2]-U2
3. Apply the principles of casting in Additive manufacturing systems.[L3]-U3
4. Develop Additive manufacturing programs using various open source softwares.[L3]-U4
5. Explain various applications of additive manufacturing using rapid tooling techniques.[L2]-U5,U6

**UNIT-I**

**Introduction:** Development of AM, Fundamentals of AM, Classification of AMS, Advantages, Standards on AM, Commonly used terms, AM process chain

**UNIT-II**

**Liquid-based Additive manufacturing Systems:** Stereo lithography Apparatus (SLA), process, working principle, photopolymers, photo polymerization, Layering technology, laser and laser scanning, Applications, Advantages and Disadvantages, 3D bio printing **Solid-based Additive manufacturing Systems:**, Laminated Object Manufacturing (LOM): process, working principle, Applications, Advantages and Disadvantages, Fused Deposition Modelling (FDM): working principle, Applications, Advantages and Disadvantages

**UNIT-III**

**Powder Based Additive manufacturing Systems:** Selective laser sintering (SLS): working principle, Applications, Advantages and Disadvantages, Colour Jet printing, working principle, Applications, Advantages and Disadvantages, **Build time calculations** – SLA, FDM, Problems

## UNIT-IV

**Additive manufacturing Data Formats:** STL Format, STL File Problems, Consequence of Building Valid and Invalid Tessellated Models, STL file Repairs: Generic Solution, Features of various AM software's like Magics, Mimics, Solid View, View Expert, 3 D View, Velocity 2, Rhino, STL View 3 Data Expert and 3 D doctor. **Design for AM** – Basic Principles and Practices

## UNIT-V

**Rapid Tooling:** Introduction to Rapid Tooling (RT), Conventional Tooling Vs RT, Need for RT. Rapid Tooling Classification, Spray Metal Deposition, Silicone rubber moulds, , Casting-Sand Casting ,Investment Casting, evaporative Casting

**Reverse engineering** – what is RE, Why use RE, RE Generic process, Overview of RE-Software and Hardware, CMMs-applications and types

## UNIT-VI

**Applications and examples :** Application - Material Relationship, Application in Design, Application in Engineering, Analysis and Planning, Aerospace Industry, Automotive Industry, Jewellery Industry, Coin Industry, Arts and Architecture. Medical and Bioengineering Applications: Planning and simulation of complex surgery, Customized Implants and Prosthesis, Design and Production of Medical Devices, Bionic ear, dentistry

## TEXT BOOKS:

1. Chua C.K., Leong K.F. and LIM C.S, Rapid prototyping; Principles and Applications, World Scientific Publications, Third Edition, 2010.
2. Reverse Engineering: An Industrial Perspective, Springer- Verlag, 2008. ISBN: 978-1-84628-855-5
3. Ian\_Gibson, David\_Rosen, Brent\_Stucker, AdditiveManufacturingTechnologies3D Printing, Rapid Prototyping, and Direct Digital Manufacturing, Springer
4. PaulF.Jacobs, Rapid Prototyping and Manufacturing ASME Press, 1996.

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**Fundamentals of Measurements and Instrumentation**  
**OPEN ELECTIVE – I**

**Code: 9AC44**

**Prerequisite: Nil**

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Course Objective:**

Understand the basic principles of all measuring instruments and in measurement of electrical and non-electrical parameters viz., Resistance, Inductance, Capacitance, voltage, current Power factor, Power, Energy, Strain, Temperature, Torque, Displacement etc. and the different types of electrical and non electrical transducers. It introduces the different signal analyzers and oscilloscopes.

**Course Outcomes:**

After completion of this course student will be able to:

1. Illustrate the required characteristics of an instrument and principle of operation of different types of instruments viz., PMMC, moving iron type of instruments. [L2]-U1
2. Explain the construction and principle of operation for Potential transformer, Current transformer, dynamometer type of Watt meter and Induction type of Energy meter. [L2]-U2,U3
3. Explain and formulate the different techniques to measure unknown values of Resistance, Inductance and Capacitance. [L6]-U4
4. Distinguish the principle of operation of Different type of digital voltmeters, wave analyzers, spectrum analyzers, Cathode ray Oscilloscope and various types of transducers. [L4]-U5,U6

**UNIT-I**

**MEASURING INSTRUMENTS- INSTRUMENT TRANSFORMERS:**

Significance of Measurement, static characteristic of system- Linearity, Sensitivity, Precision, Accuracy - Classification - Deflecting, Control and Damping torques, Ammeters and Voltmeters, PMMC, Moving iron type instruments, Expression for the Deflecting torque and Control torque, Errors and Compensations, Extension of range using Shunts and Series resistance.

**UNIT –II**

**INSTRUMENT TRANSFORMERS**

Introduction, advantages, burden of instrument transformer, Current Transformer - errors in current transformer, Effect of secondary open circuit, Potential transformer- errors in potential transformer, Testing of current transformers with silsbee's method.

Power Factor Meters: Type of P.F. Meters, Dynamometer and Moving iron type, 1- ph and 3-ph meters.

### **UNIT –III**

#### **MEASUREMENT OF POWER& ENERGY:**

Single phase dynamometer wattmeter-LPF and UPF-Double element and three element dynamometer wattmeter, Expression for deflecting and control torques, Extension of range of wattmeter using instrument transformers, Measurement of active and reactive powers in balanced and unbalanced systems, Single phase induction type energy meter, Driving and braking torques, Testing by phantom loading, Three phase energy meter .

### **UNIT - IV**

#### **MEASUREMENT OF RESISTANCE - MAGNETIC MEASUREMENTS- A.C. BRIDGES:**

Principle and operation of D.C. Crompton's potentiometer, Standardization, Measurement of unknown resistance, current, voltage. Method of measuring low- Medium and High resistance, sensitivity of Wheatstone's bridge, Carey Foster's bridge, Kelvin's double bridge for measuring low resistance, Measurement of high resistance, loss of charge method, Measurement of inductance, Quality Factor, Maxwell's bridge, Hay's bridge, Anderson's bridge, Owen's bridge. Measurement of capacitance and loss angle, Desauty Bridge, Wien's bridge, Schering Bridge.

### **UNIT-V**

#### **DIGITAL VOLTMETERS- SIGNAL ANALYZERS- CRO:**

Digital voltmeters, Successive approximation, Ramp, Dual slope integration continuous balance type, Wave Analyzers, Frequency selective analyzers, Heterodyne, Application of Wave analyzers, Harmonic Analyzers, Total Harmonic distortion, spectrum analyzers, Basic spectrum analyzers, Spectral displays, Q meter and RMS voltmeters . CRO- Cathode Ray Tube (CRT), Screens, Probes, Applications of CRO, Measurement of frequency and phase using CRO, Block diagram.

### **UNIT-VI**

#### **MEASUREMENT OF NON-ELECTRICAL QUANTITIES:**

Transducers - Classification of transducers, Advantages of Electrical transducers, Characteristics and choice of transducers, Principle operation of Resistor, Inductor, LVDT and Capacitor transducers, LVDT Applications, Strain gauge and its principle of operation, Gauge factor- Thermistors, Thermocouples, Piezo electric transducers, Photovoltaic, Photo conductive cells. Measurement of strain, Gauge Sensitivity, Displacement, Velocity, Acceleration, Force, Torque, Measurement of Temperature.

#### **TEXT BOOKS:**

1. Electrical Measurements and measuring Instruments – E.W. Golding and F.C. Widdis, 5th Edition, Wheeler Publishing.
2. Transducers and Instrumentation – D.V.S Murthy, Prentice Hall of India, 2nd Edition.
3. A course in Electrical and Electronic Measurements and Instrumentation - A.K. Sawhney, Dhanpatrai & Co. 18th Edition.

**REFERENCES:**

1. Measurements Systems, Applications and Design – D O Doebelin- Tata MC Graw-Hill.
2. Principles of Measurement and Instrumentation – A.S Morris, Pearson /Prentice Hall of India.
3. Electronic Instrumentation- H. S. Kalsi Tata MC Graw – Hill Edition, 3rd Edition.
4. Modern Electronic Instrumentation and Measurement techniques – A.D Helfrick and W.D.Cooper, Pearson/Prentice Hall of India.

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**Fundamentals of Digital Circuits and Microprocessors**  
**OPEN ELECTIVE – I**

**Code: 9CC36**

**Prerequisite:** Digital Electronics, Computer Organization

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Course Objective:**

Develop the skills for understanding the design of digital circuits, learn programming skills for 8086 Microprocessor and interfacing peripherals to it.

**Course Outcomes:**

After completion of this course student will be able to:

1. Apply Boolean algebra and K-maps to simplify the expressions.[L3]-U1
2. Design combinational and sequential circuits.[L4]-U2,U3
3. Explain the Architecture, Instructions of 8086 and write assembly language programs.[L2]-U4,U5
4. Explain interfacing of memory and I/O devices.[L2]-U6

**UNIT – I**

Number System and Boolean Algebra: Binary, decimal, octal, hexadecimal, weighted and un-weighted codes. Axiomatic definition of Boolean algebra, Binary operators, postulates of and theorems. Boolean addition, subtraction, 1's complement, 2's complement. Switching functions, Canonical forms and Standard forms, Simplification of switching functions using theorems. K-map representation, simplification of logic functions using K-map.

**UNIT - II**

Combinational Logic Design: Single output and multiple output combinational logic circuit design, Binary adders/subtractors, Encoder, Decoder, Multiplexer, Demultiplexer, Parity bit generator, Code-converters.

**UNIT - III**

Sequential circuits: Classification of sequential circuits, the clocked SR flip flop, J- K, T and D-types flip flops, triggering mechanism of flip-flops, flip-flop conversion, introduction to counters and registers

**UNIT - IV**

Architecture of 8086 Microprocessor: Memory segmentation, BIU and E.U General Purpose registers, 8086 flag register and function of 8086 Flags, Pin diagram of 8086-Minimum mode and maximum mode of operation.

**UNIT – V**

Instruction set of 8086: Addressing modes of 8086, Assembly directives, Simple programs. Assembly language programs: involving logical, Branch & Call instructions, sorting.

## **UNIT - VI**

Interfacing with 8086: Interfacing with RAM, ROM, 8255 PPI – Interfacing with key board, ADC and DAC Stepper Motor.

### **TEXT BOOKS:**

1. Digital design- Morris Mano –PHI, 2nd Edition.
2. ZviKohavi and Niraj K Jha -Switching & Finite Automata theory – Cambridge, 3rd Edition.
3. Microprocessors and interfacing – Douglas V. Hall, TMH, 2nd Edition, 1999.
4. Advanced microprocessor & Peripherals - A.K.Ray & K.M.Bhurchandi, TMH, 2000.

### **REFERENCES:**

1. Fletcher -An Engineering Approach to Digital Design – PHI.
2. Fundamentals of Logic Design, Roth, Kenny, Seventh Edition, Cengage Learning
3. R.P.Jain-Switching Theory and Logic Design- TMH Edition,2003.
4. CVS Rao -Switching Theory and Logic Design –Pearson Education, 2005
5. Micro computer systems, The 8086/8088 Family Architecture, Programming and Design – Y.Liu and G.A. Gibson, PHI, 2nd Edition.

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**Design Literacy and Design Thinking**  
**OPEN ELECTIVE – I**

**Code: 9ZC08**

**Prerequisite: Nil**

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Course Objective:**

Understand the fundamental concepts of design thinking, and to familiarize with product design process and to motivate the students to ideate new products and services.

**Course Outcomes:**

After completion of this course student will be able to:

1. Outline the inputs and concepts required for design thinking. [L1]-U1, U2
2. Illustrate the techniques of idea generation and phases of design thinking. [L4] – U3, U4
3. Construct the new product design process. [L3] – U4
4. Apply the various design thinking tools for service design. [L3] – U5
5. Demonstrate the case studies related to design thinking. [L2] – U6

**UNIT I**

Design Thinking – Introduction to Design thinking, Principles of design thinking, Benefits of design thinking, Applications of Design thinking, Social Innovation, Impact of Design thinking, Design thinking tools and techniques. Innovation and Design thinking.

**UNIT II**

Idea Generation: New Idea generation methods - Principles of Idea Generation, Techniques, Creativity thinking techniques and tools, types of creative thinking, select ideas from ideation methods.

**UNIT III**

Design Thinking Foundations: The Design Double Diamond: Discover-Define-Develop-Deliver, User-centric design approaches: Importance of user-centricity for design, Empathisation, Empathy Maps, Data collection from users and for users, Data Validation Responsible Innovation and Ethical Design:

**UNIT IV**

Product Design Process: Identification of opportunities, Problem Statement, Product planning, Characteristics of Successful product Development, New product development process, Stanford design thinking iterative model

**UNIT V**

Design Thinking for Service Design: Attributes of a good service design, service design tools – blueprint, customer journey mapping Identifying the user needs in a service-driven



economy; Process Flows and Customer Experience considerations for designing and improving services; 5 Whys; Service Delivery Pathways

## **UNIT VI**

Case Studies on Design thinking: Case 1: Arcturus IV by John E. Arnold, Case – 2: How can we make AI to make things better for humans. Case – 3: User Centered Helmet Design by Prof. B.K. Chakravarthy - Part 1 and Part 2; Case – 4: Challenges of Reaching a Million Users by Prof. Chetan Solanki and Prof Jayendran

### **TEXT BOOKS:**

1. Brown, T. (2008). Design thinking. Harvard business review, 86(6), 84.
2. “Innovation by Design”, Gerald H. (Gus) Gaynor, AMACOM {American Management Association}, NYC, 2002
3. Ansell, C., & Torfing, J. (2014). Collaboration and design: new tools for public innovation. In Public innovation through collaboration and design (pp. 19-36). Routledge.
4. Lewrick, M., & Link, P. (2015). Design thinking tools: Early insights accelerate marketer’s success. Marketing Review St. Gallen, 32(1), 40-51.

### **REFERENCES:**

1. Mæhlum, A. R. (2017). Extending the TILES Toolkit-from Ideation to Prototyping (Master's thesis, NTNU).
2. Norman, D. (2013). The design of everyday things: Revised and expanded edition. Basic books.
3. Design Thinking – A primer, Prof: Dr.Bala Ramadurai, Indian Institute of Technology, Madras.

### **WEBSITES:**

1. [www.smashingmagazine .com](http://www.smashingmagazine.com)
2. [www.ID](http://www.ID)

## Syllabus for B.Tech (CSE) III YEAR II SEM

### C# & .Net

#### (Professional Elective – II)

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Code:** 9FC11

**Prerequisite:** Object Oriented Programming through Java

#### Course Objective:

Understand .NET Interoperation services, learn Client and Server side architectures and programming technologies, .NET Remoting, also understand the significance of Web services, security, the importance of RESTful, SOAP, DISCO, and UDDI and Web Services.

#### Course Outcomes:

After completion of this course student will be able to:

1. Explain .Net Architecture and learn basic programming in C#, the object oriented programming concepts. [L2]-U1,U2
2. Describe advance features and enhance skills in writing windows applications, ADO.NET and ASP.NET. [L2]-U3
3. Identify various class libraries for different applications and data manipulation functions.[L3]-U4
4. Discuss advanced concepts in data connectivity, WPF, WCF and WWF with C# and .NET 4.5. [L2]-U4,U5
5. Develop distributed applications using .NET Framework.[L3]-U4,U6

#### UNIT I

##### INTRODUCTION TO C#

Introducing C#, Understanding .NET, overview of C#, Literals, Variables, DataTypes, Operators, checked and unchecked operators, Expressions, Branching, Looping, Methods, implicit and explicit casting, Constant, Arrays, Array Class, Array List, String, String Builder, Structure, Enumerations, boxing and unboxing.

#### UNIT II

##### OBJECT ORIENTED ASPECTS OF C#

Class, Objects, Constructors and its types, inheritance, properties, indexers, index overloading, polymorphism, sealed class and methods, interface, abstract class, abstract and interface, operator overloading, delegates, events, errors and exception, Threading.

#### UNIT III

##### APPLICATION DEVELOPMENT ON .NET

Building windows application, Creating our own window forms with events and controls, menu creation, inheriting window forms, SDI and MDI application, Dialog Box (Modal and Modeless), accessing data with ADO.NET, Dataset, typed dataset, Data Adapter, updating database using stored procedures, SQL Server with ADO.NET, handling exceptions, validating controls, windows application configuration.

#### UNIT IV

##### WEB BASED APPLICATION DEVELOPMENT ON .NET

Programming web application with web forms, ASP.NET introduction, working with XML and .NET, Creating Virtual Directory and Web Application, session management techniques,

web.config, web services.

#### **UNIT V**

**SQL Connection: Passing** datasets, returning datasets from web services, handling transaction, handling exceptions, returning exceptions from SQLServer

#### **UNIT VI**

##### **CLR AND .NET FRAME WORK**

Assemblies, Versioning, Attributes, reflection, viewing meta data, type discovery, reflection on type, marshalling, remoting, security.

##### **TEXTBOOKS:**

1. Herbert Schildt,—The Complete Reference : C# 4.0, TataMcGrawHill, 2012.
2. Christian Nageletal.—Professional C# 2012 with .NET4.5, Wiley India, 2012.

##### **REFERENCES:**

1. Andrew Troelsen ,—Pro C# 2010 and the .NET 4Platform, Fifth edition, APress,2010.
2. Ian Griffiths, Matthew Adams, Jesse Liberty, —Programming C# 4.0, Sixth Edition, O'Reilly,2010.

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**COMPUTER GRAPHICS**  
**(Professional Elective – II)**

**Code: 9FC12**

**Prerequisite:** Nil

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Explore the various primitive algorithms and the geometrical transformations useful in building computer animations.

**Course Outcomes:**

After completion of this course student will be able to:

1. Describe Computer Graphics systems, various visible surface determination algorithms.[L1]-U1
2. Explore 2D graphics algorithms using filled area primitives, clipping algorithms and geometric transformations. [L3]- U2,U3
3. Discuss Object Representation, Geometric transformations and Viewing techniques of 3D computer graphics.[L2]- U4,U5
4. Design the animation production pipeline and develop a short animation.[L3], L4,U6

**UNIT-I**

Introduction, Application areas of Computer Graphics, overview of graphics systems, video-display devices, raster-scan systems, random scan systems, graphics monitors and work stations and input devices. Output primitives : Points and lines, line drawing algorithms, mid-point circle and ellipse algorithms, Applications.

**UNIT-II**

Filled area primitives: Scan line polygon fill algorithm, boundary-fill and flood-fill algorithms  
2-D geometrical transforms: Translation, scaling, rotation, reflection and shear transformations, matrix representations and homogeneous coordinates, composite transforms, transformations between coordinate systems.

**UNIT- III**

2-D viewing: The viewing pipeline, viewing coordinate reference frame, window to view-port coordinate transformation, viewing functions, Cohen-Sutherland and Cyrus-beck line clipping algorithms, Sutherland – Hodgeman polygon clipping algorithm

**UNIT-IV**

3-D Object Representation: Polygon surfaces, quadric surfaces, spline representation, Bezier curve and B-Spline curves, polygon rendering methods. 3-D Geometric transformations: Translation, rotation, scaling, reflection and shear transformations, composite transformations.

**UNIT-V**

3-D viewing: Viewing pipeline, viewing coordinates, view volume and general projection transforms and clipping visible surface detection methods: Classification, back-face detection, depth-buffer, scan-line, depth sorting, BSP-tree methods, area sub-division and octree methods

## **UNIT-VI**

Computer animation: Design of animation sequence, general computer animation functions, raster animation, computer animation languages, key frame systems, motion specifications. (p.nos 604- 16 of text book -1, chapter 21 of text book-2).

### **TEXT BOOKS:**

1. “Computer Graphics C version”, Donald Hearn and M.Pauline Baker, Pearson Education.
2. “Computer Graphics Principles & practice”, second edition in C, Foley, VanDam, Feiner and Hughes, Pearson Education.

### **REFERENCES:**

1. “Computer Graphics”, second Edition, Donald Hearn and M.Pauline Baker, PHI/Pearson Education.
2. “Computer Graphics Second edition”, Zhigandxiang, Roy Plastock, Schaum’s outlines, Tata Mc- Graw hill edition.
3. Procedural elements for Computer Graphics, David F Rogers, Tata Mc Graw hill, 2nd edition.
4. “Principles of Interactive Computer Graphics”, Neuman and Sproul, TMH.
5. Principles of Computer Graphics, ShaliniGovil, Pai, 2005, Springer.
6. Computer Graphics, Steven Harrington, TMH

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**Artificial Intelligence and Deep Learning**  
**(Professional Elective – II)**

**Code: 9EC13**

**Prerequisite: Nil**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Apply knowledge representation, reasoning, and machine learning techniques to model the solutions for real-world problems.

**Course Outcomes:**

After completion of this course student will be able to:

1. Identify different types of agents and their relationships with the environment using searching algorithms. [L3]-U1, U2
2. Use the different data Structures for knowledge representation and derive inferences in the AI systems. [L3]-U3
3. Appraise uncertain knowledge and reasoning concepts essential to model real world solutions.[L5]-U4
4. Apply the concepts and filters of Convolutional Neural Networks for training problem domain specific algorithms.[L3]-U5
5. Examine the decision making techniques of Deep Reinforcement learning and its applications in the games. [L3]-U6

**UNIT I**

**Introduction:** AI problems, Intelligent agents: Agents and Environments, Rationality, Nature of environments, Structure of agents, Problem solving agents, Problem formulation – Planning Application – Classical Planning problem

**UNIT II**

**Searching and Game Theory:** Searching for solutions, Searching with partial information (Heuristic search), Greedy best first search, A\* search Constraint Satisfaction problem -Game Playing: Adversarial search: Games, Minimax algorithm, Optimal decisions in multiplayer games, Alpha-Beta pruning, Evaluation functions. **Case studies:** Tic-tac-toe game

**UNIT III**

**Knowledge Representation and Reasoning:** Logical Agents, Knowledge Based Agents, Wumpus world, Propositional logic, Resolution patterns in propositional Logic, First order logic, Inference in first order logic, propositional vs. First order inference, Unification and Lifting, Forward chaining, Backward chaining, Resolution

**UNIT IV**

**Uncertain Knowledge and Reasoning :** Bayes Rule, Concepts of Time and Uncertainty, Utility Functions, Value of Information, Value iteration, Policy iteration, Partially Observable MDP

## **UNIT – V**

### **BASICS OF DEEP LEARNING**

Deep learning architectures: Convolutional Neural Networks : Neurons in Human Vision-The Shortcomings of Feature Selection - Full Description of the Convolutional Layer - Max Pooling-Full Architectural Description of Convolution Networks - Closing the Loop on MNIST with Convolutional Networks- -Building a Convolutional Network for CIFAR-10 - Visualizing Learning in Convolutional Networks- Leveraging Convolutional Filters to Replicate Artistic Styles-Learning Convolutional Filters for Other Problem Domains-Training algorithms.

## **UNIT VI**

### **DEEP REINFORCEMENT LEARNING:**

Deep Reinforcement Learning Masters Atari Games - Reinforcement Learning-Markov Decision Processes (MDP)-Explore Versus Exploit - Pole-Cart with Policy Gradients-Q-Learning and Deep Q-Networks-Improving and Moving Beyond DQN.

### **TEXT BOOKS:**

1. S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, Prentice Hall, Third Edition, 2009.
2. Artificial Intelligence, 3<sup>rd</sup> Edition, Patrick Henry Winston, Pearson Education, 1992.
3. Nikhil Buduma, Nicholas Locascio, “Fundamentals of Deep Learning: Designing Next-Generation Machine Intelligence Algorithms”, O'Reilly Media, 2017.
4. I am Good fellow, Yoshua Bengio, Aaron Courville, “Deep Learning (Adaptive Computation and Machine Learning series”, MIT Press, 2017.

### **REFERENCES:**

1. M. Tim Jones, —Artificial Intelligence: A Systems Approach(Computer Science), Jones and Bartlett Publishers, Inc.; First Edition, 2008
2. Nils J. Nilsson, —The Quest for Artificial Intelligence, Cambridge University Press, 2009.
3. William F. Clocksin and Christopher S. Mellish, Programming in Prolog: Using the ISO Standard, Fifth Edition, Springer, 2003.
4. Gerhard Weiss, —Multi Agent Systems, Second Edition, MIT Press, 2013.
5. David L. Poole and Alan K. Mackworth, —Artificial Intelligence: Foundations of Computational Agents, Cambridge University Press, 2010.
6. Bratko, —Prolog: Programming for Artificial Intelligence, Fourth edition, Addison-Wesley Educational Publishers Inc., 2011.

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**DISTRIBUTED DATABASES**  
**(Professional Elective – II)**

**Code: 9EC14**

**Prerequisites:** Database Management Systems

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Get proficiency in distributed DBMS architecture and design, query processing and optimization, distributed transaction management and reliability, parallel and object database management systems.

**Course Outcomes:**

After completion of this course student will be able to:

1. Explain the Architectural Models for Distributed database design, techniques and strategies used for designing distributed database. [L2]-U1
2. Evaluate simple strategies for executing a distributed query to select the optimum strategy that minimizes the amount of data transfer and describe different protocols used to deal with committing a data transaction that accesses databases stored on multiple nodes using query optimization algorithms. [L4]-U2,U3
3. Explain reliability, fault-tolerance, and failures in distributed systems and understand parallel database system architectures. [L2]-U4
4. Describe various concepts of parallel database management system.[L1]-U5
5. Design aspects of distributed object-oriented database system and related development programming. [L2]-U5,U6

**UNIT - I**

**Introduction** Distributed Data Processing, Distributed Database System, Promises of DDBSs, Problem areas.

**Distributed DBMS Architecture:** Architectural Models for Distributed DBMS, DDMBS Architecture.

**Distributed Database Design:** Alternative Design Strategies, Distribution Design issues, Fragmentation, Allocation.

**UNIT - II**

**Query processing and decomposition:** Query processing objectives, characterization of query processors, layers of query processing, query decomposition, localization of distributed data.

**Distributed query Optimization:** Query optimization, centralized query optimization, distributed query optimization algorithms.

**UNIT - III**

**Transaction Management:** Definition, properties of transaction, types of transactions, distributed concurrency control: serializability, concurrency control mechanisms & algorithms, time - stamped & optimistic concurrency control Algorithms, deadlock Management.



#### **UNIT - IV**

**Distributed DBMS Reliability:** Reliability concepts and measures, fault-tolerance in distributed systems, failures in Distributed DBMS, local & distributed reliability protocols, site failures and network partitioning.

#### **UNIT - V**

**Parallel Database Systems:** Parallel database system architectures, parallel data placement, parallel query processing, load balancing, database clusters.

**Distributed object Database Management Systems:** Fundamental object concepts and models, object distributed design,

#### **UNIT - VI**

**Distributed object Database Management Systems:** Architectural issues, object management, distributed object storage, object query Processing.

**Object Oriented Data Model:** Inheritance, object identity, persistent programming languages, persistence of objects, comparison OODBMS and ORDBMS

#### **TEXT BOOKS:**

1. M. Tamer OZSU and PatuckValduriez: Principles of Distributed Database Systems, Pearson
2. Edn.Asia, 2001.
3. Stefano Ceri and Giuseppe Pelagatti: Distributed Databases, McGraw Hill.

#### **REFERENCES:**

1. Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom: "Database Systems: The Complete Book", Second Edition, Pearson International Edition

**Syllabus for B.Tech (CSE) III YEAR II SEM  
SEMANTIC WEB & SOCIAL NETWORKS  
(Professional Elective – II)**

**Code: 9EC15**

**Prerequisite:** Nil

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Understand Web Intelligence and Ontology. Learn basics of Semantic web, its representation issues and Social Network Analysis.

**Course Outcomes:**

After completion of this course student will be able to:

1. Appraise and Outline the role of the Web, its need and Intelligence, concepts of Machine Intelligence Ontology, Inference engines, Software Agents, Berners-Lee www and Semantic Road Map. [L4]-U1,U2
2. Interpret the relationship between Ontologies, RDF, UML and XML Schema and apply semantic web standards (e.g., RDF, OWL) to annotate web content. [L3]-U3
3. Apply Ontology Engineering using Ontology Development Tools/ Methods, Ontology Libraries, Ontology Mapping, Logic and Inference Engines.[L3]-U4
4. Illustrate Semantic Web Applications, Services and Technology. [L3]-U5
5. Assess the impact of semantic web and social networks on various domains such as e-commerce, health care or education. [L6]-U6

**UNIT I**

**Web Intelligence:** Thinking and Intelligent Web Applications, The Information Age, The World Wide Web, Limitations of Today's Web, The Next Generation Web

**UNIT II**

**Machine Intelligence:** Machine Intelligence, Artificial Intelligence, Ontology, Inference engines, Software Agents, Berners-Lee www, Semantic Road Map, Logic on the semantic Web.

**UNIT III**

**Knowledge Representation for the Semantic Web:** Ontologies and their role in the semantic web, Ontologies Languages for the Semantic Web –Resource Description Framework (RDF) / RDF Schema, Ontology Web Language (OWL), UML, XML/XML Schema.

**UNIT IV**

**Ontology Engineering:** Ontology Engineering, Constructing Ontology, Ontology Development Tools, Ontology Methods, Ontology Sharing and Merging, Ontology Libraries and Ontology Mapping, Logic, Rule and Inference Engines.

## **UNIT V**

**Semantic Web Applications, Services and Technology:** Semantic Web applications and services, Semantic Search, e-learning, Semantic Bioinformatics, Knowledge Base ,XML Based Web Services, Creating an OWL-S Ontology for Web Services, Semantic Search Technology, Web Search Agents and Semantic Methods,

## **UNIT VI**

**Social Network Analysis and Semantic web:** Development of the social networks analysis, Electronic Sources for Network Analysis – Electronic Discussion networks, Blogs and Online Communities, Web Based Networks, Building Semantic Web Applications with social network features.

### **TEXT BOOKS:**

1. Thinking on the Web - Berners Lee, Godel and Turing, Wiley interscience, 2008.
2. Social Networks and the Semantic Web, Peter Mika, Springer, 2007.

### **REFERENCES:**

1. Semantic Web Technologies, Trends and Research in Ontology Based Systems, J.Davies, RudiStuder, PaulWarren, JohnWiley & Sons.
2. Semantic Web and Semantic Web Services -Liyang Lu Chapman and Hall/CRC Publishers (Taylor & Francis Group)
3. Information Sharing on the semantic Web - Heiner Stuckenschmidt; Frank Van Harmelen, Springer Publications.
4. Programming the Semantic Web, T. Segaran, C. Evans, J. Taylor, O'Reilly, SPD.
5. A Semantic Web Primer, G. Antoniou and V. Harmelen, PHI.

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**SOFTWARE PROJECT MANAGEMENT**  
**(Professional Elective – II)**

**Code: 9FC13**

**Prerequisite:** Software Engineering

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Explain building blocks of software projects and induces the essence of project management.

**Course Outcomes:**

After completion of this course student will be able to:

1. Explain primitives of Project Planning and evolution of software economics. [L2]-U1
2. Describe software economics; reduce Software product size, improvement in software processes, improving team effectiveness, improving automation, Achieving quality. [L1]-U1,U2
3. Illustrate Life cycle phases and artifacts of the process. [L2]-U3
4. Demonstrate Model based software architectures and Work Flows. [L2]-U4
5. Apply Checkpoints for a process such as Major mile stones, Minor Milestones and apply work breakdown structures for a iterative process within cost and schedule. Describe Project Organizations and Responsibilities. [L3]-U5
6. Discuss Process Automation, Instrumentation and explain Future Software Project Management such as Modern Project Profiles and Next generation project management. [L2]-U6

**UNIT I:**

**Concept of Management:** Management Definition, Role and Responsibilities of Management, Management in Software Industry

**Types of Software Organizations:** Start-up companies, Independent Software Companies, Multi-National Software Companies.

**Conventional Software Management:** The waterfall model, conventional software Management performance.

**Evolution of Software Economics:** Software Economics, pragmatic software cost estimation.

**UNIT II:**

**Improving Software Economics:** Reducing Software product size, improving software processes, improving team effectiveness, improving automation, Achieving required quality, peer inspections.

**The old way and the new way:** The principles of conventional software Engineering, principles of modern software management, transitioning to an iterative process.

**UNIT III:**

**Life cycle phases:** Engineering and production stages, inception, Elaboration, construction, transition phases.

**Artifacts of the process:** The artifact sets, Management artifacts, Engineering artifacts, programmatic artifacts.

**UNIT IV:**

**Model based software architectures:** A Management perspective and technical perspective.

**Work Flows of the process:** Software process workflows, Iteration workflows.

**UNIT V:**

**Checkpoints of the process:** Major mile stones, Minor Milestones, Periodic status assessments. **Iterative Process Planning:** Work breakdown structures, planning guidelines, cost and schedule estimating, Iteration planning process, Pragmatic planning.

**Project Organizations and Responsibilities:** Line-of-Business Organizations, Project Organizations, evolution of Organizations.

**UNIT VI:**

**Process Automation:** Automation Building blocks, The Project Environment.

**Project Control and Process instrumentation:** The seven core Metrics, Management indicators, quality indicators, life cycle expectations,

**Future Software Project Management:** Modern Project Profiles, Next generation Software economics, modern process transitions.

**Case study:** The command center processing and display system – Replacement (CCPDS-R)

**TEXTBOOKS:**

1. Software Project Management, Walker Royce: Pearson Education, 2005.

**REFERENCES:**

1. Management Concepts and Practices, Tim Hannagan, FT Prentice Hall, 5th Edition
2. Software Project Management, Bob Hughes and Mike Cotterell: Tata McGraw-Hill Edition.
3. Software Project Management, Joel Henry, Pearson Education.
4. Software Project Management in practice, Pankaj Jalote, Pearson Education.2005

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**MOBILE APPLICATION DEVELOPMENT**  
**(Professional Elective – III)**

**Code: 9FC14**

**Prerequisite:** Operating Systems, Object Oriented Programming

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Develop applications with reasonable complexity on mobile platform.

**Course Outcomes:**

After completion of this course student will be able to:

1. Demonstrate understanding of Android OS design, features and components. [L2]-U1
2. Describe Android lifecycle and apply the knowledge of UI layouts, fragments and event handling for designing User interface. [L2, L3]- U2, U3
3. Develop applications using intents and broadcasts for activity management and notifications. [L3]- U4
4. Choose appropriate persistent storage mechanism in Android. [L3] - U5
5. Illustrate applications using alarms, internet resources, and location-based services [L2]-U6

**UNIT – I**

**Introduction to Android Operating System:** Android OS design and Features – Android development framework, SDK features, Installing and running applications on Android Studio, Creating AVDs, Types of Android applications, Best practices in Android programming, Android tools

**Android application components** – Android Manifest file, Externalizing resources like values, themes, layouts, Menus etc, Resources for different devices and languages, Runtime Configuration Changes

**UNIT - II**

**Android Application Lifecycle** – Activities, Activity lifecycle, activity states, monitoring state changes

**Android User Interface:** Measurements – Device and pixel density independent measuring units. **Layouts** – Linear, Relative, Grid and Table Layouts.

**UNIT - III**

**User Interface (UI)Components** – Editable and non editable Text Views, Buttons, Radio and Toggle Buttons, Checkboxes, Spinners, Dialog and pickers.

**Event Handling** – Handling clicks or changes of various UI components.

**Fragments** – Creating fragments, Lifecycle of fragments, Fragment states, Adding fragments to Activity, adding, removing and replacing fragments with fragment transactions, interfacing between fragments and Activities, Multi-screen Activities

## **UNIT – IV**

**Intents and Broadcasts: Intent** – Using intents to launch Activities, Explicitly starting new Activity, Implicit Intents, Passing data to Intents, Getting results from Activities, Native Actions, using Intent to dial a number or to send SMS

**Broadcast Receivers** – Using Intent filters to service implicit Intents, Resolving Intent filters, finding and using Intents received within an Activity

**Notifications** – Creating and Displaying notifications, Displaying Toasts

## **UNIT - V**

**Persistent Storage: Files** – Using application specific folders and files, creating files, reading data from files, listing contents of a directory Shared Preferences – Creating shared preferences, saving and retrieving data using Shared Preference

**Database** – Introduction to SQLite database, creating and opening a database, creating tables, inserting retrieving and deleting data, Registering Content Providers, Using content Providers (insert, delete, retrieve and update)

## **UNIT – VI**

**Advanced Topics: Alarms** – Creating and using alarms.

**Using Internet Resources** – Connecting to internet resource, using download manager

**Location Based Services** – Finding Current Location and showing location on the Map, updating location

## **TEXT BOOKS:**

1. Professional Android 4 Application Development, Reto Meier, Wiley India, (Wrox) , 2012
2. Android Application Development for Java Programmers, James C Sheusi, Cengage Learning, 2013

## **REFERENCES:**

1. Beginning Android 4 Application Development, Wei-Meng Lee, Wiley India (Wrox).

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**PATTERN RECOGNITION**  
**(Professional Elective – III)**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Code:** 9EC21

**Prerequisites:** Data Structures, Database Management Systems, Probability and Statistics

**Course Objective:**

Learn fundamental concepts, theories, and algorithms for pattern recognition and machine learning.

**Course Outcomes:**

After completion of this course student will be able to:

1. Outline basic concepts of Paradigms pattern recognition.[L4]-U1
2. Classify decision-making algorithms in pattern recognition, efficient algorithm and classification. [L2]-U2
3. Compare and contrast different Hidden Markov Models, decision trees and perform overfitting with examples.[L2]-U3
4. Apply Linear Discriminant Functions for classification of SVM and constructing Ensembles of Classifiers.[L3]-U4
5. Apply Ensemble Hierarchical and Partition clustering techniques in pattern recognition applications.[L3]-U5
6. Analyze preprocessing and Recognition techniques in pattern recognition.[L4]-U6

**UNIT - I**

**Introduction:** What is Pattern Recognition, Data Sets for Pattern Recognition, Different Paradigms for Pattern Recognition. Representation: Data Structures for Pattern Representation, Representation of Clusters, Proximity Measures, Size of Patterns, Abstractions of the Data Set, Feature Extraction, Feature Selection, Evaluation of Classifier, Evaluation of Clustering.

**UNIT - II**

Nearest Neighbour Based Classifier: Nearest Neighbour Algorithm, Variants of the NN Algorithm use of the Nearest Neighbour Algorithm for Transaction Databases, Efficient Algorithms, Data Reduction, Prototype Selection. Bayes Classifier: Bayes Theorem, Minimum Error Rate Classifier, Estimation of Probabilities, Comparison with the NNC, Naïve Bayes Classifier, Bayesian Belief Network.

**UNIT - III**

Hidden Markov Models: Markov Models for Classification, Hidden Markov Models, Classification using HMMs. Decision Trees: Introduction, Decision Tree for Pattern Classification, Construction of Decision Trees, Splitting at the Nodes, Overfitting and Pruning, Examples of Decision Tree Induction.



**UNIT - IV**

Support Vector Machines: Introduction, Learning the Linear Discriminant Functions, Neural Networks, SVM for Classification. Combination of Classifiers: Introduction, Methods for Constructing Ensembles of Classifiers, Methods for Combining Classifiers.

**UNIT - V**

Clustering: Why is Clustering Important, Hierarchical Algorithms, Partitional Clustering, Clustering Large Data Sets.

**UNIT – VI**

An Application-Hand Written Digit Recognition: Description of the Digit Data, Preprocessing of Data, Classification Algorithms, Selection of Representative Patterns, Results.

**TEXT BOOKS:**

1. Pattern Recognition: An Algorithmic Approach: Murty, M. Narasimha, Devi, V. Susheela, Springer Pub, 1st Ed.

**REFERENCES:**

1. Machine Learning - Mc Graw Hill, Tom M. Mitchell.
2. Fundamentals Of Speech Recognition: Lawrence Rabiner and Biing- Hwang Juang. Prentice-Hall Pub.

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**Big Data Analytics**  
**(Professional Elective – III)**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Code: 9FC15**

**Prerequisites:** Database Management Systems

**Course Objective:**

Explore the fundamental techniques and principles in achieving big data analytics with stream processing.

**Course Outcomes:**

After completion of this course student will be able to:

1. Comprehend the fundamentals of big data analytics using Hadoop to solve the real life problems. [L2]-U1
2. Outline the concepts of map reduce, Hive in big data environment and differentiate NoSQL and SQL databases. [L2, L4]-U2,U3
3. Develop the algorithms to process big data using Apache Spark Low Level API. [L3]-U4,U5
4. Make use of Stream Processing techniques to develop social media applications. [L3]-U6

**UNIT– I:**

Introduction to Big Data: Big Data Analytics, Characteristics of Big Data – The Four Vs, importance of Big Data, Different Use cases, Data-Structured, Semi-Structured, Un-Structured

Introduction to Hadoop and its use in solving big data problems. Comparison Hadoop with RDBMS, Brief history of Hadoop, Apache Hadoop EcoSystem, Components of Hadoop, The Hadoop Distributed File System (HDFS);, Architecture and design of HDFS in detail, Working with HDFS (Commands)

**UNIT-II**

Anatomy of Hadoop map-reduce (Input Splits, map phase, shuffle, sort, combiner, reduce phase) (theory)

Hive: Introduction to Hive, data types and file formats, HiveQL data definition(Creating Databases and Tables), HiveQL for Data loading, HiveQL data manipulation, Logical joins, Window functions, Optimization, Table partitioning, Bucketing, Indexing, Join Strategies.

**UNIT-III**

SQOOP: Introduction to SQOOP, SQOOP imports: From Database to HDFS/Hive, SQOOP exports: From HDFS/Hive to Database, Incremental imports

NoSQL & HBase: Overview, HBase architecture, CRUD operations

**UNIT-IV**

SPARK Basics: History of Spark, Spark Architecture, Spark Shell, working with RDDs in Spark: RDD Basics, Creating RDDs in Spark. RDD Operations. Passing Functions to Spark, Transformations and Actions in Spark, Spark RDD Persistence

Working with Key/Value Pairs: Pair RDDs, Transformations on Pair RDDs, Actions

Available on Pair RDDs

### **UNIT-V**

Structured API : DataFrames, SQL : Overview of Structured Spark Types, Schemas, Columns and Expressions, DataFrame Transformations, Working with different types of data, Aggregations- Aggregation Functions, Grouping, User-Defined Aggregation Functions, Joins-Inner Joins, Outer Joins, Processing CSV Files, JSON Files, Text Files and Parquet Files, Spark SQL

### **UNIT-VI**

Spark streaming: Stream Processing Fundamentals, Structured Streaming Basics - Core Concepts, Structured Streaming in Action, Transformations on Streams, Input and Output (Kafka)

Case study: Twitter Stream processing application

### **TEXT BOOKS:**

1. Tom White, "Hadoop: The Definitive Guide", Third Edition, O'Reilley, 2012
2. SPARK: The Definitive Guide, Bill Chambers & Matei Zaharia, O'Reilley, 2018 Edition

### **REFERENCES:**

1. "Hadoop Operations", O'Reilley, Eric Sammer, 2012
2. "Programming Hive", O'Reilley, E. Capriolo, D. Wampler, and J. Rutherglen, 2012
3. "HBase: The Definitive Guide", O'Reilley, Lars George, 2011
4. Big Data, Big Analytics: Emerging, Michael Minnelli, Michelle Chambers, and Ambiga Dhiraj

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**Introduction to Internet of Things**  
**(Professional Elective – III)**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Code: 9IC45**

**Prerequisite:** Computer Networks

**Course Objective:**

Describe the concepts of various transform (IoT) and develop IoT tools using python and explore about potential businesses opportunities.

**Course Outcomes:**

After completion of this course student will be able to:

1. Summarize the fundamental blocks of Internet of Things. [L2]-U1
2. Distinguish IoT and M2M network softwares. [L2]-U2
3. Compare and apply protocols in wireless sensor network [L2, L3]-U3
4. Design IoT applications in different domains and analyze their performance. [L4,L6]-U5
5. Develop and assess basic IoT applications on embedded platform using python. [L6, L5]-U6

**UNIT – I**

**Introduction to IoT:** Defining IoT, Characteristics of IoT, Physical design of IoT, Logical design of IoT, Functional blocks of IoT, Communication models & APIs

**UNIT – II**

**IoT& M2M:** Machine to Machine, Difference between IoT and M2M, Software define Network

**Unit – III**

**Network & Communication aspects:** Wireless medium access issues, MAC protocol survey, Survey routing protocols, Sensor deployment & Node discovery, Data aggregation & dissemination

**Unit – IV**

**Challenges in IoT:** Design challenges, Development challenges, Security challenges, Other challenges

**Unit – V**

**Domain specific applications of IoT:** Home automation, Industry applications, Surveillance applications, Other IoT applications.

**Human Body and IoT:** Human Sensors, Human sensors with signal transmission, Case study on Working of human sensors , Mapping of human sensors with IOT sensors

## **Unit – VI**

**Developing IoT's:** Introduction to Python, Introduction to different IoT tools, Developing applications through IoT tools, Developing sensor based application through embedded system platform, Implementing IoT concepts with python

### **REFERENCES:**

1. Vijay Madisetti, Arshdeep Bahga, “Internet of Things: A Hands-On Approach”
2. Waltenegus Dargie, Christian Poellabauer, "Fundamentals of Wireless Sensor Networks: Theory and Practice

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**ADVANCED OPERATING SYSTEMS**  
**(Professional Elective – III)**

**Code: 9EC17**

**Prerequisite: Operating Systems**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Design and implement fundamental aspects of advanced operating systems including mutual exclusion, deadlocks, multiprocessing and memory coherence.

**Course Outcomes:**

After completion of this course student will be able to:

1. Analyse the various design approaches of advanced operating systems.[L4]-U1
2. Classify Non-Token Based and Token Based Algorithm's used in mutual exclusion.[L2]-U2
3. Solve issues related to Deadlock, examine design issues of multi-processor systems.[L3]-U3,U4
4. Explain and implement Distributed File System in operating systems.[L2,L3]-U5
5. Illustrate design issues in terms of Task Migration, DSM and memory coherence in Distributed Scheduling,.[L2]-U6

**UNIT – I**

Architectures of Distributed Systems: System Architecture Types, Distributed Operating Systems, Issues in Distributed Operating Systems, Communication Primitives. Theoretical Foundations: Inherent Limitations of a Distributed System, Lamport's Logical Clocks, Vector Clocks, Causal Ordering of Messages, Termination Detection.

**UNIT - II**

Distributed Mutual Exclusion: The Classification of Mutual Exclusion Algorithms, Non-Token – Based Algorithms: Lamport's Algorithm, The Ricart - Agrawala Algorithm, Maekawa's Algorithm, Token-Based Algorithms: Suzuki-Kasami's Broadcast Algorithm, Singhal's Heuristic Algorithm, Raymond's Heuristic Algorithm.

**UNIT - III**

Distributed Deadlock Detection: Preliminaries, Deadlock Handling Strategies in Distributed Systems, Issues in Deadlock Detection and Resolution, Distributed Deadlock Detection: Control Organizations for Distributed Deadlock Detection, Centralized- Deadlock – Detection Algorithms, Distributed Deadlock Detection Algorithms, Hierarchical Deadlock Detection Algorithms

**UNIT - IV**

Multiprocessor System Architectures: Introduction, Motivation for multiprocessor Systems, Basic Multiprocessor System Architectures Multi Processor Operating Systems: Introduction, Structures of Multiprocessor Operating Systems, Operating Design Issues, Threads, Process Synchronization, Processor Scheduling.

**UNIT - V**

Distributed File Systems: Architecture, Mechanisms for Building Distributed File Systems, Design Issues, Distributed Scheduling: Issues in Load Distributing, Components of a Load Distributed Algorithm, Stability, Load Distributing Algorithms, Requirements for Load Distributing

**UNIT - VI**

Distributed Scheduling: Task Migration, Issues in task Migration Distributed Shared Memory: Architecture and Motivation, Algorithms for Implementing DSM, Memory Coherence, Coherence Protocols, Design Issues

**TEXT BOOK:**

1. Advanced Concepts in Operating Systems, Mukesh Singhal, Niranjana G. Shivaratri, TataMcGraw - Hill Edition 2001

**REFERENCES:**

1. Distributed Systems: Andrew S. Tanenbaum, Maarten Van Steen, Pearson Prentice Hall, Edition – 2, 2007

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**Soft Computing**  
**(Professional Elective – III)**

**Code: 9EC18**

**Prerequisite:** Nil

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Familiarize with characteristics and applications of Soft Computing Techniques.

**Course Outcomes:**

After completion of this course student will be able to:

1. Identify the difference between Conventional Artificial Intelligence to Computational Intelligence. [L4]-U1
2. Describe fuzzy logic and reasoning to handle and solve engineering problems. [L2]-U2
3. Illustrate the classification and clustering techniques and advanced neural networks and its application. [L4]-U3,U4
4. Use techniques to build model for various applications. [L3]-U5
5. Identify various operations of genetic algorithms and Rough Sets. [L4]-U6

**UNIT - I**

Introduction to Soft Computing: Evolutionary Computing, "Soft" computing versus "Hard" computing, Soft Computing Methods, Recent Trends in Soft Computing, Characteristics of Soft computing, Applications of Soft Computing Techniques.

**UNIT-II**

Fuzzy Systems: Fuzzy Sets, Fuzzy Relations, Fuzzy Logic, Fuzzy Rule-Based Systems

**UNIT-III**

Fuzzy Decision Making, Particle Swarm Optimization

**UNIT-IV**

Genetic Algorithms: Basic Concepts, Basic Operators for Genetic Algorithms, Crossover and Mutation Properties.

**UNIT-V**

Genetic Algorithm Cycle, Fitness Function, Applications of Genetic Algorithm.

**UNIT-VI**

Rough Sets, Rule Induction, and Discernibility Matrix, Integration of Soft Computing Techniques.

**TEXT BOOK:**

1. Soft Computing – Advances and Applications - Jan 2015 by B.K. Tripathy and J. Anuradha –Cengage Learning



**REFERENCES:**

1. S. N. Sivanandam & S. N. Deepa, "Principles of Soft Computing", 2nd edition, Wiley India, 2008.
2. David E. Goldberg, "Genetic Algorithms-In Search, optimization and Machine learning", Pearson Education.
3. J. S. R. Jang, C.T. Sun and E.Mizutani, "Neuro-Fuzzy and Soft Computing", Pearson Education, 2004.
4. G.J. Klir & B. Yuan, "Fuzzy Sets & Fuzzy Logic", PHI, 1995.
5. Melanie Mitchell, "An Introduction to Genetic Algorithm", PHI, 1998.
6. Timothy J. Ross, "Fuzzy Logic with Engineering Applications", McGraw- Hill International editions, 1995

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**Computer Science and Engineering**  
**OPERATING SYSTEMS**

**Code: 9EC06**

**Prerequisite:** Computer Organization

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Learn working principles of Operating Systems.

**Course Outcomes:**

After completion of this course student will be able to:

1. Describe the functional architecture of an Operating System with usage of system calls.[L2]-U1
2. Analyze various process scheduling algorithms & pragmatics of scheduling algorithms used by various Operating Systems.[L4]-U2
3. Solve issues related to process synchronization and Interprocess Communication (IPC) in the Operating System.[L3]-U3
4. Illustrate the concepts of Memory Management and deadlock.[L3]-U4
5. Discuss the concepts of File System with regard to Directory and Disk Management Algorithms, summarize the aspects of I/O Systems, Protection and Security.[L2]-U5,U6

**UNIT I**

**Introduction to Operating System:** Definition, Functions of Operating Systems, **Types of Operating Systems:** Batch Operating System, Multiprogramming Operating System, Multiprocessing Operating System, Multitasking Operating System, Network Operating System, Real Time Operating System, Time-Sharing Operating System and Distributed Operating System.

**Computer System Architecture:** Four Components of a Computer System, **Operating System Services, System Calls:** Definition, Types of System Calls, **Operating System Structure:** Simple Structure, Layered Approach, **Threads:** Definition, Need of threads, types of threads.

**UNIT II**

**Process Management:** Process concept: Definition of Process, Process State Diagram, PCB (Process Control Block), CPU Switch from process to process, Context Switching, Process Vs Thread.

**Process Scheduling:** Process Queues: Job Queue, Ready Queue, I/O/Device/Waiting Queue, Representation of Process Scheduling, **Types of Process Schedulers:** Long-term, Short-term and Medium-Term Scheduler.

**CPU Scheduling:** Preemptive Scheduling, Non-Preemptive Scheduling, Scheduling Criteria, **Scheduling Algorithms (Preemptive and Non-Preemptive with Arrival Time):** First Come First Serve (FCFS), Shortest-Job-First (SJF), Shortest Remaining Time First (SRTF), Priority Scheduling, Round Robin (RR), Multilevel Queue Scheduling, Multilevel Feedback Queue Scheduling.

### UNIT III

**Process Synchronization** - The Critical Section Problem: Solution to Critical Section Problem (Software and Hardware), Synchronization Hardware, Classical Problems of Synchronization, Semaphores, Monitors.

**Interprocess Communication Mechanisms:** IPC between processes on a single computer system, IPC between processes on different systems, using pipes, FIFOs, message queues, shared memory.

### UNIT IV

**Deadlocks:** Definition, Necessary Conditions for Deadlock, **Methods For Handling Deadlocks:** Deadlock Prevention, Avoidance (Bankers Algorithm), Detection and Deadlock recovery.

**Memory Management:** Logical versus Physical Address Space, Swapping, Contiguous memory allocation, Paging and Segmentation techniques, Segmentation with paging, **Virtual Memory:** Demand Paging, **Page-Replacement Algorithms:** FIFO, LRU and Optimal, Thrashing.

### UNIT V

**File System:** Different types of files and their access methods, **Directory Structures:** Various file allocation methods - Contiguous, Linked with FAT and Indexed, **Disk Scheduling Algorithms:** FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK.

### UNIT VI

**I/O Systems:** I/O Hardware: Polling, Interrupts and DMA, **Protection:** Goals of Protection, Principles of Protection, Access Matrix, Implementation of Access Matrix, Access Control List, Capability List, Program threats.

### TEXT BOOKS:

1. Operating System Concepts by Silberchatz Galvin, 8<sup>th</sup> edition.
2. Modern Operating Systems by A. Tanenbaum, 1992, Prentice-Hall.
3. Operating Systems Internals and Design Principles by William Stallings, 4<sup>th</sup> edition, 2001, Prentice-Hall

### REFERENCES:

1. Operating System by Peterson, 1985, AW.
2. Operating System by Milankovic, 1990, TMH.
3. Operating System Incorporating With Unix & Windows By Colin Ritchie, 1974, TMH.
4. Operating Systems by Mandrik & Donovan, TMH
5. Operating Systems by Deitel, 1990, AWL.
6. Operating Systems – Advanced Concepts By Mukesh Singhal , N.G. Shivaratri, 2003, T.M.H

## Syllabus for B.Tech (CSE) III YEAR II SEM MACHINE LEARNING

**Code: 9LC03**

**Prerequisite:** Introduction to Data Science

| L | T | P/D | C |
|---|---|-----|---|
| 2 | 1 | 0   | 3 |

### **Course Objective:**

Enable the students with basic knowledge on the techniques to build an intellectual machine for making decisions behalf of humans, and covers the techniques on how to make learning by a model, how it can be evaluated, what are all different algorithms to construct a learning model.

### **Course Outcomes:**

At the end of this course, the student will be able to

1. Understand basic types of machine learning and apply linear models like perceptron and regression for classification tasks. [L2]-U1
2. Design and implement machine learning models like multi-layer perceptrons, RBF networks, and support vector machines for practical applications. [L5]-U2
3. Analyze decision trees, ensemble learning, and probabilistic models like k-means and Gaussian mixtures for various learning tasks. [L4]-U3
4. Examine dimensionality reduction techniques and evolutionary strategies such as genetic algorithms to improve model performance. [L4]-U4
5. Apply advanced concepts like graphical models (Bayesian networks, Hidden Markov models) and analytical learning techniques to solve complex problems. [L3]-U5, U6

### **UNIT-I**

**INTRODUCTION:** Learning– Types of Machine Learning – Supervised Learning – The Brain and the Neuron – Linear Discriminants: Definitions of Perceptron, Linear separability, Linear Regression.

Design a Learning System– Perspectives and Issues in Machine Learning – Concept Learning Task – Concept Learning as Search – Finding a Maximally Specific Hypothesis – Version Spaces and the Candidate Elimination Algorithm.

### **UNIT-II**

#### **LINEAR MODELS:**

Multi-layer Perceptron– Going Forwards – Going Backwards: Back Propagation Error – Multi-layer Perceptron in Practice – Examples of using the MLP – Overview – Deriving Back-Propagation – Radial Basis Functions and Splines – Concepts – RBF Network – Interpolations and Basis Functions – Support Vector Machines.

### **UNIT-III**

#### **TREE AND PROBABILISTIC MODELS:**

Learning with Trees– Decision Trees – Constructing Decision Trees – Classification and Regression Trees – Ensemble Learning – Boosting – Bagging – Different ways to Combine Classifiers – Basic Statistics – Gaussian Mixture Models – Nearest Neighbor Methods – Unsupervised Learning – K means Algorithms.

## **UNIT-IV**

### **DIMENSIONALITY REDUCTION AND EVOLUTIONARY MODELS:**

Dimensionality Reduction– Linear Discriminant Analysis – Principal Component Analysis – Factor Analysis – Independent Component Analysis – Locally Linear Embedding – Isomap – Least Squares Optimization – Evolutionary Learning – Genetic algorithms – Genetic Offspring: - Genetic Operators – Using Genetic Algorithms – Reinforcement Learning – Overview – Getting Lost Example.

## **UNIT-V**

### **GRAPHICAL MODELS:**

Markov Chain Monte Carlo Methods– Sampling – Proposal Distribution – Markov Chain Monte Carlo – Graphical Models – Bayesian Networks – Markov Random Fields – Hidden Markov Models – Tracking Methods.

## **UNIT – VI**

### **ANALYTICAL LEARNING**

Learning with perfect domain theory– Explanation based Learning – Inductive analytical approach to learning – KBANN algorithm.

### **TEXT BOOKS:**

1. Stephen Marsland, —Machine Learning – An Algorithmic Perspective, Second Edition, Chapman and Hall/CRC Machine Learning and Pattern Recognition Series, 2014.
2. Tom M Mitchell, —Machine Learning, First Edition, McGraw Hill Education, 2013.

### **REFERENCES:**

1. Peter Flach, —Machine Learning: The Art and Science of Algorithms that Make Sense of Data, First Edition, Cambridge University Press, 2012.
2. Jason Bell, —Machine learning – Hands on for Developers and Technical Professionals, First Edition, Wiley, 2014.
3. Ethem Alpaydin, —Introduction to Machine Learning 3e (Adaptive Computation and Machine Learning Series), Third Edition, MIT Press, 2014.

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**Computer Science and Engineering**  
**WEB TECHNOLOGIES**

**Code: 9FC06**

**Prerequisite:** Object-Oriented Programming concepts, Computer Networks

| L | T | P/D | C |
|---|---|-----|---|
| 2 | 1 | 0   | 3 |

**Course Objective:**

Impart the concepts of HTML5, Tailwind CSS, MERN stack and design web applications.

**Course Outcomes:**

After completion of this course student will be able to:

1. Demonstrate HTML5 concepts, CSS3 syntax and Tailwind CSS framework [L2]-U1
2. Illustrate and apply JavaScript: data types, functions, objects and advanced javascript concepts. [L2,L3]-U2
3. Discuss MERN components and Node modules.[L2]-U3
4. Build HTTP web server and explain REST API.[L3,L2]-U4
5. Develop Express framework application, connecting and accessing MongoDB database with Node. [L4]-U5
6. Illustrate ReactJS features, build and deploy react application.[L2,L3]-U6

**UNIT I**

**HTML 5:** Semantic Elements, Web storage API, HTTP status codes.

**CSS 3:** Syntax structure, types, box model, Grid, Flexbox. Responsive Web Design using Media Queries, use of viewport, Transition, Animation.

**CSS Framework:** Tailwind css

**UNIT II**

**JavaScript:** Introduction to JavaScript, data types, functions, Arrays, Objects, Regular expressions

**Advanced JavaScript concepts:** let, const, arrow functions, destructuring, spread, rest, Prototypal Inheritance, Closure, understanding callbacks, Promise, Async/await.

**UNIT III**

**Introduction to MERN:** What is MERN?, MERN components, Server-Less Hello World, Server setup.

**Node JS:** Introduction to Node.js, REPL, Node Modules: events, OS, HTTP, file i/o, environment variables, dot env

**UNIT IV**

**Web Servers:** client-server architecture, request-response objects, creating a basic HTTP server

**Rest API:** Introduction to REST APIs, HTTP verbs

**UNIT V**

**Express Framework:** Introduction to Express, Installation of Express, Create first Express application, application, request, and response objects, configuring an Express application, Rendering views, sessions, forms, file upload. Connecting to an SQL database

**MongoDB:** Introduction to MongoDB, connecting to a MongoDB instance with Node, Reading from MongoDB, Writing to MongoDB.

## **UNIT VI**

**Introduction to ReactJS:** History of Front – end libraries, Motivation for using React, Key differentiators (Virtual DOM, one – way binding), React Components, JSX, props hooks, state, events, effects, fetching data from API using fetch, form validations, React Router, building and deploying react application.

### **TEXTBOOKS:**

1. Beginning HTML, XHTML, CSS, and JavaScript, Jon Duckett, Wrox Publications, 2010
2. Pro MERN Stack, Full Stack Web App Development with Mongo, Express, React, and Node, Vasan Subramanian, 2<sup>nd</sup> Edition, A Press.

### **REFERENCES:**

1. E – resource: <https://nodejs.org/en/docs/>
2. E – resource : <https://reactjs.org/>
3. E – resource : <https://tailwindcss.com/>
4. E – resource : <https://expressjs.com/>
5. E – resource : <https://web.dev/learn/css>
6. E – resource : <https://web.dev/learn/html>

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**Operating Systems and Machine Learning Lab**

**Code:** 9EC65

**Prerequisite:** Programming Language

| L | T | P/D | C   |
|---|---|-----|-----|
| 0 | 0 | 3   | 1.5 |

**Operating Systems Lab**

**Course objective:**

Understand the design aspects of operating system concepts through simulation.

**Course outcomes:**

After completion of this course student will be able to:

1. Write a program to implement scheduling algorithms, Interprocess Communication, Deadlocks, Memory management and File allocation techniques.[L3]

**Exercises**

1. Simulate the following CPU scheduling algorithms (Preemptive, With Arrival Time)  
a) FCFS b) SJF c) Priority d) Round Robin
2. Write C programs to illustrate the following IPC mechanisms  
a) Pipes b) FIFOs c) Message Queues d) Shared Memory
3. Implement Bankers Algorithm for Deadlock Avoidance.
4. Develop programs for Paging Technique of memory management.
5. Simulate all page replacement algorithms  
a) FIFO b) LRU c) LFU
6. Simulate all file allocation strategies  
a) Sequential/Contiguous b) Indexed c) Linked

**Machine Learning Lab**

**Course Objective:**

Learn various machine learning techniques and able to demonstrate them using python.

**Course Outcomes:**

After completion of this course student will be able to:

1. Generalize complexity of Machine Learning algorithms and their limitations.[L2]
2. Apply common Machine Learning algorithms in practice and implement on their own.[L3]
3. Demonstrate experiments in Machine Learning using real-world data.[L3]



## List of Experiments

1. The probability that it is Friday and that a student is absent is 3 %. Since there are 5 school days in a week, the probability that it is Friday is 20 %. What is the probability that a student is absent given that today is Friday? Apply Baye's rule in python to get the result. (Ans: 15%)
2. Extract the data from database using python
3. Implement k-nearest neighbours classification using python
4. Given the following data, which specify classifications for nine combinations of VAR1 and VAR2 predict a classification for a case where VAR1=0.906 and VAR2=0.606, using the result of k-means clustering with 3 means (i.e., 3 centroids)

VAR1 VAR2 CLASS

```
1.713 1.586 0
0.180 1.786 1
0.353 1.240 1
0.940 1.566 0
1.486 0.759 1
1.266 1.106 0
1.540 0.419 1
0.459 1.799 1
0.773 0.186 1
```

5. The following training examples map descriptions of individuals onto high, medium and low credit-worthiness.

medium skiing design single twenties no ->highRisk

high golf trading married forties yes ->lowRisk

low speedway transport married thirties yes ->medRisk

medium football banking single thirties yes ->lowRisk

high flying media married fifties yes ->highRisk

low football security single twenties no ->medRisk

medium golf media single thirties yes ->medRisk

medium golf transport married forties yes ->lowRisk

high skiing banking single thirties yes ->highRisk

low golf unemployed married forties yes ->highRisk

Input attributes are (from left to right) income, recreation, job, status, age-group, home-owner.

Find the unconditional probability of 'golf' and the conditional probability of 'single' given 'medRisk' in the dataset?

6. Implement linear regression using python.
7. Implement Naïve Bayes theorem to classify the English text
8. Implement an algorithm to demonstrate the significance of genetic algorithm
9. Implement the finite words classification system using Back-propagation algorithm.
10. Implement Classification on a sample data set using SVM

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**Computer Science and Engineering**  
**Web Technologies Lab**

| L | T | P/D | C   |
|---|---|-----|-----|
| 0 | 0 | 3   | 1.5 |

**Code: 9FC66**

**Prerequisite:** Object Oriented Programming

**Course Objective:**

Apply HTML5, CSS3, JavaScript concepts and develop full-stack web applications using MERN stack.

**Course Outcomes:**

After completion of this course student will be able to:

1. Apply HTML5, CSS3, and JavaScript technologies through practical coding exercises. [L3]
2. Develop full-stack applications using web frameworks and modern databases. [L3]

**Note: Students can select any 3 Problem Statements from each Cycle**

**Cycle 1:**

**Problem Statement 1: Responsive Personal Portfolio Website (2 hours)**

Create a responsive personal portfolio website using HTML5 and CSS3. The website should include sections for your profile, skills, experience, projects, and contact information. Use Tailwind CSS to style the website and ensure it is responsive across different screen sizes.

**Problem Statement 2: E-commerce Product Page (3 hours)**

Design and develop an e-commerce product page using HTML5, CSS3, and Tailwind CSS. The page should display product information, including images, descriptions, prices, and options for adding to cart. Use Tailwind CSS to create a visually appealing and consistent layout.

**Problem Statement 3: Interactive Blog Post with Comments (3 hours)**

Create an interactive blog post with comments using HTML5, CSS3, and JavaScript. The blog post should include a title, author, content, and a comment section. Use JavaScript to enable users to submit comments and display them on the page.

**Problem Statement 4: Adaptive Landing Page for Different Devices (2 hours)**

Develop an adaptive landing page that adjusts its layout and content based on the user's device. Use HTML5, CSS3, and JavaScript to detect the device type and display the appropriate content. Employ media queries and responsive design techniques to ensure the page looks great on all devices.

**Cycle 2:**

**Problem Statement 1: Dynamically Generated Content with JavaScript (2 hours)**

Create a web page that dynamically generates content using JavaScript. The page should include a button that, when clicked, generates a new random number and displays it on the page. Use JavaScript to manipulate the Document Object Model (DOM) to add and remove elements.

**Problem Statement 2: Interactive Shopping Cart with JavaScript (3 hours)**

Develop an interactive shopping cart using JavaScript. The cart should allow users to add and remove items, update quantities, and calculate the total price. Utilize JavaScript arrays and objects to store product information and manage cart operations.

**Problem Statement 3: Regular Expression-Based Text Manipulation (3 hours)**

Build a web application that performs text manipulation using regular expressions. The application should allow users to enter a text string and provide options for search, replace, and formatting. Implement regular expression patterns to identify and modify specific text elements.

**Problem Statement 4: Asynchronous Data Fetching and Display with JavaScript Promises and Async/await (2 hours)**

Create a web page that fetches data from an API asynchronously using JavaScript promises and Async/await. The page should display a loading indicator while the data is being fetched and then render the data in a list or table. Demonstrate the use of promises to handle asynchronous operations and improve code readability.

**Cycle 3:**

**Problem Statement 1: Building a Simple REST API with MERN Stack (2 hours)**

Create a simple REST API using the MERN stack (MongoDB, Express.js, React.js, and Node.js). The API should allow users to perform basic CRUD (Create, Read, Update, Delete) operations on a data collection, such as a list of tasks or products. Design the API endpoints, implement the server-side logic using Node.js and Express.js, and connect the API to a MongoDB database.

**Problem Statement 2: Developing a Server-Side Application with Node.js and Express.js (3 hours)**

Build a server-side application using Node.js and Express.js. The application should provide an endpoint to accept user input, perform some processing, and return a response. Implement error handling and logging mechanisms to ensure the application runs reliably. Explore using Node.js modules for common tasks, such as file I/O and data validation.

**Problem Statement 3: Working with Environment Variables and Dotenv in Node.js Applications (2 hours)**

Create a Node.js application that utilizes environment variables and dotenv to manage sensitive configuration data. Implement dotenv to load environment variables from a .env file and use them throughout the application. Demonstrate how to access and update environment variables securely.

**Problem Statement 4: Implementing Event-Driven Programming with Node.js (3 hours)**

Develop an event-driven Node.js application that utilizes the Node.js EventEmitter API. Create custom events to represent different occurrences, such as user actions or sensor readings. Register event listeners to handle these events and perform appropriate actions. Explore using modules like async/await to manage asynchronous event handling.

**Cycle 4:**

**Problem Statement 1: Building a Simple Web Server with HTTP Request Handling (2 hours)**

Create a basic HTTP server using Node.js and Express.js. The server should handle GET and POST requests, respond with appropriate HTTP status codes, and parse request payloads. Implement route handling to direct requests to specific functions for processing.

**Problem Statement 2: Designing and Implementing a REST API for Resource Management (2 hours)**

Design a REST API for managing a collection of resources, such as books or products. Define the API endpoints for each resource operation (Create, Read, Update, Delete) and map them to HTTP verbs (POST, GET, PUT, DELETE). Implement the API using Node.js and Express.js, including error handling and validation checks.

**Cycle 5:**

**Problem Statement 1: Developing a Full-Stack Web Application with Express and MongoDB (2 hours)**

Create a full-stack web application using Express.js and MongoDB. The application should allow users to create, read, update, and delete (CRUD) data stored in a MongoDB database. Implement the Express framework to handle routing, request processing, and templating. Utilize MongoDB to store and retrieve data using the Node.js MongoDB driver.

**Problem Statement 2: Building a REST API with Express and MongoDB for User Management (3 hours)**

Design and implement a REST API using Express.js and MongoDB for user management. The API should allow users to register, login, update their profiles, and manage their data. Implement authentication and authorization mechanisms to secure user access. Utilize MongoDB to store user information and session data.

**Problem Statement 3: Creating a File Upload Application with Express and MongoDB (2 hours)**

Develop a file upload application using Express.js and MongoDB. The application should allow users to upload files, store them in MongoDB, and retrieve them later. Implement file handling techniques to ensure secure and efficient file uploads and storage. Utilize MongoDB to store file metadata and references.

**Problem Statement 4: Implementing Image Processing and Display with Express and MongoDB (3 hours)**

Build an image processing and display application using Express.js and MongoDB. The application should allow users to upload images, apply image processing filters, and display the processed images. Utilize image processing libraries to manipulate images and store processed images in MongoDB. Implement image rendering techniques to display images on the web page.

**Cycle 6:**

**Problem Statement 1: Building a Dynamic ReactJS Application with State Management (2 hours)**

Create a dynamic ReactJS application that manages state using hooks. The application should display a list of items and allow users to add, remove, and update items. Implement state management techniques to keep the UI in sync with data changes. Utilize hooks like useState and useEffect to handle state updates and side effects.

**Problem Statement 2: Developing a Data-Driven ReactJS Application with API Fetching (3 hours)**

Build a data-driven ReactJS application that fetches data from an API and renders it on the UI. The application should display a list of data items retrieved from the API and allow users to filter and search for items. Implement data fetching techniques using the fetch API or libraries like Axios. Utilize state management to store and update data from the API.

**Problem Statement 3: Creating a User Interface with React Router for Navigation (2 hours)**

Develop a user interface with React Router for navigation between different components and routes. The application should have multiple pages, such as a home page, a contact page, and an about page. Implement React Router components to handle routing and provide smooth transitions between pages. Utilize nested routes to organize complex navigation structures.

**Problem Statement 4: Building and Deploying a ReactJS Application to a Hosting Platform (3 hours)**

Build a complete ReactJS application and deploy it to a hosting platform like Netlify or Vercel. The application should have a functional UI, interact with APIs, and handle user input. Implement build tools like Webpack or Parcel to bundle and optimize the application. Configure the deployment environment and deploy the application to the chosen hosting platform.

**REFERENCES:**

1. E – resource: <https://nodejs.org/en/docs/>
2. E – resource : <https://reactjs.org/>
3. E – resource : <https://tailwindcss.com/>
4. E – resource : <https://expressjs.com/>
5. E – resource : <https://web.dev/learn/css>
6. E – resource : <https://web.dev/learn/html>

**Syllabus for B.Tech (CSE) III YEAR II SEM**  
**Comprehensive Viva Voce**

**Code: 9E686**

| <b>L</b> | <b>T</b> | <b>P</b> | <b>C</b> |
|----------|----------|----------|----------|
| <b>0</b> | <b>0</b> | <b>0</b> | <b>1</b> |

**Course Objective:**

Assess a student's overall knowledge of the field of engineering.

**Course Outcomes:**

After completion of this course, students will be able to:

1. Discuss and summarize Core, and advanced subjects that they have studied till the completion of that academic year.[L2]

Comprehensive Viva Voce will be conducted in the third year second semester for 100 marks.

Two Internal Exams (Oral) of 50 marks each will be considered for CIE. There will be no Semester End Exam for this course

A student must secure 40% of the total marks to obtain a pass grade.

**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**Advanced Entrepreneurship**  
**(Open Elective – II)**

**Code: 9ZC23**

**Prerequisite:** Basics of Entrepreneurship

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Course Objective:**

Impart the necessary managerial skills and tactics required for an emerging Entrepreneur for the Engineering students to enhance their prospects as an Entrepreneur.

**Course Outcomes:**

After completion of this course student will be able to:

1. Explains the students to gain knowledge on the need of revisiting business model and also learning about types of business models. [L2]-U1
2. Discovers the various types of business planning that are needed to implement the business model [L4]-U2
3. Identify various funding options for an entrepreneur and also the team members involved in it [L3]-U3
4. Explains the digital platforms and tools available for marketing and also for online storage of documents [L2]-U4
5. Evaluate the growth of the start-ups using performance and financial metrics and seeking support and final project[L5]-U5,U6

**UNIT I**

**Fundamentals of Entrepreneurship & Refining Business Model and Product:**

Fundamentals and key concepts of entrepreneurship, refining the business model, products and services, pivoting, types of business models, business model evolution, generating new business models, analyzing the business model, adding new customer segment, product manager, significance and role of product manager.

**UNIT II**

**Business Planning & Exploring Revenue:**

Business plan, sales plan, hiring sale team, people plan, financial planning, financial forecasting, create a procurement plan, negotiating role play, understanding primary revenue sources, exploring customer lifecycle for growth customers, exploring and identify secondary sources of revenue.

**UNIT III**

**Funding the Growth & Building the A-Team:**

Overview of funding, funding options for an entrepreneur, explore the right funding options, create funding plan, pitch deck, introduction to building A-Team, pitching to attract the talent, setting your team, defining roles, hiring the A-Team members.

**UNIT IV**

**Brand and Channel Strategy & Leveraging Technologies:**

Introduction to branding, drawn the venture's golden circle, positioning and positioning statements, creating brand name, logo, social media handle, Identify right channels, leaping ahead with technology, digital marketing for startups, plan a social media campaign, digital collaboration, store documents online, other technology platforms, make tech plan, platform wish list.

## **UNIT V**

### **Measuring Progress and Legal Matters:**

Metrics for customer acquisition (CAC, CLV, and ARPU), metrics for customer retention and satisfaction, find CAC, CLV and ARPU, key financial metrics, communicate metrics, new revenue stream through key financial metrics, re-forecasting of financial plan, identify professional help for legal and compliance requirements, searching of trademark and brand name and company name.

## **UNIT VI**

### **Seeking Support and Final Project:**

Mentors help to create successful startups, identify mentors and advisors, importance of mentors and advisors, scout the board of directors, overview on final project, capstone project presentation, contents of capstone project.

### **TEXTBOOKS:**

1. Entrepreneurship Rajeev Roy, oxford, 2012
2. Entrepreneurship Development Khanka, S.Chand 2012

### **REFERENCES:**

1. Small Scale industries and Entrepreneurship Vasanth Desai “Himalya publishing 2012
2. Robert Hisrich et al “entrepreneurship TMH 2012
3. Entrepreneurship Development Khanka, ,S.Chand 2012
4. Entrepreneurship Development B.Janikairam and M Rizwana
5. e-source: - [www.learnwise.org](http://www.learnwise.org)



**Syllabus for B.Tech (CSE) IV YEAR I SEM  
FINANCIAL MARKETS AND SERVICES  
(Open Elective – II)**

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Code: 9ZC15**

**Prerequisite:** Business Economics and Financial Analysis

**Course Objective:**

Understand Financial Markets and the Services offered in Indian Financial System within this framework.

**Course Outcomes:**

After completion of this course student will be able to:

1. Explain the structure of Indian Financial System and role of money market and its instruments.[L2]-U1,U2
2. Illustrate about Indian Capital Markets and Fund based financial services.[L2]-U3,U4
3. Demonstrate knowledge on Indian Mutual Fund Industry.[L3]-U5
4. Describe various fee based financial services provided in India.[L1]-U6

**UNIT I**

**INTRODUCTION:** Investment – Investment Attributes, Structure of Indian Financial System, Financial Markets, Classification of Financial Markets, Financial Sector reforms-1991.

**UNIT II**

**FINANCIAL AND SECURITIES MARKETS:** Role and functions of SEBI, Structure and functions of Call Money Market, Government Securities Market – T-bills Market, Commercial Bills Market, Commercial paper and Certificate of Deposits.

**UNIT III**

**CAPITAL MARKETS:** Securities Market – Organization and Structure, Listing, Trading and Settlement, Buying and Selling of shares, Stock Market Quotations, Stock Market Indices, SEBI and Regulation of Primary and Secondary Markets.

**UNIT IV**

**FUND BASED FINANCIAL SERVICES:** Lease Finance, Hire purchase Finance, Factoring - Definition, Functions, Advantages, Evaluation, Forfeiting, Bills Discounting, Venture Capital Financing.

**UNIT V**

**MUTUAL FUNDS:** Definition, Structure and functioning of Unit Trust of India and Mutual Funds, Types of Mutual Fund Schemes, Growth of Indian Mutual funds and their Regulation, Role of AMFI.

**UNIT VI**

**FEE BASED FINANCIAL SERVICES:** Stock Broking, Credit Rating, Underwriting, Banker to an Issue, Debenture Trustees and Portfolio managers, Challenges faced by Investment Bankers. Merchant Banking: Definition, Services of Merchant Banks.

**TEXTBOOKS:**

1. L.M. Bhole: Financial Institutions and Markets, TMH, 2009.
2. E. Gordon, K. Natarajan: Financial Markets and Services, Himalaya Publishing House, 2013.
3. Vasant Desai: Financial Markets and Financial Services, Himalaya, 2009
4. Pathak: Indian Financial Systems, Pearson, 2009
5. M.Y. Khan: Financial Services, TMH, 2009.

**REFERENCES:**

1. S. Gurusamy: Financial Services and System, Cengage, 2009
2. Justin Paul and Padmalatha Suresh: Management of Banking and Financial Services, Pearson, 2009.
3. Gomez, Financial Markets, Institutions and Financial Services, PHI, 2012.
4. R M Srivatsava: Dynamics of Financial Markets and Institutions in India, Excel, 2013.

**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**Principles of Operations Research**  
**(Open Elective – II)**

**Code: 9BC52**

**Prerequisite: Nil**

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Course Objective:**

Analyze different situations in the industrial/ business scenario involving limited resources and finding the optimal solution within constraints.

**Course Outcomes:**

After completion of this course student will be able to:

1. Build mathematical model (linear programming problem) for production, distribution of goods and economics.[L3]-U1
2. Solve the problem of transportation involving a large number of shipping routes with optimal transportation cost.[L3]-U2
3. Use Johnson's rule to create the optimal sequencing schedule for a sequencing problem and make decisions about replacing an item using replacement policy.[L3]-U3
4. Analyze the performance measures of Queuing system and Calculate the EOQ for minimizing the total inventory cost.[L4]-U4,U5
5. Apply simulation techniques for solving various types of problems and general idea development about Markov chains.[L3]-U6

**UNIT – I**

**INTRODUCTION:** Definition, Characteristics and Phases and Types of models, applications.

**LINEAR PROGRAMMING PROBLEM-** Formulation – Graphical solution, Simplex method-Types of variables, Unique and Multiple optimal solution, Redundancy & Degeneracy in LPP, Unbounded solution, Artificial variables techniques - Big-M method with feasible and infeasible solutions, Two-phase method, Primal to Dual formation with Duality Principle.

**UNIT – II**

**TRANSPORTATION PROBLEM** – Formulation – methods of finding initial solution (NW corner, VAM, Least cost Method) Optimal solution (Stepping stone Method, MODI method) Special cases in TP: unbalanced, Degeneracy, Restriction and maximization case.

**ASSIGNMENT PROBLEM** – Formulation – Optimal solution (Hungarian Method) - Variants of Assignment Problem-Unbalanced, Restriction, Maximization, Airlines layover case, Travelling Salesman problem.

**UNIT – III**

**SEQUENCING** – Introduction – Terminology, Assumptions, Johnson's procedure-Processing n jobs through two machines – Processing n jobs through three machines – Processing two jobs through 'm' machines (Gantt Chart).

**REPLACEMENT:** Introduction – Types of failure, Replacement of items that deteriorate with time – when money value is not counted and counted – Replacement of items that fail completely, Group replacement.

#### **UNIT – IV**

**THEORY OF GAMES:** Introduction and Terminologies, Criterion and optimal strategy – Solution of games with saddle points: Mixed Strategies-Rectangular games without saddle points, Dominance principle, Average Relational Dominance,  $m \times 2$  &  $2 \times n$  games - Graphical method and Sub Game Method, Matrix Method, Application of LPP in game theory.

#### **UNIT – V**

**WAITING LINES:** Introduction, Terminology, Structure of a queue, calling population characteristics-size, behaviour, pattern of arrivals, Kendall-Lee notation, Queuing Models: Single Channel: Poisson arrivals: exponential service times: with finite and infinite population, Multichannel: Poisson arrivals: exponential service times with infinite population

**INVENTORY :** Introduction, Inventory costs, Concept of EOQ, Single item Deterministic models with and without shortages, Single item inventory models with one price break and multiple price breaks, Stochastic models – Instantaneous demand and no set up cost.

#### **UNIT – VI**

**SIMULATION:** Definition – Types of simulation – phases of simulation– applications of simulation – Inventory and Queuing problems – Advantages and Disadvantages

Markov chains: Introduction to Markov chains, Analysis Assumptions, Input output probabilities, Applications (Only basic understanding)

#### **TEXT BOOKS:**

1. Operations research / Hira & Gupta
2. Operation Research /J.K.Sharma/Macmillan Publishers.

#### **REFERENCES:**

1. Quantitative Techniques in Management: N D Vohra, TMH

**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**FUNDAMENTALS OF RENEWABLE ENERGY SOURCES**  
**(Open Elective – II)**

**Code: 9AC45**

**Prerequisite: Nil**

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Course Objective:**

Familiar with solar energy, its radiation, Collection, storage and application and also gets introduced to other forms of Renewable Energy sources viz., the Wind energy, Biomass energy, geothermal energy and ocean energy.

**Course Outcomes:**

After completion of this course student will be able to:

1. Illustrate the role and potential of new and renewable energy sources, the potential of solar energy and use of different techniques for solar collection and storage.[L2]-U1,U2
2. Describe different types of horizontal, vertical axis wind mills and geothermal energy, ocean energy.[L2]-U3,U4,U5
3. Demonstrate the knowledge of Direct Energy Conversion in different phenomena viz., Joule Thomson effect, Seebeck effect, Peltier effect etc. and the principle of operation of Fuel Cells.[L3]-U6

**UNIT I**

**PRINCIPLES OF SOLAR RADIATION:**

Role and potential of new and renewable source, The solar energy option, Environmental impact of solar power, Physics of the sun, the solar constant, Extraterrestrial and terrestrial solar radiation, Solar radiation on tilted surface, Instruments for measuring solar radiation and sun shine, Solar radiation data.

**UNIT II**

**SOLAR ENERGY COLLECTION STORAGE AND APPLICATIONS:**

Flat plate and concentrating collectors, Classification of concentrating collectors, orientation and thermal analysis, advanced collectors.

Different methods, Sensible, Latent heat and stratified storage, solar ponds. Solar Applications- solar heating/cooling technique, solar distillation and drying, photovoltaic energy conversion.

**UNIT III**

**WIND ENERGY:**

Sources and potentials, Horizontal and vertical axis windmills, Performance characteristics, Betz criteria

BIO-MASS: Principles of Bio-Conversion, Anaerobic/aerobic digestion, Types of Bio-gas digesters, Gas yield, Combustion characteristics of bio-gas, Utilization for cooking, I.C.Engine operation and economic aspects.

**UNIT IV**

**GEO THERMAL ENERGY:**

Resources, types of wells, methods of harnessing the energy, Potential in India.

## **UNIT V**

### **OCEAN ENERGY:**

OTEC, Principles utilization, Setting of OTEC plants, Thermodynamic cycles. Tidal and wave energy, Potential and conversion techniques, Mini-hydel power plants and their economics.

## **UNIT VI**

### **DIRECT ENERGY CONVERSION:**

Need for DEC, Carnot cycle, Limitations, principles of DEC. Thermoelectric generators, seebeck, Peltier and joule Thomson effects, Figure of merit, materials, Applications, MHD generators, Principles, Dissociation and ionization, Hall effect, Magnetic flux, MHD accelerator, MHD Engine, Power generation systems, Electron gas dynamic conversion, economic aspects. Fuel cells – principles - Faraday's law's - Thermodynamic aspects - selection of fuels and operating conditions.

### **TEXT BOOKS:**

1. Non-Conventional Energy Sources - G.D. Rai
2. Renewable Energy Technologies - Ramesh & Kumar /Narosa.

### **REFERENCES:**

1. Renewable energy resources - Tiwari and Ghosal/ Narosa.
2. Non-Conventional Energy - Ashok V Desai /Wiley Eastern.
3. Non-Conventional Energy Systems - K Mittal /Wheeler.

**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**Fundamentals of Communication**  
**(Open Elective – II)**

**Code: 9CC37**

**Prerequisite:** Digital Electronics, Computer Networks

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Course Objective:**

Classify and describe the different types of technologies and their components for wired and wireless communication networks.

**Course Outcomes:**

After completion of this course student will be able to:

1. Illustrate various types of modulation techniques.[L2]-U1,U2
2. Demonstrate the various communication modules in radio, antenna and wave propagation.[L2]-U3
3. Explain basics of tele, satellite, optical, cellular and mobile communications techniques.[L2]-U4,U5,U6
4. Outline different wireless technologies.[L3]-U6

**UNIT - I**

**Introduction:** Need for Modulation, Frequency translation, Electromagnetic spectrum, Gain, Attenuation and decibels. Fundamentals of antenna and wave propagation.

**UNIT - II**

**Simple description on Modulation:** Analog Modulation-AM, FM, Pulse Modulation-PAM, PWM, PCM, Digital Modulation Techniques-ASK, FSK, PSK, QPSK modulation and demodulation schemes.

**UNIT - III**

**Radio System:**

Transmitter fundamentals, Power amplifier, and Typical transmitter circuit. Super heterodyne receiver, Typical receiver circuit and Noise.

**Antenna and Wave Propagation:**

Antenna fundamentals, commonly used antenna, wave propagation and transmission line.

**UNIT - IV**

**Telecommunication Systems:** Telephones Telephone system, Paging systems, Internet Telephony.

**Networking and Local Area Networks:** Network fundamentals, LAN hardware, Ethernet LANs, Token Ring LAN.

**UNIT - V**

**Satellite Communication:** Satellite Orbits, satellite communication systems, satellite subsystems, Ground Stations Satellite Applications, Global Positioning systems.

**Optical Communication:** Optical Principles, Optical Communication Systems, Fiber –Optic Cables, Optical Transmitters & Receivers, Wavelength Division Multiplexing.

## **UNIT - VI**

**Cellular and Mobile Communications:** Cellular telephone systems, AMPS, GSM, CDMA, and WCDMA.

**Wireless Technologies:** Wireless LAN, PANs and Bluetooth, Zig Bee and Mesh Wireless networks, Wimax and MANs, Infrared wireless, RFID communication, UWB.

### **TEXT BOOKS:**

1. Principles of Electronic Communication Systems, Louis E. Frenzel, 3e, McGraw Hill publications, 4th edition, 2016.
2. Electronic Communications systems, Kennedy, Davis 4e, MC GRAW HILL EDUCATION, 1999

### **REFERENCES:**

1. Theodore Rapp port, Wireless Communications - Principles and practice, Prentice Hall, 2002.
2. Roger L. Freeman, Fundamentals of Telecommunications, 2e, Wiley publications.
3. Introduction to data communications and networking, Wayne Tomasi, Pearson Education, 2005.



**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**Co-Creation and Product Design**  
**(Open Elective – II)**

**Code: 9ZC09**

**Prerequisite:** Intellectual Property Rights

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Course Objective:**

Understand the fundamental concepts of design thinking, and to familiarize with product design process and to motivate the students to ideate new products and services.

**Course Outcomes:**

After completion of this course student will be able to:

1. Outline the significance of Human Centered Design from customer perspective. [L1]-U1,U2
2. Examine the scope of emerging technologies in the field of product design. [L3]-U3
3. Design the product prototypes and develop the testing solutions. [L4]-U4
4. Determine the relevance of reverse engineering and ergonomics at work place. [L3]-U5
5. Apply the knowledge into practice with regard to IPR's.[L3]-U6

**UNIT I:**

**HUMAN CENTERED DESIGN:** Understanding user and Customer perspectives, Identify insights and opportunities, Interviewing, User Experience design. Frame your design challenge. Empathy tools and techniques.

**UNIT II:**

**IDEATION PROCESS:** Articulation of Problem Statement, Visualizing Ideas, Communicating ideas and compelling story telling, Brainstorming, Divergent thinking in exploring solutions, 3- box thinking, 3-box framework and Box-3 ideation.

**UNIT III:**

**EMERGING TECHNOLOGIES AND DESIGN:** Emerging technologies, utilization and growth, Automation through Industry 4.0, IOT for Network and Intelligent world, efficient and effective manufacturing aided by Robotics, Custom manufacturing by Additive / 3D printing, Augmented reality for product and process.

**UNIT IV:**

**PROTOTYPING:** Introduction to Prototype, types of prototype, prototyping strategies, Design consideration in the five stages of the product life cycle. Prototype building by different engineering disciplines. Testing Solution and taking the solution to the users. Create a pitch for your design.

**UNIT V:**

**REVERSE ENGINEERING IN PRODUCT DEVELOPMENT:** Reversing engineering methods, identifying the bad features in a product, reduction in size and weight, usage of new materials, importance of ergonomics in product development, environmental considerations in design, and safety considerations in design.

## **UNIT VI:**

**INTELLECTUAL PROPERTY RIGHTS:** Introduction to IPR, Patents – Types of Patents, elements of patentability, Patents registration Procedure, Patent office and Appellate Board, Rights and Duties of Patentee, Restoration of Lapsed patents.

## **TEXT BOOKS:**

1. Philip Kosky, Robert T. Balmer, William D. Keat, George Wise, “Exploring Engineering: An Introduction to Engineering and Design”, 4th edition, Elsevier, 2016.
2. David Ralzman, “History of Modern Design”, 2nd edition, Laurence King Publishing Ltd., 2010
3. An AVA Book, “Design Thinking”, AVA Publishing, 2010.
3. Ingle, B. R. (2013). Design thinking for entrepreneurs and small businesses: Putting the power of design to work. Apress.
4. Norman, D. A. (2016). Living with complexity. MIT press.
5. Chapman, J. (2017). Routledge handbook of sustainable product design. Taylor & Francis.
6. Nithyananda, K.V. (2019), IPR, protection and Management, India, Cengage learning India.

## **REFERENCES:**

1. G. Pahl, W.Beitz, J. Feldhusen, KH Grote, “Engineering Design: A Systematic Approach”, 3rd edition, Springer, 2007.
2. Tom Kelley, Jonathan Littman, “Ten Faces in Innovation”, Currency Books, 2006.
2. Kumar, V. (2012). 101 design methods: A structured approach for driving innovation in your organization. John Wiley & Sons.
3. Chapman, J. (2012). Designers Visionaries and Other Stories: A Collection of Sustainable Design Essays. Taylor & Francis.
4. Garrett, J. J. (2010). The elements of user experience: user-centered design for the web and beyond. Pearson Education.
5. Neeraj, P. & Khusdeep, D (2014), IPR, India, IN: PHI Learning.

**Syllabus for B.Tech (CSE) IV YEAR I SEM  
FULL STACK DEVELOPMENT  
(Professional Elective – IV)**

**Code: 9EC19**

**Prerequisite:** Web Technologies

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Course Objective:**

Provide with essential skills in modern software development, including version control with Git, frontend development with JavaScript and jQuery, and backend development using Spring Boot for RESTful APIs, also learn containerization with Docker, orchestration with Kubernetes, and principles of micro services architecture.

**Course Outcomes:**

After completion of the course, students should be able to

1. Use Git for version control and demonstrate advanced branching and collaboration techniques. [L4]-U1
2. Create interactive web applications using core JavaScript, AJAX, and jQuery to make front-end development effectively productive. [L6]-U2
3. Build secure and scalable backend applications using Spring Boot, including creating REST APIs and managing data access. [L6]-U3
4. Deploy and manage containerized applications with Docker, using best practices for container security, networking, and orchestration. [L4]-U4
5. Use Kubernetes to orchestrate applications packaged in containers and scale your application with potential clusters and services effectively for reliability and scalability. [L4]-U5,U6

**UNIT - I:**

**Version Control Systems:** In-depth exploration of Git and its features, Advanced Git branching and merging strategies, Git collaboration and work flow management.

**UNIT - II:**

**Frontend Development:** Javascript basics, OOPS Aspects of JavaScript Memory usage and Functions in JS AJAX for data exchange with server jQuery Framework jQuery events, UI component setc. JSON data format.

**UNIT III:**

**Spring Boot Fundamentals:** Introduction to Spring Boot, Spring Boot application structure and components, Dependency management with Spring Boot, Creating and running Spring Boot applications, Implementing REST APIs with Spring Boot, Data access with Spring Boot, User authentication and security in Spring Boot

**UNIT IV:**

**Containerization with Docker:** Introduction to Docker and containerization, Docker image creation and management, Docker container execution and management, Docker networking and volume management, Orchestrating containerized applications with Docker Compose,

Docker security and best practices.

#### **UNIT V:**

**Kubernetes for Orchestration:** Introduction to Kubernetes and container orchestration, Kubernetes cluster architecture and components, Kubernetes deployments, pods, and services  
Kubernetes labels, selectors, and annotations, Managing Kubernetes applications with kubectl, Kubernetes monitoring, logging, and alerting, Kubernetes security and best practices.

#### **UNIT VI:**

**Microservices Architecture:** Principles and concepts, Design patterns, Development with Spring Boot, Docker, and Kubernetes, Testing and deployment, Monitoring and observability.

#### **TEXT BOOKS:**

1. Web Design with HTML, CSS, Java Script and JQuery Set Book by Jon Duckett  
Professional JavaScript for Web Developers Book by Nicholas C. Zakas
2. Learning PHP, MySQL, JavaScript, CSS & HTML5: A Step-by-Step Guide to Creating Dynamic Websites by Robin Nixon
3. Full Stack JavaScript: Learn Backbone.js, Node.js and MongoDB. Copyright © 2015  
BY AZATMARDAN

#### **REFERENCES:**

1. "ProGit" by Scott Chacon and Ben Straub
2. "Spring Boot in Action" by Craig Walls
3. "Docker: Up and Running" by Nigel Poulton
4. "Kubernetes in Action" by Marko Luksa
5. "Microservices: From Design to Deployment" by Sam Newman

**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**NATURAL LANGUAGE PROCESSING**  
**(Professional Elective – IV)**

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Code: 9LC18**

**Prerequisites:** Data Structures, Automata Theory and Compiler Design, Machine Learning

**Course Objective:**

Discuss about word and document structure, semantic parsing, and syntax analysis. Know how different language models and representation systems hinder the processing of concepts.

**Course Outcomes:**

After completion of this course student will be able to:

1. Describe the Structure of Words and Documents. [L1]-U1
2. Analyze and Parse Syntactic Structures. [L4]-U2
3. Illustrate and Implement Semantic Parsing Techniques. [L2]-U3
4. Assess Predicate-Argument Structures and Meaning Representation. [L5]-U4
5. Develop Language Models, overall structure of discourse and evaluate their performance. [L4]-U5, U6

**UNIT-I**

**Finding the Structure of Words:** Words and Their Components, Issues and Challenges, Morphological Models.

**Finding the Structure of Documents:** Introduction, Methods, Complexity of the Approaches, Performances of the Approaches.

**UNIT-II**

**Syntax Analysis:** Parsing Natural Language, Tree banks: A Data-Driven Approach to Syntax, Representation of Syntactic Structure, Parsing Algorithms, Models for Ambiguity Resolution in Parsing, Multilingual Issues

**UNIT-III**

**Semantic Parsing:** Introduction, Semantic Interpretation, System Paradigms, Word Sense Systems, Software.

**UNIT-IV**

Predicate-Argument Structure, Meaning Representation Systems, Software.

**UNIT-V**

**Discourse Processing:** Cohesion, Reference Resolution, Discourse Cohesion and Structure.

**UNIT-VI**

**Language Modeling:** Introduction, N-Gram Models, Language Model Evaluation, Parameter Estimation, Language Model Adaptation, Types of Language Models, Language-Specific Modeling Problems, Multilingual and Cross lingual Language Modeling.

**TEXT BOOKS:**

1. Multilingual Natural Language Processing Applications: From Theory to Practice–

- Daniel M.Bikel and ImedZitouni, Pearson Publication.
2. Natural Language Processing and Information Retrieval: Tanvier Siddiqui, U.S.Tiwary.

## **REFERENCES:**

1. Speech and Natural Language Processing-Daniel Jurafsky&James H Martin, Pearson Publications.
2. Charniak, E.: Statistical Language Learning. The MIT Press.
3. Jelinek, F.: Statistical Methods for Speech Recognition. The MIT Press.
4. Lutz and Ascher - "Learning Python", O'Reilly.

**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**DEVOPS**  
**(Professional Elective – IV)**

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Code: 9LC17**

**Prerequisite:** Web Technologies

**Course Objective:**

1. Discuss the DevOps concepts and software development models.
2. Apply the concepts of project management and system integration.
3. Implement automated system update and DevOps lifecycle.

**Course Outcomes:**

After completion of this course student will be able to:

1. Explain DevOps Practices. [L2]-U1
2. Analyze Software Architecture and DevOps Integration. [L4]-U2
3. Apply Source Code Management Techniques. [L3]-U3
4. Design and Implement Automated Build Pipelines. [L6]-U4
5. Assess Testing Automation and Deployment Tools and Techniques. [L5]-U5, U6

**UNIT – I**

**Introduction:** Introduction, Agile development model, DevOps, and ITIL. DevOps process and Continuous Delivery, Release management, Scrum, Kanban, delivery pipeline, bottlenecks, examples

**UNIT – II**

**Software development models and DevOps:** DevOps Lifecycle for Business Agility, DevOps, and Continuous Testing.

DevOps influence on Architecture: Introducing software architecture, The monolithic scenario, Architecture rules of thumb, The separation of concerns, Handling database migrations, Microservices, and the data tier, DevOps, architecture, and resilience.

**UNIT – III**

**Introduction to project management:** The need for source code control, The history of source code management, Roles and code, source code management system and migrations, Shared authentication, Hosted Git servers, Different Git server implementations, Docker intermission, Gerrit, The pull request model, GitLab.

**UNIT – IV**

**Integrating the system:** Build systems, Jenkins build server, Managing build dependencies, Jenkins plugins, and file system layout, The host server, Build slaves, Software on the host, Triggers, Job chaining and build pipelines, Build servers and infrastructure as code, Building by dependency order, Build phases, Alternative build servers, Collating quality measures.

**UNIT– V**

**Testing Tools and automation:** Various types of testing, Automation of testing Pros and cons,

Selenium - Introduction, Selenium features, JavaScript testing, Testing backend integration points, Test-driven development, REPL-driven development

## **UNIT – VI**

**Deployment of the system:** Deployment systems, Virtualization stacks, code execution at the client, Puppet master and agents, Ansible, Deployment tools: Chef, Salt Stack and Docker.

### **TEXT BOOKS:**

1. Joakim Verona. Practical Devops, Second Edition. Ingram short title; 2nd edition (2018). ISBN-10: 1788392574.
2. Deepak Gaikwad, Viral Thakkar. DevOps Tools from Practitioner's Viewpoint. Wiley publications. ISBN: 9788126579952.

### **REFERENCES:**

1. Len Bass, Ingo Weber, Liming Zhu. DevOps: A Software Architect's Perspective. Addison Wesley; ISBN-10.



**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**BLOCK CHAIN TECHNOLOGIES**  
**(Professional Elective – IV)**

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Code: 9JC05**

**Prerequisite:** Information Security

**Course Objective:**

Understand the concepts and technology of Block Chain systems.

**Course Outcomes:**

At the end of this course the student will be able to

1. Summarize the conceptual working of blockchain using Cryptographic principles. [L2]-U1, U2
2. Explore the Payment Verification protocol, its lifecycle and the different consensus algorithms.[L4]-U2, U3
3. Classify the different types of crypto currencies alongside their features. [L4]-U4, U5
4. Outline the legal issues of consideration in the development of block chain. [L4]-U5
5. Examine the proofs and protocols used in the blockchain technology. [L3]-U6

**UNIT I**

**Basics:** Distributed Database, Two General Problem, Byzantine General problem and Fault Tolerance, Hadoop Distributed File System, Distributed Hash Table, ASIC resistance, Turing Complete.

**Cryptography:** Hash function, Digital Signature - ECDSA, Memory Hard Algorithm, Zero Knowledge Proof.

**UNIT II**

**Blockchain:** Introduction, Advantage over conventional distributed database, Blockchain Network, Mining Mechanism, Distributed Consensus, Merkle Patricia Tree, Gas Limit, Transactions and Fee, Anonymity, Reward, Chain Policy, Life of Blockchain application, Soft & Hard Fork, Private and Public blockchain.

**UNIT III**

Distributed Consensus: Nakamoto consensus, Proof of Work, Proof of Stake, Proof of Burn, Difficulty Level, Sybil Attack, Energy utilization and alternates.

**UNIT IV**

Cryptocurrency: History, Distributed Ledger, Bitcoin protocols- Mining strategy and rewards, Ethereum - Construction, DAO, Smart Contract, GHOST, Vulnerability, Attacks, Sidechain, Namecoin

**UNIT V**

Cryptocurrency Regulation: Stakeholders, Roots of Bitcoin, Legal Aspects-Cryptocurrency Exchange, Black Market and Global Economy. Applications: Internet of Things, Medical Record Management System, Domain Name Service and future of Blockchain.

## **UNIT VI**

(Trends and Topics)-Zero Knowledge proofs and protocols in Blockchain-Succinct non interactive argument for Knowledge (SNARK)- pairing on Elliptic curves-Zcash.

### **TEXT BOOKS:**

1. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder, Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction, Princeton University Press (July19, 2016).

### **REFERENCES:**

1. Antonopoulos, Mastering Bitcoin: Unlocking Digital Crypto currencies
2. Satoshi Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System
3. DR. Gavin Wood, "ETHEREUM: A Secure Decentralized Transaction Ledger,"Yellow paper.2014.
4. Nicola Atzei, Massimo Bartoletti, and Tiziana Cimoli, A survey of attacks on Ethereum smart contracts

**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**MOBILE COMPUTING**  
**(Professional Elective – IV)**

**Code: 9EC20**

**Prerequisite:** Operating Systems, Mobile Application Development

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Course Objective:**

Understand the concepts and protocols of mobile computing.

**Course Outcomes:**

After completion of this course student will be able to:

1. Identify application areas for mobile / wireless communications and understand GSM Architecture, Services. [L2]-U1
2. Examine hidden and exposed terminals, near and far terminals and differentiate Medium Access Control methods for wireless communication. [L4]-U2
3. Demonstrate IP packet delivery, Mobile Network and Transport Layer protocols. [L3]-U3,U4
4. Identify emerging mobile operating systems and use of applications of MANETs routing algorithms, data hoarding, and client server computing.[L2]-U5
5. Illustrate data delivery mechanisms, WAP, Bluetooth and Android tools. [L2]-U6

**UNIT - I**

**Introduction to Mobile Communications and Computing:** Mobile Computing(MC): Introduction to MC, novel applications, limitations, introduction to mobile architecture-UMTS, GSM.

**Architecture GSM:** Mobile services, System architecture, Radio interface, Protocols, Localization and calling, Handover, Security, and New data services.

**UNIT - II**

**(Wireless)Medium Access Control:** Motivation for a specialized MAC(Hidden and exposed terminals, Near and far terminals), SDMA, FDMA, TDMA, CDMA.

**UNIT - III**

**Mobile Network Layer:** Mobile IP (Goals, assumptions, entities and terminology, IP packet delivery, agent advertisement and discovery, registration, tunneling and encapsulation, optimizations), Dynamic Host Configuration Protocol (DHCP).

**UNIT - IV**

**Mobile Transport Layer:** Traditional TCP, Indirect TCP, Snooping TCP, Mobile TCP, Fast retransmit/fast recovery, Transmission/time-out freezing, Selective retransmission, Transaction oriented TCP.

**UNIT - V**

**Mobile Ad hoc Networks (MANETs):** Overview, Properties of a MANET, spectrum of MANET applications, routing algorithms.

**Database:** Hoarding techniques, caching invalidation mechanisms, client server computing with adaptation, data delivery mechanism

## UNIT - VI

**Protocols and Tools:** Wireless Application Protocol-WAP. (Introduction, protocol architecture, and treatment of protocols of all layers), Bluetooth (User scenarios, physical layer, MAC layer, networking, security, link management), introduction to mobile operating systems- Android: Android versions, Typographical Conventions, the Necessary Tools for Android.

### TEXTBOOKS:

1. Jochen Schiller–Mobile Communications, *Addison-Wesley*. (Chapters 1,2,3,4,7,8 and 9). Second edition, 2004.
2. Stojmenovic and Cacute, –Handbook of Wireless Networks and Mobile Computing, *Wiley*, 2002, ISBN 0471419028. (Chapters 6,11,15,17,18,19,26 and 27)
3. *Android Programming: The Big Nerd Ranch Guide* by Bill Phillips, Chris Stewart, Brian Hardy and Kristin Marsicano, second edition.

### REFERENCES:

1. Reza Behravan far, –Mobile Computing Principles: Designing and Developing Mobile Applications with UML and XML, ISBN: 0521817331, Cambridge University Press, October 2004,
2. Adelstein, Frank, Gupta, Sandeep KS, Richard III, Golden, Schwiebert, Loren, –Fundamentals of Mobile and Pervasive Computing, ISBN: 0071412379, McGraw-Hill Professional, 2005.
3. Hansmann, Merk, Nicklous, Stober–Principles of Mobile Computing, *Springer*, second edition, 2003.
4. Martyn Mallick–Mobile and Wireless Design Essentials, *Wiley DreamTech*, 2003  
A. Tanenbaum—Computer Networks, 4<sup>th</sup> edition.
5. *Android Programming (Big Nerd Ranch Guide)*, by Phillips, Stewart, Hardy and Marsicano
6. *Android Programming– Pushing the limits* by Hellman

**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**Agile Software Development**  
**(Professional Elective – IV)**

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Code: 9FC16**

**Prerequisite:** Software Engineering and OOAD

**Course Objective:**

Understand Agile Software Development and examine their applications in the real world and addresses their impacts on developing software.

**Course Outcomes:**

After completion of this course student will be able to:

1. Summarize the various agile methodologies.[L2]-U1
2. Apply the XP practices, pair programming and agile modelling for real time projects.[L3]-U2,U3
3. Examine the process of Feature-Driven Development and Regaining Control.[L3]-U4
4. Relate Agile Modelling and RUP and choose appropriate tools to help with Agile Development.[L2]-U5
5. Identify obstacles in Agile software development and gain familiarity with agility.[L3]-U6

**UNIT - I**

Introduction: Agile Methods, Agile Manifesto, and Agile Modelling Introduction, What Is Agile, The Agile Manifesto, Agile Methods, XP: Extreme Programming, DSDM, SCRUM, Feature-Driven Development, Modelling Misconceptions, Agile Modelling, Tools of Misconceptions, Updating Agile Models

**UNIT - II**

Extreme Programming: Introduction, Core XP Values, The Twelve XP Practices, About Extreme Programming, Planning XP Projects, Test First Coding, Making Pair Programming Work

**UNIT - III**

Agile Modelling and XP: Introduction, The Fit, Common Practices, Modelling Specific Practices, XP Objections to Agile Modelling, Agile Modelling and Planning XP Projects, XP Implementation Phase

**UNIT - IV**

Feature-Driven Development: Introduction, Incremental Software Development, Regaining Control: The Motivation behind FDD, Planning an Iterative Project, Architecture Centric, FDD and XP

**UNIT - V**

Agile Methods with RUP and PRINCE2 and Tools and Obstacles: Agile Modelling and RUP, FDD and RUP, Agile Methods and PRINCE2, Tools to Help with Agile Development, Eclipse: An Agile IDE.

## **UNIT-VI**

Obstacles to Agile Software Development, Management Intransigence, The Failed Project Syndrome, Contractual Difficulties, Familiarity with Agility

### **TEXT BOOKS:**

1. Agile software construction, 1/e, John hunt, springer, 2005
2. Agile and Iterative Development: a manager & #39;s guide, Addison-Wesley Craig Larman, Pearson Education - 2004.

### **REFERENCES:**

1. The Art of Agile Development, Pearson, Robert C. Martin, Juli, James Shore, Chromatic, 2013, O'Reilly Media.
2. Agile Testing, Elisabeth Hendrickson, Quality Tree Software Inc 2008.

**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**AUGMENTED REALITY AND VIRTUAL REALITY**  
**(Professional Elective – V)**

**Code: 9LC08**

**Pre-Requisites:** Data Structures

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Course Objective:**

Provides an opportunity to explore the research issues in Augmented Reality and Virtual Reality (AR &VR), also makes the students know the basic concepts and framework of virtual reality.

**Course Outcomes:**

After completion of this course student will be able to:

1. Explain the Concepts and Components of Virtual Reality. [L2]-U1
2. Apply input and output models in VR Systems.[L3]-U2
3. Analyze Visual Computation and Real-Time rendering techniques. [L4]-U3
4. Develop interactive VR Experiences using Hand Gestures and Object Manipulation. [L6]-U4
5. Elaborate Augmented Reality Systems and their applications. [L6]-U5, U6

**UNIT-I**

Introduction of Virtual Reality: Fundamental Concept and Components of Virtual Reality. Primary Features and Present Development on Virtual Reality.

**UNIT-II**

Multiple Models of Input and Output Interface in Virtual Reality: Input -- Tracker, Sensor, Digital Glove, Movement Capture, Video-based Input, 3D Menus & 3DScanner etc. Output-Visual / Auditory Haptic Devices.

**UNIT-III**

Visual Computation in Virtual Reality: Fundamentals of Computer Graphics. Software and Hardware Technology on Stereoscopic Display. Advanced Techniques in CG: Management of Large scale Environments & Real Time Rendering.

**UNIT -IV**

Interactive Techniques in Virtual Reality: Body Track, Hand Gesture, 3D Manus, And Object Grasp.

**UNIT-V**

Development Tools and Frameworks in Virtual Reality: Frameworks of Software Development Tools in VR. X3D Standard; Vega, MultiGen, Virtools etc. Application of VR in Digital Entertainment: VR Technology in Film & TV Production. VR Technology in Physical Exercises and Games. Demonstration of Digital Entertainment by VR.

**UNIT-VI**

Augmented and Mixed Reality, Taxonomy, technology and features of augmented reality, difference between AR and VR, Challenges with AR, AR systems and functionality,

Augmented reality methods, visualization techniques for augmented reality, wireless displays in educational augmented reality applications, mobile projection interfaces, marker-less tracking for augmented reality, enhancing interactivity in AR environments, evaluating AR systems.

**TEXTBOOKS:**

1. Burdea, G. C. and P. Coffet. Virtual Reality Technology, Second Edition. Wiley-IEEE Press, 2003/2006.
2. Alan B. Craig, Understanding Augmented Reality, Concepts and Applications, Morgan Kaufmann, 2013.

**REFERENCES:**

1. Alan Craig, William Sherman and Jeffrey Will, Developing Virtual Reality Applications, Foundations of Effective Design, Morgan Kaufmann, 2009.



**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**COMPUTER VISION**  
**(Professional Elective – V)**

**Code: 9EC22**

**Prerequisite:** Computer Graphics, Machine learning

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Learn basic principles of image formation, image processing algorithms and different algorithms for 3D reconstruction and recognition from single or multiple images (video).

**Course Outcomes:**

After completion of this course student will be able to:

1. Use the fundamental principles and applications of computer vision also analyze different camera models, multi-view geometry and structures to produce 3D models. [L3,L4]-U1
2. Apply image pre-processing techniques and various feature extraction techniques to construct 3D models from images. [L3]-U2,U3
3. Apply motion estimation techniques to identify and classify moving objects within a video stream.[L3]-U4
4. Demonstrate various image shape representations and explore segmentation methods. [L3]-U5
5. Classify various object recognition methodologies, apply Hough transforms and showcase mastery in shape matching techniques. [L2,L3]-U6

**UNIT I**

Introduction : Computer Vision and Computer Graphics , What is Computer Vision - Low-level, Mid-level, High-level , Overview of Diverse Computer Vision Applications: Document Image Analysis, Biometrics, Object Recognition, Tracking, Medical Image Analysis, Content-Based Image Retrieval, Video Data Processing, Multimedia.

**UNIT II**

Image Formation Models : Monocular imaging system , Radiosity: The ‘Physics’ of Image Formation, Radiance, Irradiance, BRDF, color etc, Orthographic & Perspective Projection,• Camera model and Camera calibration, Binocular imaging systems, Multiple views geometry, Structure determination, shape from shading , Photometric Stereo, Depth from Defocus , Construction of 3D model from images

**UNIT III**

Image Processing and Feature Extraction: Image pre-processing, Image representations (continuous and discrete), Edge detection

**UNIT IV**

Motion Estimation: Regularization theory, Optical computation, Stereo Vision, Motion estimation, Structure from motion

**UNIT V**

Shape Representation and Segmentation: Contour based representation, Region based representation, Deformable curves and surfaces, Snakes and active contours, Level set

representations, Fourier and wavelet descriptors, Medial representations, Multire solution analysis

## **UNIT VI**

Object recognition: Hough transforms and other simple object recognition methods, Shape correspondence and shape matching, Shape priors for recognition

### **TEXT BOOKS:**

1. Computer Vision - A modern approach, by D. Forsyth and J. Ponce, Prentice Hall Robot Vision, by B. K. P. Horn, McGraw-Hill.
2. Introductory Techniques for 3D Computer Vision, by E. Trucco and A. Verri, Publisher: Prentice Hall.

### **REFERENCES:**

1. R. C. Gonzalez, R. E. Woods. Digital Image Processing. Addison Wesley Longman, Inc., 1992.
2. D. H. Ballard, C. M. Brown. Computer Vision. Prentice-Hall, Englewood Cliffs, 1982.
3. Richard Szeliski, Computer Vision: Algorithms and Applications (CVAA). Springer, 2010
4. Image Processing, Analysis, and Machine Vision. Sonka, Hlavac, and Boyle. Thomson.
4. E. R. Davies, Computer & Machine Vision, Fourth Edition, Academic Press, 2012
5. Simon J. D. Prince, Computer Vision: Models, Learning, and Inference, Cambridge University Press, 2012
6. Mark Nixon and Alberto S. Aquado, Feature Extraction & Image Processing for Computer Vision, Third Edition, Academic Press, 2012

**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**CLOUD COMPUTING**  
**(Professional Elective – V)**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Code : 9FC17**

**Prerequisite:** Computer Networks

**Course Objective:**

Understand and analyze the basic cloud computing models, applications and security issues in cloud technology.

**Course Outcomes:**

After completion of this course student will be able to:

1. Summarize the characteristics of cloud technologies and differentiate the cloud service and deployment models and to explore vendor specific services.[L2]-U1,U2
2. Illustrate different architectures for cloud applications, create and run Amazon ec2 instance through python programs.[L2]-U3
3. Classify the performance of cloud services and summarize the innovative applications of IOT on cloud.[L2]-U4
4. Demonstrate to deploy the cloud based applications.[L3]-U5
5. Discuss various security aspects in cloud computing.[L2]-U6

**UNIT-I**

Introduction to Cloud Computing: Introduction, characteristics, Cloud Models and examples, Applications of Cloud Services .Cloud Concepts and Technologies.

**UNIT-II**

Cloud Services and Platforms : Compute Services, Storage Services, Database Services, Application Services, Content Delivery Services, Analytics Services, Deployment and Management Services, Identity and Access Management Services, Open Source Private cloud Software.

**UNIT-III**

Cloud Application Design: Design Considerations for Cloud Application, Reference Architectures for Cloud Applications .Cloud Application Design Methodologies, Data Storage Approaches. Python For Cloud: Python for Amazon Web Services, Map Reduce

**UNIT – IV**

**Book 2.**Cloud and the Internet of Things: Performance of Distributed Systems and the Cloud- Enabling Technologies for the Internet of Things- Innovative Applications of the Internet of Things- Online Social and Professional Networking

**UNIT-V**

Cloud Application Development in Python: Design Approaches, Image Processing App, Document Storage App, Map Reduce App, Social Media Analytics App.

**UNIT-VI**

Cloud Security: Introduction, Cloud Security Architecture (CSA), Authentication, Authorization, Identity Access Management (IAM), Data Security, Key Management, Auditing.

Cloud for Industry, Healthcare and Education.

**TEXT BOOKS:**

1. Cloud Computing –A Hands on Approach, Arshdeep, Vijay Mediseti, University Press.
2. Distributed and Cloud Computing,1st Edition, From Parallel Processing to the Internet of Things, Authors: Kai Hwang Jack Dongarra Geoffrey Fox(Unit4)
3. Cloud Computing: Raj Kumar Buyya, James Broberg, Andrzej GOscinski, Wiley.

**REFERENCES:**

1. Cloud Computing: Dr.Kumar Saurab Wiley India 2011.
2. Code in the cloud computing: K Chandrasekharan CRC Press.
3. Cloud Computing: John W. Rittinghouse, James Ransome, CRC press.
4. Virtualization Security: Dave Shackleford2013, SYBEX a Willy Brand.
5. Cloud Computing and Software Service: Ahson, iiyas.2011.
6. Cloud Computing Bible: Sosinsky 2012 Wiley India.

**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**ETHICAL HACKING**  
**(Professional Elective – V)**

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Code: 9JC04**

**Prerequisite:** Knowledge in information security, Web application

**Course Objective:**

Learn the methodologies and framework of ethical hacking for enhancing the security.

**Course Outcomes:**

After completion of this course student will be able to:

1. Distinguish the components of Information Security Models. [L2]-U1, U2
2. Plan for a controlled attack and interpret the results. [L5, L6]-U3, U4
3. Apply the knowledge of tools to support ethical hacking to meet the business challenges. [L3]-U2, U4
4. Enumerate comprehensive deliverables and integrate security findings. [L4]- U5, U6

**UNIT- I**

Introduction: Hacking Impacts, The Hacker Framework: Planning the test, Sound Operations, Reconnaissance, Enumeration, Vulnerability Analysis, Exploitation, Final Analysis, Deliverable, Integration

Information Security Models: Computer Security, Network Security, Service Security, Application Security, Security Architecture

**UNIT - II**

Information Security Program: The Process of Information Security, Component Parts of Information Security Program, Risk Analysis and Ethical Hacking

The Business Perspective: Business Objectives, Security Policy, Previous Test Results, Business Challenges

**UNIT - III**

Planning for a Controlled Attack: Inherent Limitations, Imposed Limitations, timing is Everything, Attack Type, Source Point, Required Knowledge, Multi-Phased Attacks, Teaming and Attack Structure, Engagement Planner, The Right Security Consultant, The Tester, Logistics, Intermediates, Law Enforcement

**UNIT - IV**

Preparing for a Hack: Technical Preparation, Managing the Engagement

Reconnaissance: Social Engineering, Physical Security, Internet Reconnaissance

**UNIT - V**

Enumeration: Enumeration Techniques, Soft Objective, Looking Around or Attack, Elements of Enumeration, Preparing for the Next Phase

Exploitation: Intuitive Testing, Evasion, Threads and Groups, Operating Systems, Password Crackers, Root Kits, applications, War dialing, Network, Services and Areas of Concern

**UNIT - VI**

Deliverable: The Deliverable, The Document, Overall Structure, Aligning Findings, Presentation Integration: Integrating the Results, Integration Summary, Mitigation, Defense Planning, Incident Management, Security Policy, Conclusion

**TEXT BOOK:**

1. James S. Tiller, “The Ethical Hack: A Framework for Business Value Penetration Testing”, Auerbach Publications, CRC Press

**REFERENCE BOOKS:**

1. EC-Council, “Ethical Hacking and Countermeasures Attack Phases”, Cengage Learning
2. Michael Simpson, Kent Backman, James Corley, “Hands-On Ethical Hacking and Network Defense”, Cengage Learning

**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**QUANTUM COMPUTING**  
**(Professional Elective – V)**

| L | T | P | C |
|---|---|---|---|
| 3 | 0 | 0 | 3 |

**Code: 9LC21**

**Prerequisite:** Mathematics

**Course Objective:**

Discover the principles of quantum computing and problem-solving using finite dimensional mathematics.

**Course Outcomes:**

After completion of this course student will be able to:

1. Illustrate essential linear algebra, including matrix math and complex numbers, for foundational quantum computing concepts. [L2]-U1
2. Explain basic quantum physics concepts like quantum states, uncertainty, and entanglement, and their role in quantum computing. [L2]-U2
3. Apply quantum mechanics theories, including quantum electrodynamics and entanglement, to understand quantum communication and cryptography. [L3]-U3
4. Analyze quantum architecture and hardware, including qubits, quantum gates, and circuits, for building quantum systems. [L4]-U4
5. Assess the impact of quantum computing on cryptography by evaluating quantum algorithms like Deutsch's and Shor's. [L5]-U4, U5

**UNIT – I**

Introduction to Essential Linear Algebra: Some Basic Algebra, Matrix Math, Vectors and Vector Spaces, Set Theory. Complex Numbers: Definition of Complex Numbers, Algebra of Complex Numbers, Complex Numbers Graphically, Vector Representations of Complex Numbers, Pauli Matrices, Transcendental Numbers.

**UNIT – II**

Basic Physics for Quantum Computing: The Journey to Quantum, Quantum Physics Essentials, Basic Atomic Structure, Hilbert Spaces, Uncertainty, Quantum States, Entanglement.

**UNIT-III**

Basic Quantum Theory: Further with Quantum Mechanics, Quantum Decoherence, Quantum Electrodynamics, Quantum Chromodynamics, Feynman Diagram Quantum Entanglement and QKD, Quantum Entanglement, Interpretation, QKE.

**UNIT – IV**

Quantum Architecture: Further with Qubits, Quantum Gates, More with Gates, Quantum Circuits, The D-Wave Quantum Architecture. Quantum Hardware: Qubits, How Many Qubits Are Needed? Addressing Decoherence, Topological Quantum Computing, Quantum Essentials.

## **UNIT – V**

Quantum Algorithms: What Is an Algorithm? Deutsch's Algorithm, Deutsch-Jozsa Algorithm, Bernstein- Vazirani Algorithm, Simon's Algorithm, Shor's Algorithm, Grover's Algorithm.

## **UNIT – VI**

Current Asymmetric Algorithms: RSA, Diffie-Hellman, Elliptic Curve. The Impact of Quantum Computing on Cryptography: Asymmetric Cryptography, Specific Algorithms, Specific Applications.

### **TEXT BOOKS:**

1. Nielsen M. A., Quantum Computation and Quantum Information, Cambridge University Press.
2. Dr. Chuck Easttom, Quantum Computing Fundamentals, Pearson.

### **REFERENCES:**

1. Quantum Computing for Computer Scientists by Noson S. Yanofsky and Mirco A. Mannucci.
2. Principles of Quantum Computation and Information, Volume II: Basic Tools and Special Topics By Benenti G., Casati G. and Strini G., World Scientific Publishing Co Pte Ltd.



**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**ADVANCED ALGORITHMS**  
**(Professional Elective – V)**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Code: 9EC23**

**Pre-requisites:** Computer Programming, Data Structures & Algorithms

**Course Objective:**

Learn major algorithmic techniques of searching, sorting networks, string matching, NP Complete and approximations.

**Course Outcomes:**

After completion of this course student will be able to:

1. Select the appropriate algorithm by analyzing its efficiency in real time scenario.[L2]-U1,U2
2. Apply suitable techniques for sorting and searching to achieve optimal solutions.[L3]-U1,U2
3. Compare various sorting networks by applying zero-one principle and solve networking issues using matrix operations. [L4]-U3
4. Apply various string matching algorithms in real time problems.[L3]-U4
5. Apply NP-Completeness and Approximation Algorithms.[L3]-U5,U6

**UNIT – I**

Introduction: Role of Algorithms in computing, Order Notation, Recurrences, Probabilistic Analysis and Randomized Algorithms. Sorting and Order Statistics: Heap sort, Quick sort and Sorting in Linear Time.

Advanced Design and Analysis Techniques: Dynamic Programming- Matrix chain Multiplication, Longest common Subsequence and optimal binary Search trees.

**UNIT - II**

Greedy Algorithms - Huffman Codes, Activity Selection Problem. Amortized Analysis.

Graph Algorithms: Topological Sorting, Minimum Spanning trees, Single Source Shortest Paths, Maximum Flow algorithms.

**UNIT - III**

Sorting Networks: Comparison Networks, Zero-one principle, bitonic Sorting Networks, Merging Network, Sorting Network.

Matrix Operations- Strassen's Matrix Multiplication, Inverting matrices, Solving system of linear Equations

**UNIT - IV**

String Matching: Naive String Matching, Rabin-Karp algorithm, matching with finite Automata, Knuth Morris - Pratt algorithm.

**UNIT- V**

NP-Completeness Algorithms: Polynomial time, polynomial time verification, NP-Completeness and reducibility, NP-Complete problems.

## **UNIT- VI**

Approximation Algorithms- Vertex covers Problem, Travelling Sales person problem

### **TEXT BOOK:**

1. Introduction to Algorithms," T.H. Cormen, C.E. Leiserson, R.L. Rivest, and C. Stein, Third Edition, PHI.

### **REFERENCES:**

1. Fundamentals of Computer Algorithms, Ellis Horowitz, SatrajSahni and Rajasekharam, Galgotia publications pvt.Ltd.
2. Design and Analysis Algorithms - Parag Himanshu Dave, Himanshu Bhalchandra Dave Publisher: Pearson
3. Algorithm Design: Foundations, Analysis and Internet examples, M.T. Goodrich and R.Tomassia, John Wiley and sons.
4. Data structures and Algorithm Analysis in C++, Allen Weiss, Second edition, Pearson Education

**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**Automata Theory and Compiler Design**

**Code : 9EC07**

**Prerequisite : Nil**

| L | T | P/D | C |
|---|---|-----|---|
| 2 | 1 | 0   | 3 |

**Course Objective:**

Understand the working of a compiler.

**Course Outcomes:**

After completion of this course student will be able to:

1. Design finite Automata for a given regular expressions, and derive strings with suitable examples. Conceptualize context free grammars and normal forms. [L4]-U1,U2
2. Design the push down automata and Turing Machine for complex languages. [L4]-U3
3. Use of LEX tool and relate parsing techniques. [L3]-U4
4. Demonstrate and solve problems on SLR, CLR, LALR, operator precedence parser, LR (0) grammar and use YACC tool.[L3]-U5
5. Discuss Semantic Analysis concepts to design compiler, and describe Intermediate code generation. [L2]-U6

**UNIT-I:**

**Introduction:** Strings, Alphabet, Language, Operations, finite automaton model, acceptance of strings and languages, Chomsky hierarchy of languages. Deterministic finite automaton and non-deterministic finite automaton, NFA to DFA conversion, equivalence of DFA's

**UNIT-II:**

**Regular Languages,** Regular sets, regular expressions, Constructing finite Automata for a given regular expressions, Conversion of Finite Automata to Regular expressions. Closure properties of regular sets (proofs not required).

**Context Free Grammars:** Context free grammar, derivation trees, Right most and leftmost derivation, ambiguity in context free grammars, simplification of Context Free Grammars (removal of  $\epsilon$ -productions, UNIT productions and useless symbols).Chomsky normal form (CNF), Greibach normal form (GNF).

**UNIT-III:**

**Push down automata:** Definition, model, acceptance of CFL, introduction to Deterministic PDA and Non Deterministic PDA, design of PDA.

**Turing Machine:** Turing Machine, definition, model, design of TM, recursively enumerable languages.

**UNIT IV:**

**Overview of compiler** – Environment, pass, phase, phases of compiler, LEX tool.

**Top Down Parsing:** Top down parsing technique, Recursive decent parsing with back tracking, Ambiguous grammar, Elimination of left recursion, Left factoring, LL (1) Parser.

**UNIT V:**

**Bottom up parsing:** shift reduce parser, LR(0), SLR, CLR, LALR, operator precedence parser, YACC tool.

**UNIT VI:**

**Semantic Analysis:** Syntax directed translation, S- Attributed, L Attributed definition, Symbol table format.

**Intermediate Code Generation:** 3-address code form, DAG.

**Code Optimization:** Local Optimization, loop optimization, peep-hole optimization.

**TEXTBOOKS:**

1. Introduction to Automata Theory Languages and Computation. Hopcroft H.E. and Ullman J. D. Pearson Education
2. Introduction to Theory of Computation? Sipser 2nd edition Thomson
3. Compilers Principles, Techniques and Tools Aho, Ullman, Ravisethi, Pearson Education

**REFERENCES:**

1. Introduction to Computer Theory, Daniel I.A. Cohen, John Wiley.
2. Introduction to languages and the Theory of Computation, John C Martin, TMH
3. Elements of Theory of Computation?, Lewis H.P. & Papadimitriou C.H. Pearson /PHI.
4. Theory of Computer Science Automata languages and computation-Mishra and Chandrashekar, 2nd edition, PHI Course Requirements.
5. Modern Compiler Construction in C, Andrew W.Appel Cambridge University Press. Compiler Construction, LOUDEN, Thomson

## Syllabus for B.Tech (CSE) IV YEAR I SEM

### Linux Programming

**Code : 9EC08**

**Prerequisite :** Operating Systems

| L | T | P/D | C |
|---|---|-----|---|
| 2 | 1 | 0   | 3 |

#### **Course Objective:**

Understand working of Linux operating system.

#### **Course Outcomes:**

After completion of this course student will be able to:

1. Explain Linux Environment, Utilities, Categories of Linux Utilities, Sed and Awk Scripts and Applications. [L2]-U1
2. Demonstrate proficiency in shell scripting to enhancing their ability to solve real-world problems in system administration and software development. [L3]-U2,U3
3. Develop the programs that interact with the file system and develop system programs that manage processes, handle signals, process management in the Linux environment. [L5]-U4
4. Apply Inter-process Communication and use semaphores for process synchronization.[L3]-U5,U6

#### **UNIT-I**

Linux Utilities-File handling utilities, Security by file permissions, Process utilities, Disk utilities, Networking commands, Filters, Text processing utilities and Backup utilities, sed – scripts, operation, addresses, commands, applications, awk – execution, fields and records, scripts, operation, patterns, using system commands in awk.

(Applications: Determining what types of files are present in a system, debugging issues with file accessibility, finding a process troubling for a task and discarding from its existing, Write and extract necessary information from huge test files.)

#### **UNIT – II**

Working with the Bourne again shell(bash): Introduction, shell responsibilities, pipes and input Redirection, output redirection, running a shell script, the shell as a programming language, shell meta characters, file name substitution, shell variables, command substitution, shell commands, the environment, quoting, test command, control structures, arithmetic in shell, shell script examples, interrupt processing, functions, debugging shell scripts.

(Applications: Writing shell scripts for automating most of the regular jobs, taking backup on regular basis and restoring the same)

#### **UNIT-III**

Files: File Concept, File System Structure, I nodes, File Attributes, File types, Library functions, the standard I/O and formatted I/O in C, stream errors, kernel support for files, System calls, file descriptors, low level file access – File structure related system calls (File APIs), file and record locking, file and directory management – Directory file APIs, Symbolic links & hard links. (Applications: write some system programs to interact with file system, developing small system software's to work with files and devices, developing program's on directory management system)

## **UNIT-IV**

Process – Process concept, Kernel support for process, process attributes, process control - process creation, waiting for a process, process termination, zombie process, orphan process, Process APIs. Signals– Introduction to signals, Signal generation and handling, Kernel support for signals, Signal function, unreliable signals, reliable signals, kill, raise, alarm, pause, abort, sleep functions.

(Applications: Applications to find number of typical processes are under different context and controlling them in synchronous manner. Develop user defined modules for handling a signal and controlling several issues with signals.)

## **UNIT-V**

Inter-process Communication: Introduction to IPC, Pipes, FIFOs, Introduction to three types of IPC-message queues, semaphores and shared memory. Message Queues Kernel support for messages, UNIX system V APIs for messages, client/server example.

(Applications: Developing applications complying with IPC mechanisms, developing an application that exchanges a set of messages among different processes. Write a client server application to go with any concurrent approach)

## **UNIT- VI**

Semaphores-Kernel support for semaphores, UNIX system V APIs for semaphores. Shared Memory- Kernel support for shared memory, UNIX system V APIs for shared memory, semaphore and shared memory example.

(Applications: Develop critical section handling mechanisms to deal with any real problems. Building applications to share a piece of memory resource among processes concurrently)

## **TEXT BOOKS:**

1. Unix System Programming using C++, T.Chan, PHI.
2. Unix Concepts and Applications, 4th Edition, Sumitabha Das, TMH, 2006.
3. Beginning Linux Programming, 4th Edition, N.Matthew, R.Stones, Wrox, Wiley India Edition, rp-2008

## **REFERENCES:**

1. Linux System Programming, Robert Love, O'Reilly, SPD.
2. Advanced Programming in the Unix environment, 2nd Edition, W.R.Stevens, Pearson Education.
3. Unix Network Programming, W.R. Stevens, PHI.
4. Unix for programmers and users, 3rd Edition, Graham Glass, King Ables, Pearson Education

**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**Cryptography and Network Security**

**Code : 9EC09**

**Prerequisite :** Computer Networks

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Learn various cryptographic techniques and protocols to thwart against security attacks.

**Course Outcomes:**

After completion of this course student will be able to:

1. Classify the security attacks, security services and also describe the model for network Security.[L2]-U1
2. Analyze conventional and Public cryptographic techniques.[L4]-U2
3. Classify and analyze Message Authentication, Secure Hash functions and outline the concepts of Kerberos and email privacy.[L3]-U3
4. Examine architecture, key management and header formats of IPsec and Outline the various web security threats and protocols.[L3]-U4,U5
5. Interpret Intrusion Detection System and Design principles of Firewalls.[L3]-U6

**UNIT – I**

Security Attacks (Interruption, Interception, Modification and Fabrication), Security Services (Confidentiality, Authentication, Integrity, Non-repudiation, access Control and Availability) and Mechanisms, A model for Internetwork security, Internet Standards and RFCs.

**UNIT – II**

Conventional Encryption Principles, Conventional encryption algorithms: DES, TDES, AES, RC4, cipher block modes of operation, location of encryption devices, key distribution, Public key cryptography principles, public key cryptography algorithms: RSA, DIFFIE HELL MAN

**UNIT – III**

Digital signatures, digital Certificates, Certificate Authority and key management, Approaches of Message Authentication, Secure Hash Functions: SHA1 and HMAC

Kerberos, X.509 Directory Authentication Service. Email privacy: Pretty Good Privacy (PGP) and S/MIME.

**UNIT - IV**

IP Security Overview, IP Security Architecture, Authentication Header, Encapsulating Security Payload, Combining Security Associations and Key Management.

**UNIT – V**

Web Security Requirements, Secure Socket Layer (SSL) and Transport Layer Security (TLS), Secure Electronic Transaction (SET). Intruders, Viruses and related threats

## **UNIT – VI**

Firewall Design principles, Trusted Systems. Intrusion Detection Systems.

### **TEXT BOOKS:**

1. Network Security Essentials (Applications and Standards) by William Stallings  
Pearson Education, 4<sup>th</sup> Edition.
2. Hack Proofing your network by Ryan Russell, Dan Kaminsky, Rain Forest Puppy,  
Joe Grand, David Ahmad, Hal Flynn Ido Dubrawsky, Steve W.Manzuik and Ryan  
Permech, wiley Dreamtech

### **REFERENCES:**

1. Fundamentals of Network Security by Eric Maiwald (Dreamtech press)
2. Network Security - Private Communication in a Public World by Charlie Kaufman,  
Radia Perlman and Mike Speciner, Pearson/PHI.
3. Cryptography and network Security, Third edition, Stallings, PHI/Pearson
4. Principles of Information Security, Whitman, Thomson.
5. Network Security: The complete reference, Robert Bragg, Mark Rhodes, TMH
6. Introduction to Cryptography, Buchmann, Springer.



**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**Compiler Design and Information Security Lab**

**Code: 9EC66**

**Prerequisite:** Computer Networks

| L | T | P/D | C   |
|---|---|-----|-----|
| 0 | 0 | 3   | 1.5 |

**Compiler Design Lab**

**Course Objective:**

Experiment on the basic techniques of compiler construction and tools that can be used to perform syntax-directed translation of a high-level programming language into an executable code.

**Course Outcomes:**

After completion of this course student will be able to:

1. Develop C program for a given Languages/ Regular Expression. [L4]
2. Use of LEX of tool to implement lexical analyzer in compiler design and implementation of Top-Down Parser. [L3]
3. Use of YACC tools for implementing bottom up parser. [L3]

**Exercises:**

1. Implement DFA accepting the language containing even binary numbers.
2. Implement DFA that accept all the strings of a's and b's 3<sup>rd</sup> symbol from is RHS always a.
3. Implement DFA accepting the language of strings not ending with 00 over the input (0,1)
4. Implement the DFA that accepts all the string of a's and b's where number of a's is divisible by 3 and number of b's is divisible by 2.
5. Write lex program to implement lexical analyzer functionality.
6. Write a lex program to count the number of words and number of lines in a given file or program.
7. Write a 'C' program to implement lexical analyzer using c program.
8. Write recursive descent parser for the grammar  $E \rightarrow E+T$   $E \rightarrow T$   $T \rightarrow T * F$   $T \rightarrow F$ 
  - a.  $F \rightarrow (E)/id$ .
9. Write recursive descent parser for the grammar  $S \rightarrow (L)$   $S \rightarrow a$   $L \rightarrow L, S$   $L \rightarrow S$
10. Write a C program to calculate first function for the grammar
  - a.  $E \rightarrow E+T$   $E \rightarrow T$   $T \rightarrow T * F$   $T \rightarrow F$
  - b.  $F \rightarrow (E)/id$
11. Write a YACC program to implement top down parser for the given grammar.
12. Write a YACC program to evaluate algebraic expression.

**Information Security Lab**

**Prerequisite :** Nil

**Course Objective:**

Understand the various cryptographic techniques and tools to secure the data.

**Course Outcomes:**

After completion of this course student will be able to:

1. Implement various Authentication and Encryption Algorithms and use NS3 Tool.[L3]

**List of Programs**

1. Implement Substitution Cipher.
2. Implement Transposition Cipher.
3. Implement DES
  - a. Generate Cipher text for the given Plaintext.
  - b. Retrieve the Plaintext from the given Cipher text.
4. Implement Diffie Hellman Algorithm and generate Secret Key.
5. Implement RSA algorithm
  - a. Generate Public key and Private key pair
  - b. Generate Cipher text for the Plaintext
  - c. Obtain the Plaintext from the Cipher text
6. Implement Hash Algorithm.
7. Generate Digital Signature
8. Implement Digital Envelope.
9. Installation of NS3.
10. Demonstration of NS3.
11. Executing simple projects in NS3.
12. Implement a program to transfer file between Client-Server using Socket.

**Syllabus for B.Tech (CSE) IV YEAR I SEM**  
**Linux Programming Lab**

**Code : 9EC67**

**Prerequisite:** Operating Systems

| L | T | P/D | C   |
|---|---|-----|-----|
| 0 | 0 | 3   | 1.5 |

**Course Objective:**

Understand working of Linux Operating System.

**Course Outcomes:**

After completion of this course student will be able to:

1. Describe the basic commands of Linux operating system and can write shell scripts. [L2]
2. Write shell scripts that perform file operations using system calls and utilize process-related utilities for monitoring, diagnosing, and managing processes on Linux systems. [L3]
3. Develop robust and efficient applications that comply with IPC mechanisms, enabling seamless communication and collaboration between processes in a Linux environment. [L5]

**List of Experiments**

1. Basic Linux Commands File handling utilities, Security by file permissions, Process utilities, Disk utilities, sed, awk, grep.
2. Write a shell script that accepts a file name, starting and ending line numbers as arguments and displays all the lines between the given line numbers.
3. Write a shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.
4. Write a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.
5. C programming examples using Linux Operating systems.
  - i. a) wc   b) cat   c) cp
6. Write a shell script that receives any number of file names as arguments checks if every argument supplied is a file or a directory and reports accordingly. Whenever the argument is a file, the number of lines on it is also reported.
7. Write the following Shell scripts:
  - a) To accept a list of file names as its arguments, counts and reports the occurrence of each word that is present in the first argument file on other argument files.
  - b) To list all of the directory files in a directory.

8. To find factorial of a given integer.
  - a. Write an awk script to count the number of lines in a file that do not contain vowels.
  - b. Write an awk script to find the number of characters, words and lines in a file.
9. Implement in C the following UNIX commands using System calls a) rename b) link
10. Write a C program to emulate the UNIX ls – l command.
11. Write a C program on zombie process
12. Write a C program that illustrates the following. a) Creating a message queue. b) Writing to a message queue. c) Reading from a message queue.
13. Write a C program that illustrates file locking using semaphores.
14. Write a C program to implement record locking.
15. Write a C program to implement data communication between two processes using PIPE.

## Syllabus for B.Tech (CSE) IV YEAR I SEM Summer Industry Internship - II

**Code: 9E787**

| L | T | P/D | C |
|---|---|-----|---|
| 0 | 0 | 0   | 1 |

### Course Objective:

Enhance the skills and knowledge in multidisciplinary and thrust areas to apply and solve complex problems.

### Course Outcomes:

After completion of this course, students will be able to:

1. Apply advanced technologies and modern tools to build multidisciplinary products. [L3]
2. Inculcate an enthusiasm to use creative ideas to build innovative projects and prototypes that meet the current needs of the market and society as a whole. [L4]
3. Improve their communicative skills and team skills. [L6]

A summer industry internship project shall be carried out by a group of students consisting of 1 or 2 to 3 in number during the summer second-year semester at industries or institutions. This work shall be carried out under the guidance of the faculty assigned as an internal guide as well as an external guide at the industry where students are carrying out summer industry internship projects. The project shall consist of the design, fabrication, software development, or building of a prototype or application app. This can also be of an interdisciplinary nature.

There will be 100 marks in total with 40 marks for internal evaluation and 60 marks for external

The internal evaluation shall consist of:

|                      |          |
|----------------------|----------|
| Presentation:        | 15 marks |
| Internal Evaluation: | 15 marks |
| Report:              | 10 marks |
|                      | 40 marks |

End examination: 60 Marks.

External Evaluation of the project (viva voce) shall be conducted by a committee appointed by the Chief Superintendent. The end examination will be carried out by a committee consisting of an external examiner, the head of the department, a senior faculty member, and the internal guide.

|                            |          |
|----------------------------|----------|
| Presentation:              | 10 marks |
| Demonstration/ Technology: | 40 marks |
| Report:                    | 10 marks |
|                            | 60 marks |

**Syllabus for B.Tech (CSE) IV YEAR II SEM**  
**PRODUCT AND SERVICES**  
**( Open Elective – III )**

**Code: 9ZC24**

**Prerequisite: Nil**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Study the concepts of Product management and Services Design.

**Course Outcomes:**

After completion of this course student will be able to:

1. Explain the students about basic concepts of product [L2]-U1
2. Demonstrate ten the students with the process of new product development and stages in the process. [L2]-U2
3. Inspect the new product using testing, product planning and the preparatory groundwork for launching a new product [L4]-U3
4. Explains the students to understand the nature of services, its differences with the goods and the application of marketing principles for services [L2]-U4,U5
5. Examines the quality of services and and also helps in knowing various forms of delivering services [L4]-U6

**UNIT- I**

**INTERPRETATIONS ON PRODUCT**

Meaning of Product, Product Market Fit Analysis, Product Levels, Product Mix, Product Pruning, and Product Cannibalization. Types of Product Classification

**UNIT- II**

**PRODUCT INNOVATION and VALIDATION**

New products-What is a new product, New Product Development Process, Idea generation Systems, Concept test, Product testing, Pre-launch, Market test, Final evaluation using “Stage / Gate Process. Product Validity, Break Even Analysis, Financial and Market Analysis.

**UNIT- III**

**LAUNCHING PRODUCT**

Cost Analysis, Steps to fix the final price, Promotion planning, Digital Marketing and Methods, Retailing, Types of Retailing online Retailing, Post Market Analysis of the Launch

**UNIT - IV**

**INTRODUCTION TO SERVICE:**

Meaning and Definition of Service, Characteristics of Services, Classification of Service, Five levels of Service, Service verses Physical Goods, 7 P’s for Marketing of Services, Service scape

**UNIT – V**

**SERVICE PROCESS DESIGN and SERVICE DISTRIBUTION:**

Service Design Meaning, Tools for Service Design, Attributes of a Good Design Customer involvement in the Production Process, Role of Intermediaries, Methods of Distribution of Services

#### **UNIT – VI:**

**QUALITY OF SERVICE:** Definition of Service Quality, Elements of Service Quality, Service Quality Measuring Tools; SERVQUAL Scale, Service Quality Gap Analysis, Objective Service Metrics, Cost of Quality in Service. Challenges and Problems of Service Quality in India.

#### **TEXT BOOKS:**

1. Dr. S.L. Gupta, Product Management, Wisdom Publications
2. C.Merle Crawford ,New Product Management
3. Valarie A.Zeithaml & Mary Jo-Bitner: Services Marketing—Integrating Customer Focus Across the Firm, 3/e, Tata McGraw Hill, 2007.

#### **REFERENCES:**

1. Thomas J.Delong & Asish Nanda: Managing Professional Services—Text and Cases, McGraw-Hill International, 2006.
2. Christopher Lovelock: Services Marketing People, Technology, Strategy, Fourth Edition, Pearson Education, 2006

**Syllabus for B.Tech (CSE) IV YEAR II SEM**  
**PROJECT AND RISK MANAGEMENT**  
**( Open Elective – III )**

**Code: 9ZC19**

**Prerequisite: Software Engineering**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Use knowledge of Project Management and Risk management avenues for making better financial decisions.

**Course Outcomes:**

After completion of this course student will be able to:

1. Describe the concepts of project management and project formulation.[L1]-U1,U2
2. Demonstrate the procedure for project implementation.[L2]-U3
3. Identify the risk management process and methods.[L3]-U4
4. Analyze the role of Derivatives and Options and Swaps as a risk management tool.[L4]-U5,U6

**UNIT I**

**BASICS OF PROJECT MANAGEMENT:** Concept and characteristics of a project - types of projects - Project life cycle - Challenges and problems of project management - Qualities & functions of a project manager.

**UNIT-II**

**PROJECT FORMULATION:** Generation of Project Ideas; Monitoring the environment; Preliminary Screening of Projects; Feasibility study; Project Selection, Detailed Project Report: Market, Technical, Financial and Economic aspects

**UNIT III**

**PROJECT IMPLEMENTATION:** Pre-requisites for Successful Project Implementation; Control of in-progress Projects (Gantt chart, PERT, CPM); Financing of Projects, Project Risk Management Process, Post-audit; Abandonment Analysis

**UNIT-IV**

**RISK MANAGEMENT:** Risk – Definition, Types of Risk, Approaches to Risk Management, Risk Management Process and Techniques.

**UNIT-V**

**DERIVATIVES – I:** Introduction to Financial Derivatives, Types of Derivative Instruments – Forwards, Futures, Differences between Forwards and Futures, Margin Limits.

**UNIT-V1**

**DERIVATIVES –II:**

- a) Options: Definition, Operation of Option contracts, Call Option, Put Option, Factors determining Option values.
- b) Swaps: Definition, Types of Swap agreements, Mechanics of Interest rate swaps, Mechanics of Currency swaps.



**TEXT BOOKS:**

1. H. Nandan, Fundamentals of Entrepreneurship, Prentice Hall of India, First Edition, New Delhi, 2007.
2. Jeffrey K. Pinto “Project Management”, 2<sup>nd</sup> edition, Pearson
3. Dhandapani Alagiri “Structured Finance – Concepts & Perspectives”, ICFAI University press.
4. Projects by Prasanna Chandra, McGraw-Hill Publishing Co. Ltd

**REFERENCES:**

1. Project Management: Systems approach to Planning Scheduling and Controlling, H. Kerzner.
2. The Complete Real Estate Documents by Mazyar M. Hedayat, John J. Oleary
3. The Fundamentals of Listing and Selling Commercial Real Estate - By Keim K. Loren (Author)

**Syllabus for B.Tech (CSE) IV YEAR II SEM**  
**PRINCIPLES OF AUTOMATION AND ROBOTICS**  
**Open Elective – III**

**Code: 9BC53**

**Prerequisite: Nil**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Course Objective:**

Understand intelligent machines that can assist humans in a variety of ways.

**Course Outcomes:**

After completion of this course student will be able to:

1. Explain production system that includes pneumatic and hydraulic components.[L2]-U1
2. Analyze flow lines with various methods of work part transfer mechanical buffer storage control functions.[L4]-U2
3. Build up various automated material handling and storage system models.[L3]-U3
4. Apply adaptive control systems to material requirement plans I and II.[L3]-U4
5. Demonstrate industrial robots with the help of kinematics analysis and type of applications.[L2]-U5,U6

**UNIT – I**

Introduction: Production system, Automated manufacturing systems, Reasons, Principles and strategies of automation, Basic elements of automated system, pneumatic and hydraulic circuit components, Assembly system and line balancing: Manual Assembly process, and work transport systems, Line pacing, Analysis of manual assembly lines, line balancing methods-problems, ways of improving line balance lines.

**UNIT – II**

Analysis of Automated flow lines: System configuration, Workpart transfer, General terminology and analysis of transfer lines without and with buffer storage.

Automated Assembly systems: Fundamentals and Design of assembly systems.

**UNIT – III**

Automated material handling: Principles, Types of equipment, functions, analysis and design of material handling systems, conveyor systems, automated guided vehicle systems-technology, Analysis of material transport systems.

Automated storage systems: Basic terminology, AS/RS; Carousel storage, work in process storage,

**UNIT – IV**

Adaptive control systems: Introduction, Adaptive control with optimization, Adaptive control with constraints, Application of A.C. in Machining operations. Use of various parameters such as cutting force, Temperature, vibration and acoustic emission. Concept of Concurrent Engineering, MRP, MRP II, Techniques of Rapid Proto typing.

## **Unit – V**

### **Robotics:**

Classification and structure of Robotic systems, structure of continuous path robot systems, drives and control systems, control approaches for robots.

## **Unit – VI**

Robot arm kinematics, the direct kinematics problem and inverse kinematic solutions, planning of manipulator trajectories, robot sensors, range sensors, proximity sensors, touch sensors, force and torque sensors, programming, manual teaching, lead through teaching, programming languages, storing and operating task programmes, robot selection and application.

### **TEXT BOOKS:**

1. Automation, Production Systems and Computer Integrated Manufacturing: M.P. Groover./PE/PHI
2. Mittal and Nagrath, 'Robotics and Control', Tata Mc Graw Hill.

### **REFERENCES:**

1. Computer control of Manufacturing Systems by Yoram Coreom.
2. CAD / CAM/ CIM by Radhakrishnan.

**Syllabus for B.Tech (CSE) IV YEAR II SEM**  
**POWER ELECTRONIC DEVICES AND CONVERTERS**  
**(Open Elective – III)**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Code: 9AC47**

**Prerequisite:** Basic Electrical and Electronics Engineering

**Course Objective:**

Learn the basic concepts of power semiconductor devices, Converters and choppers and their analysis.

**Course Outcomes:**

After completion of this course student will be able to:

1. Illustrate the construction and operation of various power semiconductor devices and analyze about the series and parallel operation of SCRs.[L2]-U1
2. Analyze the operation of different configurations of single phase and three phase converters for different loads.[L4]-U2,U3
3. Explain the operation of different type's choppers and inverters with its applications.[L6]-U4,U5
4. Explain the working of an AC voltage controller and Cyclo-Converters for different configurations.[L6]-U6

**UNIT – I**

**POWER SEMI CONDUCTOR DEVICES:**

Thyristors, Silicon Controlled Rectifiers (SCR's), BJT, Power MOSFET, Power IGBT, DIAC, TRIAC, GTO and their characteristics. Basic theory of operation of SCR, Static characteristics, Two transistor analogy, Turn on and turn off methods, Dynamic characteristics of SCR, Turn on and Turn off mechanism., SCR, UJT firing circuit, Series and parallel connections of SCR's, Snubber circuit details, Specifications and Ratings of SCR's, BJT, IGBT.

**UNIT – II**

**SINGLE PHASE CONTROLLED CONVERTERS:**

Phase control technique, Single Phase Line commutated converters, Midpoint and Bridge connections; Half controlled and Fully controlled converters, Derivation of average load voltage and current with R and RL loads.

**UNIT – III**

**THREE PHASE CONTROLLED CONVERTERS:**

Three phase half controlled and fully controlled bridge converters with R and RL loads, Effect of Source inductance, Waveforms, Numerical Problems.

**UNIT – IV**

**CHOPPERS:**

Choppers, Time ratio control and Current limit control strategies, Step down choppers Derivation of load voltage and currents with R, RL and RLE loads, Step up Chopper, load

voltage expression, Jones chopper and waveforms, Problems, Buck, Boost, Buck-Boost choppers.(Qualitative treatment).

## **UNIT – V**

### **INVERTERS:**

Inverters, Single phase inverter, Half and Full bridge VSI & CSI inverters, Waveforms, Voltage control techniques for inverters, Three phase inverters with 120degrees and 180 degrees mode of conduction, Pulse width modulation techniques (Multiple Pulse and Sinusoidal), Numerical problems.

## **UNIT –VI**

### **AC VOLTAGE CONTROLLERS & CYCLO CONVERTERS:**

AC voltage controllers, Single phase two SCR's in anti parallel with R and RL loads, Derivation of RMS load voltage, current and power factor wave forms, Firing circuits, Numerical problems, Cyclo converters, Single phase midpoint cyclo converters with Resistive and inductive load (Principle of operation only), Bridge configuration of single phase cyclo converter (Principle of operation only), Waveforms

### **TEXT BOOKS:**

1. Power Electronics - P.S.Bimbhra, Khanna Publishers.
2. Power Electronics Circuits, Devices and Applications - M. H. Rashid, Prentice Hall of India, 2nd edition.

### **REFERENCES:**

1. Power Electronics – Vedam Subramanyam, New Age International (P) Limited, Publishers.
2. Power Electronics - V.R.Murthy 1st edition, OXFORD University Press.
3. Power Electronics - P.C.Sen, Tata Mc Graw Hill Publishing.
4. Power Electronics - M. D. Singh & K. B. Kanchandhani, Tata Mc Graw Hill Publishing Company.

**Syllabus for B.Tech (CSE) IV YEAR II SEM**  
**Embedded Systems**  
**(Open Elective – III)**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Code: 9CC38**

**Prerequisite:** Digital Electronics, Computer Organization

**Course Objective:**

Demonstrate embedded system design architecture, communication interface and realtime operating systems.

**Course Outcomes:**

After completion of this course student will be able to:

1. Identify the design techniques of a modern embedded system.[L1]-U1
2. Develop assembly language programs for 8051 architecture and Design hardware interface of I/O devices.[L4]-U2,U3
3. Demonstrate interfaces for Embedded System using various protocols and hardware modules.[L3]-U4,U5
4. Explain the concepts and design requirements related to a real time systems.[L2]-U6

**UNIT – I**

**Introduction to Embedded Systems**

Embedded Systems, Comparing Embedded and General Computing, Complex System Design and Processors, Classification of Embedded Systems, Embedded System Design Process, Formalization of System Design, Embedded SOC and VLSI Circuit Technology, Application examples of Embedded Systems.

**UNIT – II**

**8051 Architecture, Memory Organization and Programming**

8051 Architecture, features, Addressing modes, Instruction set, Input/Output Ports and Circuits, External Memory, Counter and Timers, Serial data, Input/Output, Interrupts; The Assembly Language programming Process, Programming the 8051, Data Transfer and Logical Instructions. Arithmetic Operations, Decimal Arithmetic. Jump and Call Instructions, use of C programming for 8051.

**UNIT – III**

**8051 Real World Interfacing**

Part A - Real World Interfacing, Performance metrics, Memory map, Processor and Memory selection,

Part B - IO Subsystem, Sensors and Actuators, LED and LCD Interfacing, Keyboard Interfacing, Stepper Motor Interfacing, DC motor Interfacing Using PWM

**UNIT – IV**

**Embedded Communication Interface**

Serial and Parallel Communication, Timer and Counting Devices, Watchdog Timer, Real Time Clock, I<sup>2</sup>C, SPI protocol, ISA, PCI, Internet Enabled Systems, Wireless and Mobile Systems Protocols

## **UNIT – V**

### **Introduction to Real - Time Operating Systems**

Tasks and Task States, Tasks and Data, Semaphores, and Shared Data; Message Queues, Mailboxes and Pipes, Timer Functions, Events, Memory Management, Interrupt Routines in an RTOS Environment.(Chapter 6 and 7 from Text Book 3, Simon).

## **UNIT – VI**

### **Basic Design Using a Real-Time Operating System:**

Principles, Semaphores and Queues, HardReal-Time Scheduling Considerations, Saving Memory and Power, An example RTOS like uC-OS (Open Source);

**Embedded Software Development Tools:** Host and Target machines, Linker! Locators for Embedded Software, Getting Embedded Software into the Target System; Debugging

### **TEXT BOOKS:**

1. Embedded Systems- Architectuer, Programming and Design 2E, Raj Kamal, TMH
2. Introduction to Embedded Systems, K.Shibu, Tata McGraw-Hill
3. The 8051 Microcontroller And Embedded Systems Using Assembly And C – Mazidi, Pearson Education India,2<sup>nd</sup> edition, 2008.
4. An Embedded Software Primer, David E. Simon, Pearson Education

### **REFERENCES:**

1. Computers and Components: principles of embedded *computing* system design, Wayne Wolf, Elseveir.
2. 8051 Application Notes by Atmel.

**Syllabus for B.Tech (CSE) IV YEAR II SEM  
ENTREPRENEURSHIP & BUSINESS DESIGN  
(Open Elective – III)**

| L | T | P/D | C |
|---|---|-----|---|
| 3 | 0 | 0   | 3 |

**Code: 9ZC10**

**Prerequisite:** Basics of Entrepreneurship

**Course Objective:**

Understand the essentials of building their startups and to familiarize with business design process develop business models, and market their product.

**Course Outcomes:**

After completion of this course student will be able to:

1. Explain the essential concepts of entrepreneurship.[L2]-U1
2. Distinguish the different phases of UI/UX.[L2]-U2
3. Develop the attentiveness on design of business strategy and services.[L3]-U3,U4
4. Illustrate the concept of business model and its validation.[L4]-U5
5. Identify the sources of funding and risk assessment. [L3]-U6

**UNIT I**

**Introduction to Entrepreneurship:** Meaning of Entrepreneurship. Reasons feeding the Entrepreneurial fire. Understanding Entrepreneurship as a Process. Multiple roles of Entrepreneur: Intrapreneur, Inventor, Coordinator, Manager and Controller. Psychological and behavioural aspects of First-Generation Entrepreneur.

**UNIT II**

**Introduction to UI/UX:** Human centered design and benefits, the distinction between UX and UI, UX process – user research, prototyping strategies, UI principles, UI analysis, UI design, UI components and Responsive design.

**UNIT III**

**Designing a Business Strategy:** Define a problem and frame a strategic question, map the lives of users, journey mapping and ideation, color theory, killing the ideas through Stage Gate Models, pitching of full-fledged, idea, choosing the Start-Up Team.

**UNIT IV**

**Designing Services and Services Delivery:** Services as solutions, Service delivery pathways, rapid branding and marketing strategies, key metrics for Design thinking. Types of New services, Mix of core services and secondary and enhancing services, service flower and service design matrix.

**UNIT V**

**Business Model:** Meaning of business model, Difference between business model and business planning, the business model canvas, Risks and Assumptions, Validation of business models, building solution demo and MVP, revenue streams and pricing strategies.



## **UNIT VI**

**Entrepreneurial Funding and Risk Management:** Bootstrapping, Angel Investors, Venture capitalists, Private equity funding, customer acquisition, return on equity and Break even analysis, Risk propensity Vs. Risk avoidance, Locus of control of entrepreneur, Risk estimation techniques, risk avoidance strategies.

### **TEXT BOOKS:**

1. Adrian McEwen, Hakim Cassimally – “Designing the Internet of Things”, Wiley Publications, 2012
2. Hedman, J., &Kalling, T. (2003). The business model concept: theoretical underpinnings and empirical illustrations. European journal of information systems, 12(1), 49-59.
3. Cabrera, J. (2017). Modular Design Frameworks: A Projects-based Guide for UI/UX Designers. Apress.

### **REFERENCES:**

1. J. Chris Leach & Ronald W. Melicher “Entrepreneurial Finance, Fourth Edition”, South Western, Cengage Learning, 2012.
2. Robert D. Hisrich & Veland Ramadani – “Effective Entrepreneurial Management, Strategy, Planning, Risk Management, and Organization” , Springer, 2017.
3. Mæhlum, A. R. (2017). Extending the TILES Toolkit-from Ideation to Prototyping (Master's thesis, NTNU).
4. Norman, D. (2013). The design of everyday things: Revised and expanded edition. Basic

**Syllabus for B.Tech (CSE) IV YEAR II SEM**  
**Computer Science and Engineering**  
**CYBER SECURITY AND CYBER LAWS**

**Code: 9FC07**

**Prerequisite:** Intellectual Property Rights, Cryptography

| L | T | P/D | C |
|---|---|-----|---|
| 2 | 0 | 0   | 2 |

**Course Objective:**

Illustrate cybercrime violations and penalties, and to interpret security policies and laws associated with it.

**Course Outcomes:**

After completion of this course student will be able to:

1. Summarize the cryptographic procedures and understand its primitives. [L2]-U1
2. Outline Security policy in Legislation and Comprehend E-Commerce frame work, models and its associated threats.[L4]-U2
3. Demonstrate the role of electronic signatures in E-Commerce and summarize the various Laws relating to it.[L2,L3]-U3
4. Categorize international cyber laws and cybercrimes and explore Penalties, Compensation and Adjunction of violations of provisions of IT Act 2000.[L4]-U4,U5
5. Classify and outline the offences under the Cyberspace law and the Internet in India.[L2]-U6

**UNIT-I**

**Introduction to cyber Security, cryptography, Types of Attacks, Secret Key Cryptography**

Introduction: Cyber attacks, Defence Strategies and Techniques Mathematical background for Cryptography: Modulo arithmetic, The greatest common divisor, Useful Algebraic Structures, Chinese Remainder Theorem Basics of Cryptography: Secret versus Public key Cryptography, Types of attacks, Elementary substitution Ciphers, Elementary Transposition Ciphers, Other Cipher Properties Secret Key Cryptography: Product Ciphers, DES Construction, Modes of Operation, MAC and other Applications, Attacks, Linear Crypt analysis.

**UNIT-II**

**Introduction to Computer Security, Internet, E-commerce and E-governance with reference to Free Market Economy**

Definition, Threats to security, Government requirements, Information Protection and Access Controls, Computer security efforts, Standards, Computer Security mandates and legislation, Privacy considerations, International security activity, Conceptual Framework of E-commerce: governance, the role of Electronic Signatures in E-commerce with Reference to Free Market Economy in India.

### **UNIT-III**

#### **Law relating to electronic records and intellectual property rights in India**

Legal aspects of Electronic records / Digital signatures, Cyber laws, the roles and regulations of Certifying Authorities in India, Protection of Intellectual Property Rights in Cyberspace in India.

### **UNIT-IV**

#### **International Efforts relating to Cyberspace laws and Cyber crimes**

International efforts related to Cyber laws, Council of Europe (COE) convention on Cyber Crimes.

### **UNIT-V**

#### **Penalties, Compensation**

Penalties, Compensation and Adjunction of violations of provisions of IT Act 2000 and judicial review.

### **UNIT-VI**

#### **Offences under the Cyberspace, Internet in India and Miscellaneous provisions of IT Act and Conclusions**

Some important offences under the Cyberspace law and the Internet in India, Other offences under the Information Technology Act in India, The role of Electronic Evidence and miscellaneous provisions of the IT Act.

#### **TEXT BOOKS:**

1. Network security and Cryptography by Bernard Menezes CENGAGE Learning Publications, 2010.
2. Cyber Laws and IT Protection, Harish Chander, PHI, 2012.

#### **REFERENCES:**

1. Debby Russell and Sr. G.T Gangemi, "Computer Security Basics (Paperback)", 2nd Edition, O' Reilly Media, 2006.
2. Wenbo Mao, "Modern Cryptography – Theory and Practice", Pearson Education, New Delhi, 2006.
3. Cyberspace and Cybersecurity, George Kostopoulos, Auerbach Publications, 2012.
4. Cyber Forensics: A Field Manual for Collecting, Examining, and Preserving Evidence of Computer Crimes, Second Edition, Albert Marcella, Jr., Doug Menendez, Auerbach Publications, 2007

**Syllabus for B.Tech (CSE) IV YEAR II SEM  
MAJOR PROJECT**

**Code: 9E888**

| L | T | P  | C  |
|---|---|----|----|
| 0 | 0 | 20 | 10 |

**Course Objective:**

To enhance the knowledge on selecting a domain, identify problem statements, usage of tools to develop products or software to solve industrial needs or societal problems.

**Course Outcomes:**

After completion of this course, students will be able to:

1. Apply engineering principles and other relevant disciplines to solve complex problems. [L3]
2. Use research-based knowledge to design and develop the software or prototype using modern software tools wherever applicable. [L3]
3. Defend effectively in professional contexts [L5]
4. Identify the importance of professional responsibility and making informed judgments based on legal and ethical principles. [L3 & L5]
5. Function effectively and efficiently as a member of the team or a team leader [L5]

A project shall be carried out by a group of students consisting of 1 or 2 to 3 in number in the fourth year second semester. This work shall be carried out under the guidance of the faculty assigned as an internal guide and shall involve design, fabrication, software/ product development, or any other significant activity. This can also be of an interdisciplinary nature.

Out of a total of 100 marks for project work (in the final year second semester), 40 marks shall be for Continuous Internal Evaluation and 60 marks for the Semester External Evaluation at the end of the Semester.

Semester End External Evaluation of the project (viva voce) shall be conducted by a committee appointed by the Chief Superintendent. The committee consists of an external examiner, an HOD, a Senior Faculty Member, and an Internal Guide.

### **Division of Marks for Continues Internal Evaluation –40 marks**

| <b>S1.No</b> | <b>Description</b>                                               | <b>Marks</b> |
|--------------|------------------------------------------------------------------|--------------|
| 1            | Abstract Seminar at the end of 3 weeks                           | 5            |
| 2            | Design Seminar at the end of 8 weeks                             | 5            |
| 3            | Implementation / Demonstration of Seminar at the end of 12 weeks | 10           |
| 4            | Evaluation of Project by Internal Guide                          | 10           |
| 5            | Project Report                                                   | 10           |
| 6            | Total                                                            | 40           |

### **Division of Marks for Semester End External Evaluation –60 Marks**

| <b>Sl.No</b> | <b>Description</b>               | <b>Marks</b> |
|--------------|----------------------------------|--------------|
| 1            | Project Report                   | 10           |
| 2            | Presentation                     | 20           |
| 3            | Demonstration/Defense of project | 30           |
|              | Total                            | 60           |

A student must secure 40% of the total marks to obtain a pass grade.